QUESTIONS ON HARDY SPACES AND HANKEL OPERATORS

ØISTEIN SØVIK

Answer. Open means no informed opinion. Expected yes or Expected no gives a belief with reasons, without a proof. Answered gives the proposed answer; Partial names what remains open.

Evidence. Heuristic means reasons or numerics; Sketch means an argument by Søvik, with or without AI assistance; arXiv means a posted preprint or a public institutional report; Published means a peer-reviewed result. Sketch arguments have not been through arXiv or peer review and should be checked by the reader. Conditional on names an unrefereed result or standing hypothesis used by the argument.

The labels describe the evidence collected here as of 13 October 2026; an Open label is not a claim that no solution exists.

Contents

Part 1. Replacing factorization: point evaluations 2
1. Embeddings and boundary values 2
2. Hellinger–Bessel inequalities 5
3. Nevanlinna classes 6
4. Paley–Wiener spaces 7
5. Bernstein–Nikolskii inequalities 14
6. Nehari theorems and Hankel forms 16
7. Composition operators 17

Part 2. Where Nehari's theorem survives: structured Helson matrices 17
8. Embeddings and boundary values 17
9. Nehari theorems and Hankel forms 17
10. Hilbert points 22
11. Fock-space spectral optimization 23
12. Hankel forms and Schatten classes 24
13. Coefficient inequalities 26
14. Contractivity and hypercontractivity 26

Part 3. Where Nehari's theorem fails: Schatten classes 27
15. Paley–Wiener spaces 27
16. Hankel forms and Schatten classes 28

Part 4. The borderline: the multiplicative Hilbert matrix 33
17. Paley–Wiener spaces 33
18. Nehari theorems and Hankel forms 33
19. Hilbert-type forms 34
20. Coefficient inequalities 35
21. Dirichlet series and frequencies 37

Part 5. Related function-theoretic questions 37

Date: 13 October 2026.

22. Extremal coefficient problems 37
23. Hardy and Bergman extremals 38
24. Contractivity and hypercontractivity 41
25. Bohr radii and monomial convergence 48
26. Random multiplicative questions 51
27. Composition operators 58
28. Dirichlet series and frequencies 64
29. Vertical limits 68
30. Optimal polynomial approximants and cyclicity 69

Part 1. Replacing factorization: point evaluations

1. Embeddings and boundary values

1.1. Removing a spectrum restriction in the infinite polydisc

Is it possible to remove the restriction on the Fourier spectrum in Theorem 1?

Aleman, Olsen, and Saksman, *Fatou and brother Riesz theorems in the infinite-dimensional polydisc*, Question 1, p. 16.
Answer: Partial. **Evidence:** Sketch.

The source's unrestricted part (i) is not the issue; the open endpoint content is the finite-measure weak $(1,1)$ bound and full almost-everywhere convergence in parts (ii) and (iii). Mixed-sign frequencies remain outside the established restricted-spectrum theorem.

1.2. Radial failure along decreasing coordinate radii

Does there exist a bounded analytic function $f \in H^\infty(\mathbb{T}^\infty)$ such that almost surely the radial convergence fails even if the approach is limited by assuming decreasing radii in n, i.e. $r_{n,k} \geq r_{n+1,k}$ for all n, k?

Aleman, Olsen, and Saksman, *Fatou and brother Riesz theorems in the infinite-dimensional polydisc*, Question 2, p. 16.
Answer: Answered: yes. **Evidence:** arXiv.

Sun, Lu, and Zu's latest preprint constructs the requested bounded analytic function and verifies failure of almost-sure convergence along an ordered approach.
Conditional on: Sun, Lu, and Zu, arXiv:2608.03164, latest unrefereed version.

1.3. Boundary-independent radial failure

What is the answer to the above question under the added condition that the radial approach does not depend on the point on the boundary?

Aleman, Olsen, and Saksman, *Fatou and brother Riesz theorems in the infinite-dimensional polydisc*, Question 3, p. 16.
Answer: Answered: yes. **Evidence:** arXiv.

The same current preprint gives a boundary-independent construction. Its theorem and deduction from the source's good ordered path provide the claimed nonconvergence example.
Conditional on: Sun, Lu, and Zu, arXiv:2608.03164, latest unrefereed version.

1.4. Nontangential Fatou theory in infinitely many variables

What can one say about non-tangential approach for functions in $H^\infty(\mathbb{T}^\infty)$, or more generally, in $H^p(\mathbb{T}^\infty)$?

Aleman, Olsen, and Saksman, *Fatou and brother Riesz theorems in the infinite-dimensional polydisc*, Question 4, p. 16, corrected arXiv version.
Answer: Partial. **Evidence:** Sketch.

Aleman, Olsen, and Saksman establish Fatou limits in correlated one-parameter tube regions and sharpness for broader unrestricted approaches. A classification for general coordinatewise nontangential regions remains open.

1.5. **Uniqueness for radial approaches**

For which radial approaches is the brothers Riesz theorem on the uniqueness of boundary values true? That is, when is it true for all $f \in H^\infty(\mathbb{T}^\infty)$ that having vanishing radial boundary values on a set of positive Lebesgue measure implies $f \equiv 0$?

Aleman, Olsen, and Saksman, *Fatou and brother Riesz theorems in the infinite-dimensional polydisc*, Question 5, p. 16.
Answer: Partial. **Evidence:** Sketch.

Aleman, Olsen, and Saksman prove that a predictable-tail class of approaches is sufficient, and classify a factorized model. Arbitrary boundary-dependent approaches are not classified.

1.6. **Carleson interpolation beyond reciprocal-even exponents**

We would like to know if this result extends to other values of p.

Saksman and Seip, *Some open questions in analysis for Dirichlet series*, §2, paragraph after Problem 2.4. The inferred target is a necessary-and-sufficient criterion for interpolation with the standard normalized data norm.
Answer: Partial. **Evidence:** Sketch.

Saksman and Seip prove the reciprocal-even cases; later necessary product conditions cover additional ranges. No full criterion for arbitrary finite p and bounded distinct sequences is known.

1.7. **Local vertical embeddings for $\mathscr{H}^p$**

Is the L^p integral of a Dirichlet polynomial over any segment of fixed length on the vertical line $\operatorname{Re} s = 1/2$ bounded by a universal constant times its $\mathscr{H}^p$ norm?

Saksman and Seip, *Some open questions in analysis for Dirichlet series*, Problem 2.1, pp. 3–4.
Answer: Answered: exactly for $p \geq 2$. **Evidence:** arXiv.

Chen, Fang, Guo, Hou, Shao, and Zhou prove the inequality for every finite $p \geq 2$ and rule it out below 2.
Conditional on: Chen and collaborators, arXiv:2609.11560, latest unrefereed version.

1.8. **Carleson measures for $\mathscr{H}^p$ below two**

Characterize Carleson measures for $\mathscr{H}^p$ on $\{\operatorname{Re} s > 1/2\}$ for $p < 2$.

Saksman and Seip, *Some open questions in analysis for Dirichlet series*, Problem 2.2.
Answer: Partial. **Evidence:** Sketch.

Necessary Carleson-square estimates and failures for bounded-support examples are known. They do not give a necessary-and-sufficient condition for arbitrary finite measures.

1.9. **The Blaschke condition below two**

Do the zero sets of functions in $\mathscr{H}^p$ for $p < 2$ satisfy the Blaschke condition in the half-plane $\operatorname{Re} s > 1/2$?

Saksman and Seip, *Some open questions in analysis for Dirichlet series*, Problem 2.3.
Answer: Answered: no. **Evidence:** Sketch.

A lacunary sum of long Dirichlet blocks belongs to every $\mathscr{H}^p$ with $p < 2$ and has zeros violating the global half-plane Blaschke sum. The full construction is in the answers entry. *Answers*, "The Blaschke condition below two", p. 2.

1.10. **Local Nevanlinna membership below two**

Are elements of $\mathscr{H}^p$ for $p < 2$ locally in the Nevanlinna class?

Saksman and Seip, *Some open questions in analysis for Dirichlet series*, Problem 2.4.
Answer: Answered: no. **Evidence:** Sketch.

The same block construction has unbounded local logarithmic means at the critical line, so it fails the classical boundary-local Nevanlinna condition. The full argument is in the answers entry.
Answers, "Local Nevanlinna membership below two", p. 2.

1.11. **Weighted critical-line trace inequalities**

An interesting open problem is to describe the non-negative weights ω for which there is a finite constant $C(\omega)$ such that $\int_{\mathbb{R}} |f(1/2+it)|^2\omega(t)\ \mathrm{d}t \le C(\omega)\|f\|^2_{\mathscr{H}^2}$.

Brevig and Perfekt, *A mean counting function for Dirichlet series and compact composition operators*, §7, remark following Lemma 7.2 and equation (7.4).
Answer: Partial. **Evidence:** Sketch.

Necessarily $\omega \in L^1 \cap L^\infty$, and the source proves sufficiency for even weights decreasing on $[0,\infty)$. A characterization without monotonicity remains open.

1.12. **Extension of Smirnov functions to the square-summable polydisc**

Does there exist $f \in N_*(\mathbb{T}^\infty)$ that cannot be extended to a holomorphic function on $\mathbb{D}^\infty \cap \ell^2$?

Guo, Ni, and Zhou, *Nevanlinna class, Dirichlet series and Szegő's problem*, Question 2.28, §2, p. 20.
Answer: Answered: no. **Evidence:** Sketch.

Every such function extends holomorphically to $\mathbb{D}^\infty \cap \ell^2$. A full proof is in the public note. The proof

1.13. **Holomorphic Carleson measures on finite polydiscs**

Nonetheless, a complete characterization of Carleson measures for the holomorphic Hardy spaces in higher dimensions has yet to emerge. In particular it is not clear whether the sufficient condition of Chang–Stein is also necessary, or if the class of $H^p(\mathbb{D}^d)$ Carleson measures is the same for every $p \ge 1$ for that matter.

Chalmoukis and Dayan, *Simply interpolating and Carleson sequences for Hardy spaces in the polydisc*, §1, p. 3. We isolate those two questions for finite-dimensional polydiscs.
Answer: Partial. **Evidence:** Sketch.

Chalmoukis and Dayan identify the same Carleson class for $p = 1, 2$ and prove Chang–Stein necessity and sufficiency under a structured sector hypothesis. The arbitrary-measure characterization and all $p \ge 1$ remain open.

1.14. **Small-exponent Carleson classes for Dirichlet series**

Below $p = 2$, Carleson measures don't depend on p.

Søvik, *Where Nehari Survives*, Chapter 3, question B4. The scoped statement asks whether one subcritical Carleson exponent forces all exponents.
Answer: Answered: no. **Evidence:** Sketch.

For each $q \in (1,2)$, a block construction gives a positive measure that is $\mathscr{H}^q$-Carleson but not $\mathscr{H}^1$-Carleson. The complete construction is given in the answers entry.
Answers, "Small-exponent Carleson classes for Dirichlet series", p. 3.

1.15. **Vertical blow-up rates below $p = 2$**

Determine $N_p(\varepsilon) = \sup_{f\neq 0} \int_0^1 |f(1/2+\varepsilon+it)|^p\ \mathrm{d}t/\|f\|^p_{\mathscr{H}^p}$ for $p < 2$.

Søvik, the open question on vertical lines in the critical-line chapter. At $p = 1$ the proposed order is $(\log(1/\varepsilon))^{1/4}$.
Answer: Partial. **Evidence:** Sketch.

Harper's first-moment estimate supplies a lower-order input, and the checked argument gives complementary upper and lower estimates. The exact growth for $p < 2$, including the proposed $p = 1$ order, remains open.

1.16. **Window lengths for mean values of Dirichlet polynomials**

How does $T_p(N, C)$ depend on p? In particular, is $T_p(N, C) = O(N)$ for $1 \le p < 2$?

Seip, *Two Problems on Dirichlet Polynomials*, in *Dirichlet Series and Function Theory in Polydiscs*, Oberwolfach Report 06/2014, Problem 4(b), p. 387.
Answer: Partial. **Evidence:** Published.

For $p = 2$, the Montgomery–Vaughan mean-value theorem gives the linear window bound. Seip leaves the range $1 \le p < 2$ open.

1.17. **A logarithmic trace to the critical line**

Does a universal constant bound $\int_0^1 \log(1 + |F(r(\sigma))|^2)\, \mathrm{d}\sigma$ by $\int_{\mathbb{T}^\infty} \log(1 + |F|^2)\, \mathrm{d}m$ for analytic Dirichlet polynomials?

Søvik, the centered-defect version of the logarithmic embedding question.
Answer: Partial. **Evidence:** Sketch.

A necessary centered-defect estimate is proved with a finite lower bound on its constant. Neither that bound nor the separate completed-Ne-vanlinna trace extension resolves the uniform inequality.

1.18. **Bounded-support Carleson exponents above two**

Theorem 22 invites the following question. Question. If μ defines an H^p-Carleson measure on $\mathbb{D}^\infty$ for some $0 < p < \infty$, does it also define an H^q-Carleson measure for every $p < q < \infty$?

Bayart, Brevig, Haimi, Ortega-Cerdà, and Perfekt, *Contractive inequalities for Bergman spaces and multiplicative Hankel forms*, §5, question after Theorem 22, p. 26. The bounded-support implication below is a separate inferred scope.
Answer: Partial. **Evidence:** arXiv.

The latest preprint proves the implication for bounded-support measures. It does not settle the original all-measure question or the full range of exponents below two.
Conditional on: Chen and collaborators, arXiv:2609.11560, latest unrefereed version.

2. Hellinger–Bessel inequalities

2.1. **The sharp real-curve constant for general exponents**

For $p > 0$, let

$$C_p = \sup_{f \neq 0} \frac{\int_0^1 |f(1/2 + \sigma)|^p \, \mathrm{d}\sigma}{\|f\|_{\mathcal{H}^p}^p},$$

where f ranges over Dirichlet polynomials including the constant coefficient.

Determine the sharp constant for general $p > 0$.

Søvik, *Where Nehari Survives*, Hellinger–Bessel chapter, final open-problem discussion, second question after the result on C_2.
Answer: Partial. **Evidence:** Sketch.

Our proof establishes the full-space value $C_2 = \pi$ and the comparisons $C_{2m} \le \pi$ and $C_{2/m} \ge \pi$ for positive integers m. The sharp constants at other exponents, including $p = 1$, remain open.

3. Nevanlinna classes

3.1. **Interchanging mean-counting limits**

For $f \in \mathcal{N}_u$, can the $\sigma \to 0$ and $T \to \infty$ limits in $M_f(w)$ be interchanged for every $w \neq f(+\infty)$?

Brevig and Perfekt, *A mean counting function for Dirichlet series and compact composition operators*, §8.1, Problem 1, p. 33.
Answer: Answered: no. **Evidence:** Published.

Brevig, Kouroupis, and Perfekt give a counterexample to the interchange in their 2025 published paper on mean-counting limits.

3.2. **A logarithmic local embedding for $\mathcal{N}_u$**

Does there exist a constant $C > 0$ such that, for every $f \in \mathcal{N}_u$ with $\sigma_u(f) \leq 0$, $\sup_{\sigma>0,\,\tau\in\mathbb{R}} \int_\tau^{\tau+1} \log(1 + |f(\sigma + it)|)\, \mathrm{d}t \leq C\|f\|_0$?

Allu and Mondal, *Composition operators and Carleson embeddings for the Nevanlinna class of Dirichlet series*, Problem 4.13, p. 25.
Answer: Answered: no. **Evidence:** Sketch.

A prime-clustered construction violates the proposed estimate. The proof is in the public local-embedding note; it concerns the stated local logarithmic bound. The proof

3.3. **Characterizing composition operators on $\mathcal{N}_u$**

It is natural to investigate which analytic symbols Φ generate bounded composition operators C_Φ on the Dirichlet–Nevanlinna class $\mathcal{N}_u$.

Allu and Mondal, *Composition operators and Carleson embeddings for the Nevanlinna class of Dirichlet series*, §1. The curated question asks for correct criteria for continuity or Lipschitz continuity in d_0; Theorems 1.3–1.4 motivate but do not settle that inferred target.
Answer: Partial. **Evidence:** Sketch.

Explicit symbols show that the displayed half-plane inclusion and a compact-image condition do not suffice for continuity. The exact correct conditions for boundedness or continuity remain undetermined.

3.4. **Removing a compact-image assumption in boundary composition**

It would be desirable to eliminate the assumption that $\Phi(\mathbb{C}_0)$ is a compact subset of $\mathbb{C}_0$ in our Lemma 4.4, following the approach of Brevig and Perfekt [12].

Allu and Mondal, *Composition operators and Carleson embeddings for the Nevanlinna class of Dirichlet series*, Remark 4.5, p. 27. The curated question infers whether the lemma remains true without that compact-image hypothesis.
Answer: Answered: no. **Evidence:** Sketch.

A boundary-approaching symbol makes the proposed unrestricted push-forward identity fail. The question is inferred from the compact-image hypothesis in the quoted source passage.

3.5. **A counting formula for the completed Nevanlinna class**

However, for the Nevanlinna–Dirichlet class $\mathcal{N}$, defined as the completion of $\mathcal{N}_u$, the existence of a corresponding counting function yielding a Stanton-type formula remains an open question.

Allu and Mondal, *Composition operators and Carleson embeddings for the Nevanlinna class of Dirichlet series*, Remark 4.2, p. 22.
Answer: Partial. **Evidence:** Sketch.

A conditional potential identity is established for a restricted completion-level setup. A general counting function and Stanton formula for the completed class remain open.

3.6. **Extending boundary composition to the completion**

It remains an open question whether Lemma 4.4 can be extended to the full Nevanlinna class $\mathcal{N}$.

Allu and Mondal, *Composition operators and Carleson embeddings for the Nevanlinna class of Dirichlet series*, Remark 4.5, pp. 27–28.

Answer: Answered: no. **Evidence:** Sketch.

A lacunary H^1 function composed with a suitable characteristic-zero symbol has divergent local logarithmic means, contradicting any finite completion-valued extension. The full proof is in the answers entry.

Answers, "Extending boundary composition to the completed Nevanlinna class", p. 5.

4. Paley–Wiener spaces

4.1. **A quantitative lower bound below exponent two**

There is a positive constant A such that $\mathcal{C}_p \geq p/2(1 - A(p-2))$ for $1 \leq p \leq 2$.

Conjecture 9.2, printed p. 56, of Brevig, Chirre, Ortega-Cerdà and Seip, *Point evaluation in Paley–Wiener spaces*. The source's stated interval and sign convention are part of the question.

Answer: Answered: yes. **Evidence:** Sketch.

A computer-assisted proof gives the bound with $A = 7/100$ throughout $1 \leq p \leq 2$. The certificate and proof are in the point-evaluation note.

4.2. **Uniform boundedness of Fourier interpolation basis functions**

There exists $C > 0$ such that $|f_n(x)| < C$ holds for all $n \in \mathbb{Z}_{\geq 0}$, $x \geq 0$.

Conjecture 8.1, §8.2, printed p. 38, of David Berghaus, Andriy Bondarenko, Danylo Radchenko, Kristian Seip and Qihang Sun, *The basis functions of Fourier interpolation*, arXiv: 2512.18677v1. Here $f_n(x) = b_n^+(\sqrt{x})$ is the Fourier-invariant Radchenko–Viazovska basis function in the source's notation.

Answer: Open. **Evidence:** Heuristic.

The source reports numerical values mostly below two, but its discussion supplies no uniform analytic bound in both n and x.

4.3. **Statistical distribution of the basis-function values**

Does the sequence $\{n^{1/4} f_n(x)\}_{n \geq 0}$ have a statistical distribution with finite moments? If not, is this true for $\{n^{1/4} \log^{\alpha}(n) f_n(x)\}_{n \geq 2}$ for some $\alpha \in \mathbb{R}$?

The unnumbered question in §8.2, printed p. 38, of Berghaus, Bondarenko, Radchenko, Seip and Sun, *The basis functions of Fourier interpolation*, arXiv:2512.18677v1. The source fixes x and asks about the sequence of values as n varies.

Answer: Open. **Evidence:** Heuristic.

The plotted values suggest a limiting distribution, while the authors' moment calculations do not support normality. No finite-moment distribution is established.

4.4. **Moving-point decay of the basis functions**

For any $\varepsilon, \delta > 0$ we have $f_n(x) = O_{\varepsilon,\delta}(n^{-1/2+\delta})$ for $\varepsilon n \leq x \leq n - \sqrt{n}$.

Conjecture 8.2, §8.2, printed p. 39, of Berghaus, Bondarenko, Radchenko, Seip and Sun, *The basis functions of Fourier interpolation*, arXiv:2512.18677v1. In this range the conjecture is equivalent to subpower growth of the source's cosine sum Ψ.

Answer: Open. **Evidence:** Heuristic.

The source proves an $O(n^{-0.392})$ estimate on the smaller range $n/8 < x < n - \sqrt{n}$ and notes that an exponent-pair conjecture would imply the stated decay. Neither gives the requested bound unconditionally.

4.5. **Polylogarithmic growth of the cosine sum**

There exists $k > 0$ such that

$$\Psi(x) = \sum_{n\geq 1} \frac{2}{\sqrt{n}} \cos(\pi(\frac{3n-1}{4} - \frac{x}{n}))$$

satisfies

$$\Psi(x) = O(\log^k |x|), \qquad x \to \pm\infty.$$

Conjecture 8.3, §8.2, printed p. 39, of Berghaus, Bondarenko, Radchenko, Seip and Sun, *The basis functions of Fourier interpolation*, arXiv:2512.18677v1. The sum is denoted by Ψ in their formula (4.6).

Answer: Open. **Evidence:** Heuristic.

Their exponent-pair reduction gives subpower growth under an additional conjecture, but the checked source proves no polylogarithmic estimate.

4.6. **Fourier-invariant functions vanishing on near-square-root samples**

"this result fails when $\beta > 1/2$"; "explore further the finer details of the transition".

The first passage is in the abstract and the second in the introduction, printed p. 2, of Andriy Bondarenko and Kristian Seip, *Fourier-invariant functions with dense zero sets*, arXiv: 2608.13468v1. Together they motivate two targets: whether a nonzero Fourier-invariant continuous $L^1 \cap L^2$ function can vanish at $\lambda_n = \sqrt{n}/(\log n)^\beta$ for $\beta > 1/2$, and the regularity threshold for Fourier uniqueness at $\lambda_n = \sqrt{n}$.

Answer: Partial. **Evidence:** Sketch.

A written argument proves uniqueness in Fourier-invariant H^s for every $s > 1/2$ and every $\beta > 0$. The $L^1 \cap L^2$ existence question above $\beta = 1/2$, and the endpoint regularity threshold at $\beta = 0$, remain open.

Conditional on: The endpoint examples for $0 \leq \beta \leq 1/2$ use Bondarenko–Seip's unrefereed Theorem 2.3.

4.7. **Complex interpolation of ball Paley–Wiener spaces**

"possible to interpolate between"

The discussion before Theorem 1.3, printed p. 4, of Konstantinos Bampouras and Karl-Mikael Perfekt, *Boundary-weighted Fourier inequalities for convex domains*, arXiv:2608.19806v1, says the authors do not know whether the ball spaces interpolate. The reconstructed question is whether $[PW^{p_1}(B), PW^{p_2}(B)]_\theta = PW^{p_\theta}(B)$ with equivalent norms for $1 < p_1 < p_2 < \infty$ and $1/p_\theta = (1-\theta)/p_1 + \theta/p_2$.

Answer: Partial. **Evidence:** Sketch.

A written argument proves interpolation for radial subspaces when both exponents lie in a common power-weight strip. It does not establish the full nonradial ball identity.

4.8. **Subcritical weighted Fourier inequalities on the ball**

"We do not know whether the hypothesis of Theorem 1.5 is ever satisfied."

The passage after Theorem 1.5, printed p. 4, of Bampouras and Perfekt, *Boundary-weighted Fourier inequalities for convex domains*, arXiv:2608.19806v1. The inferred target is whether for every $n \geq 3$ some $1 \leq q < 2$ satisfies

$$\int_B |\widehat{f}(x)|^q \omega_B(x)^{-d}\, \mathrm{d}x \leq C_{q,d} \|f\|_q^q, \qquad d < 2 - q, \quad f \in PW^q(B),$$

where B is the unit ball in $\mathbb{R}^n$ and $\omega_B(x) = |B \cap (2x - B)|$. For each fixed n, the same q must work for every strict subcritical d.

Answer: Partial. **Evidence:** Sketch.

Stein–Tomas gives a smaller range. The public institutional report *Diagonal Fourier Extension for Positively Curved Surfaces in Three Dimensions* (Theorem 1.1) conditionally

supplies the full target for $n = 3$ and $3/2 \le q < 2$; the all-function problem in dimensions $n \ge 4$ remains open.
Conditional on: The three-dimensional claim depends on that unreviewed report's named inputs, *Elliptic Capacity Propagation and Fourier Restriction to the Sphere* (Theorem 6.4) and *The Kakeya Maximal Conjecture in Three Dimensions* (Theorem 1.1).

4.9. **Monotonicity of the normalized evaluation constant**

The function $p \mapsto \mathcal{C}_p/p$ is strictly decreasing on $(0, \infty)$.

Conjecture 9.1, printed p. 56, of Ole Fredrik Brevig, Daniel Chirre, José Ortega-Cerdà and Kristian Seip, *Point evaluation in Paley–Wiener spaces*, J. Anal. Math. 153 (2024), 595–670. Here $\mathcal{C}_p$ is the sharp constant in $|f(0)|^p \le \mathcal{C}_p \|f\|_p^p$ for entire functions of type at most π whose real restrictions belong to $L^p(\mathbb{R})$.
Answer: Partial. **Evidence:** Sketch.

A full written proof gives differentiability of $\mathcal{C}_p$ on $(1, \infty)$, strict monotonicity near $p = 2$, and eventual monotonicity. The conjecture for all pairs of positive exponents remains open.

4.10. **The imaginary-axis evaluation envelope**

Conjecture 9.3 asserts $K_p = \mathcal{C}_p$ for $0 < p < 2$ and $K_p = p/2$ for $2 \le p < \infty$.

Conjecture 9.3, printed p. 57, of Brevig, Chirre, Ortega-Cerdà and Seip, *Point evaluation in Paley–Wiener spaces*. Here K_p is the supremum over $y \ge 0$ of the best normalized point-evaluation constant at iy, with the source's factor $\operatorname{sinc}(i\pi p y)$.
Answer: Partial. **Evidence:** Sketch.

The source's asymptotic as $y \to \infty$ and the power trick settle integer exponents. A written argument also gives a strict gap at bounded heights for even exponents; the full envelope identity remains open.

4.11. **Uniqueness of the low-exponent extremal**

The extremal problem (1.6) has a unique solution for $0 < p < 1$.

Conjecture 9.4, printed p. 58, of Brevig, Chirre, Ortega-Cerdà and Seip, *Point evaluation in Paley–Wiener spaces*. Problem (1.6) minimizes $\int_{\mathbb{R}} |\varphi(x)|^p \, dx$ over PW^p subject to $\varphi(0) = 1$.
Answer: Open. **Evidence:** Heuristic.

Existence, reality and structural zero identities are known, and uniqueness holds for $p \ge 1$. At low exponents the known arguments do not synchronize the signs of two candidate extremizers.

4.12. **Evenness of low-exponent extremizers**

Any solution of (1.6) for $0 < p < 1$ is even.

Conjecture 9.5, printed p. 58, of Brevig, Chirre, Ortega-Cerdà and Seip, *Point evaluation in Paley–Wiener spaces*. The assertion concerns every minimizer, not merely the existence of an even one.
Answer: Open. **Evidence:** Heuristic.

Reflection preserves the minimizing problem, but averaging the reflected functions need not preserve minimality when $p < 1$. The written variation calculations give necessary conditions without proving symmetry of the zero divisor.

4.13. **Uniform separation of extremal zero sets**

The zero set of any solution of (1.6) is uniformly discrete for all $0 < p < \infty$.

Conjecture 9.6, printed p. 58, of Brevig, Chirre, Ortega-Cerdà and Seip, *Point evaluation in Paley–Wiener spaces*. Uniform discreteness here is for the distinct zero locations of each fixed extremizer.
Answer: Answered: yes. **Evidence:** Sketch.

The remaining range below $p = 1/2$ follows from a cluster argument using the canonical-product identity and a uniform polynomial estimate with real exponential weight. The full written proof is in the proof note.
Answers, "Uniform separation of extremal zero sets", p. 6.

4.14. **The full-range distributional reproducing identity**

Theorem 8.1 extends to the full range $0 < p < \infty$.

Conjecture 9.7, printed p. 58, of Brevig, Chirre, Ortega-Cerdà and Seip, *Point evaluation in Paley–Wiener spaces*. For every extremizer φ, the source defines $\Phi = |\varphi|^{p-2}\varphi/\|\varphi\|_p^p$ and asks that Φ be tempered and $f(0) = \int f\Phi$ for every $f \in \mathcal{S} \cap PW^2$.
Answer: Answered: yes. **Evidence:** Sketch.

For $p < 1/2$, the written proof combines uniform zero separation with simplicity of the zeros to obtain polynomial lower bounds, then differentiates the extremal problem to establish the identity. The remaining exponents are covered in the source. A full proof appears in the proof note.
Answers, "The distributional reproducing identity", p. 6.

4.15. **Monotonicity of positive zeros**

The function $p \mapsto t_n(p)$ is strictly decreasing for all $n \geq 1$.

Conjecture 9.8, printed p. 58, of Brevig, Chirre, Ortega-Cerdà and Seip, *Point evaluation in Paley–Wiener spaces*. The source assumes uniqueness of the extremizer for $p < 1$ when defining $t_n(p)$.
Answer: Partial. **Evidence:** Sketch.

A horizontal-strip expansion near $p = 2$ proves the comparison for any fixed finite set of zero indices on a two-sided neighborhood of 2. It does not give a neighborhood uniform over all indices or monotonicity at arbitrary exponents.
Conditional on: Uniqueness of the extremizer for $0 < p < 1$, as assumed by the source.

4.16. **Asymptotic unit spacing of extremal zeros**

We have $\lim_{n\to\infty}(t_{n+1} - t_n) = 1$ for all $p > 0$.

Conjecture 9.9, printed p. 58, of Brevig, Chirre, Ortega-Cerdà and Seip, *Point evaluation in Paley–Wiener spaces*. Below $p = 1$ the source continues to use its standing uniqueness assumption.
Answer: Partial. **Evidence:** Heuristic.

The unit-spacing limit is known for the sinc extremizer at $p = 2$ and follows from the published zero-location estimate at $p = 1$. A constructed family of nonextremizers shows why the coarse zero bounds and moment information alone do not force convergence.
Conditional on: Uniqueness of the extremizer for $0 < p < 1$, as assumed by the source.

4.17. **Endpoint limits of the first positive zero**

We have $\lim_{p\to\infty} t_1(p) = 1/2$ and $\lim_{p\to 0^+} p\, t_1(p) > 0$.

Conjecture 9.10, printed p. 59, of Brevig, Chirre, Ortega-Cerdà and Seip, *Point evaluation in Paley–Wiener spaces*. The source assumes uniqueness for small p when it uses $t_1(p)$.
Answer: Answered: yes. **Evidence:** Sketch.

The large-exponent limit follows from local convergence of extremizers to $\cos(\pi z)$. Under the stated low-exponent uniqueness assumption, a rescaling and zero-potential argument proves that the second limit exists and is finite and positive. Both proofs are written out in the proof note and the point-evaluation note.
Conditional on: Uniqueness of the extremizer for all sufficiently small $p > 0$.
Answers, "Limits of the first positive extremal zero", p. 7.

4.18. **Extremal zeros as a uniqueness set**

$Z(\varphi_p) \cup \{0\}$ is a uniqueness set for PW^p if and only if $p \geq 1$.

Conjecture 9.11, printed p. 59, of Brevig, Chirre, Ortega-Cerdà and Seip, *Point evaluation in Paley–Wiener spaces*. The source assumes a unique extremizer below $p = 1$ when using the notation φ_p.

Answer: Answered: yes. **Evidence:** Sketch.

For $p \geq 1$ the published uniqueness theorem gives the assertion. For $0 < p < 1$, a dilation-times-sinc competitor gives a tail estimate implying $x\varphi(x) \in PW^p$; this nonzero function vanishes on $Z(\varphi) \cup \{0\}$. The argument works for any extremizer; the source extquotesingle s uniqueness convention is only used to name φ_p.

Answers, "Extremal zeros as a uniqueness set", p. 7.

4.19. **An A_p weight from the extremal zeros**

Suppose that $1 < p < \infty$. The function $\left(\frac{|x\varphi_p(x)|}{\operatorname{dist}(x, Z(\varphi_p)\cup\{0\})}\right)^p$ is a Muckenhoupt A_p weight.

Conjecture 9.12, printed p. 59, of Brevig, Chirre, Ortega-Cerdà and Seip, *Point evaluation in Paley–Wiener spaces*. The A_p condition is uniform over all finite intervals, as in the standard Muckenhoupt definition.

Answer: Partial. **Evidence:** Sketch.

The $p = 2$ weight is bounded above and below. A written criterion reduces the general question to bounded midpoint logarithmic slopes and a discrete weighted A_p condition; near-minimizers show that several tempting geometric estimates alone do not imply this criterion.

4.20. **Fourier regularity of the extremal**

If $1 < p < \infty$, then the Fourier transform $\widehat{\varphi_p}$ is a continuous integrable function on $(-\pi, \pi)$.

Conjecture 9.13, printed p. 59, of Brevig, Chirre, Ortega-Cerdà and Seip, *Point evaluation in Paley–Wiener spaces*. Continuity at the endpoints is not part of the assertion.

Answer: Partial. **Evidence:** Sketch.

The source settles $p \leq 1$. A written argument establishes the Fourier conclusion for $1 < p \leq 2$ and Fourier integrability for $2 < p < 4$; the remaining range is open. A workshop announcement claims interior continuity at $p = 4$, without a public proof manuscript.

4.21. **Positivity and shape of the Fourier transform**

Let φ be a solution of (1.6). Then $\widehat{\varphi}$ is nonnegative. Moreover, (a) if $0 < p < 2$, it is decreasing on $(0, \pi)$ and tends to 0 as $\xi \to \pi^-$; (b) if $2 < p < \infty$, it is increasing on $(0, \pi)$ and tends to ∞ as $\xi \to \pi^-$.

Conjecture 9.14, printed p. 60, of Brevig, Chirre, Ortega-Cerdà and Seip, *Point evaluation in Paley–Wiener spaces*. The assertions concern every solution of (1.6).

Answer: Partial. **Evidence:** Sketch.

A written certificate proves the full Fourier-shape conclusion at $p = 1$. The general ranges, including the endpoint behavior for other exponents, remain open; a workshop announcement about $p = 4$ has no public proof manuscript.

4.22. **Real-valued solutions of the convolution equation**

(a) There is exactly one real-valued admissible solution of (9.5) when n is an even integer, namely $\widehat{\psi} = \widehat{\varphi}_{n+2}/\|\varphi_{n+2}\|_{n+2}^{1+(n+1)^{-1}}$. (b) There are no real-valued admissible solutions of (9.5) when n is an odd integer.

Conjecture 9.15, printed p. 62, of Brevig, Chirre, Ortega-Cerdà and Seip, *Point evaluation in Paley–Wiener spaces*. For $n \geq 0$, admissible $\widehat{\psi}$ is integrable, supported on $[-\pi, \pi]$ and continuous in the open interval; (9.5) uses the paper's iterated convolution and normalization. The real-valued restriction is part of the source conjecture.
Answer: Partial. **Evidence:** Sketch.

For odd n, a written argument proves that every solution must violate a certain global sector bound on an open interval, extending the source's even-solution obstruction. It does not exclude all real-valued solutions or prove the even-n uniqueness assertion.

4.23. **Monotonicity in the imaginary-axis height**

> For fixed p, the best constant $K_p(y)$ "may increase with y when $p > 2$ and decrease with y when $0 < p < 2$."

The paragraph before Conjecture 9.3, printed p. 57, of Brevig, Chirre, Ortega-Cerdà and Seip, *Point evaluation in Paley–Wiener spaces*. The inferred question is whether $K_p(y)$ is monotone in y in the indicated direction for each fixed p.
Answer: Partial. **Evidence:** Sketch.

A written argument proves the proposed order for each fixed pair of heights when p is sufficiently close to 2. Full monotonicity for fixed $p \neq 2$, including collapsing height pairs, remains open.

4.24. **Sharp bounds for positive zeros**

> "$n - 1/2 + 1/p \leq t_n \leq n$" for $2 \leq p < \infty$, and "$n \leq t_n \leq n - 1/2 + 1/p$" for $0 < p \leq 2$.

The paragraph following (9.4), printed p. 58, of Brevig, Chirre, Ortega-Cerdà and Seip, *Point evaluation in Paley–Wiener spaces*. The source presents these inequalities as a reasonable expectation for the positive zeros of the normalized extremizer.
Answer: Partial. **Evidence:** Sketch.

The bounds are known at $p = 1$ and $p = 2$. A written perturbation argument proves both inequalities strictly for each fixed finite set of indices in a two-sided neighborhood of $p = 2$; no all-index neighborhood or global range is known.
Conditional on: Uniqueness of the extremizer for $p < 1$, as assumed by the source.

4.25. **The small-exponent evaluation constant**

> "It remains an interesting problem to determine the constant c_0" of Theorem 1.3.

The paragraph before Conjecture 9.1, printed p. 56, of Brevig, Chirre, Ortega-Cerdà and Seip, *Point evaluation in Paley–Wiener spaces*. The source places c_0 in $[1.1393830, 1.1481785]$.
Answer: Partial. **Evidence:** Sketch.

A written, computer-assisted construction improves the lower bound to $c_0 > 57/50$. The exact value and a matching upper bound remain open.

4.26. **Schwartz-convergent Fourier interpolation formulas**

> "we do not know whether there exists a similar interpolation formula convergent in the topology of the Schwartz space"

The final paragraph of §2.2, arXiv p. 13, of Aleksei Kulikov, Fedor Nazarov and Mikhail Sodin, *Fourier uniqueness and non-uniqueness pairs*, arXiv:2306.14013v1; the paragraph follows formula (2.1) in the published version and (2.2.1) in the preprint. The source asks whether its interpolation formula for supercritical pairs can converge in the Schwartz topology. Here $1 < p, q < \infty$, $1/p + 1/q = 1$, and supercriticality is the strict weighted-spacing condition in their Theorem 1.
Answer: Partial. **Evidence:** Sketch.

Published formulas converge in weighted Hilbert spaces, and later work treats a smaller dense class of supercritical pairs. A written criterion reduces Schwartz convergence to all-order off-diagonal decay of a range projection; that decay has not been proved in the general supercritical range.

4.27. **Higher-multiplicity Fourier interpolation uniqueness**

> Let d and k be positive integers. If $f \in \mathcal{S}_{\mathrm{rad}}(\mathbb{R}^d)$ satisfies $f^{(j)}(\sqrt{kn}) = \widehat{f}^{(j)}(\sqrt{kn}) = 0$ for all integers $n \geq 0$ and $0 \leq j < k$, then must f vanish identically?

Open Problem 7.1, printed p. 93, of Henry Cohn, Abhinav Kumar, Stephen D. Miller, Danylo Radchenko and Maryna Viazovska, *Universal optimality of the E_8 and Leech lattices and interpolation formulas*, Ann. Math. 196 (2022), 983–1082.
Answer: Partial. **Evidence:** Heuristic.

The $k = 1$ case is proved in all dimensions, and $k = 2$ is known for $d = 8, 24$. The general multiplicity and dimension problem remains open; Rolle-type arguments do not provide the required spacing of the extra zeros.

4.28. **Interpolation at Gaussian-optimal radii**

> Let $r_1, r_2, \ldots$ be the radii from Conjecture 7.2 for some d. For which d is every $f \in \mathcal{S}_{\mathrm{rad}}(\mathbb{R}^d)$ uniquely determined by the values $f(r_n), f'(r_n), \widehat{f}(r_n), \widehat{f}'(r_n)$ for $n \geq 1$ through an interpolation formula?

Open Problem 7.3, printed p. 94, of Cohn, Kumar, Miller, Radchenko and Viazovska, *Universal optimality of the E_8 and Leech lattices and interpolation formulas*. The question remains conditional on the radii provided by Conjecture 7.2.
Answer: Partial. **Evidence:** Sketch.

A written criterion reduces the question to polynomial separation of the given squared radii and a supported order-one inverse-Laplace kernel. The criterion is conditional: neither hypothesis has been proved for the conjectural radii outside the known dimensions.
Conditional on: Existence of the radii in Conjecture 7.2 for the dimension in question.

4.29. **Hexagonal radii and nonuniqueness**

> Let $r_1, r_2, \ldots$ be the positive real numbers of the form $(4/3)^{1/4}\sqrt{j^2 + jk + k^2}$, where j and k are integers. Then radial Schwartz functions $f : \mathbb{R}^2 \to \mathbb{R}$ are not uniquely determined by the values $f(r_n), f'(r_n), \widehat{f}(r_n), \widehat{f}'(r_n)$ for integers $n \geq 1$.

Conjecture 7.5, printed p. 95, of Cohn, Kumar, Miller, Radchenko and Viazovska, *Universal optimality of the E_8 and Leech lattices and interpolation formulas*.
Answer: Answered: yes. **Evidence:** arXiv.

Sardari's Theorem 1.7 constructs infinitely many linearly independent radial Schwartz functions with the stated vanishing data. The paper is listed as in press in the *Journal d'Analyse Mathématique*.

4.30. **Finite perturbations of Fourier interpolation nodes**

> For example, does moving (but not removing) finitely many interpolation points preserve the interpolation property?

The paragraph after Conjecture 7.5, printed p. 96, of Henry Cohn, Abhinav Kumar, Stephen D. Miller, Danylo Radchenko and Maryna Viazovska, *Universal optimality of the E_8 and Leech lattices and interpolation formulas*, Ann. Math. 196 (2022), 983–1082. The canonical data are $f, f', \widehat{f}, \widehat{f}'$ at radii $\sqrt{2n}$ in dimensions 8 and 24.
Answer: Answered: no. **Evidence:** Sketch.

A full written proof constructs, in each of these two systems, a collision-free finite move for which the data fail to determine the radial Schwartz function.
Answers, "Finite perturbations can destroy Fourier interpolation", p. 8.

4.31. **Orthogonality at smaller exponential type in a zeta-weighted space**

"orthogonal to all functions in H of strictly smaller exponential type"

Problem 5, p. 32, proposed by Jean-François Burnol in the open-problems session of *Mini-Workshop: Interplay between Number Theory and Analysis for Dirichlet Series*, Oberwolfach Report 51/2017. The question asks whether there are $\tau > 0$ and a nonzero entire function f of exponential type τ in $H = L^2(\mathbb{R}, |\zeta(1/2 + ix)|^2 \, \mathrm{d}x)$ orthogonal in H to every function of strictly smaller type.

Answer: Open. **Evidence:** Heuristic.

The report gives an entire function in the weighted space but does not establish the required orthogonality.

5. Bernstein–Nikolskii inequalities

5.1. **A finite-q Levin–Lubinsky limit formula**

Prove a Levin–Lubinsky type result for $q \neq \infty$.

Problem 1, Section 7, printed p. 36, of Sergei Gorbachev, *Sharp Bernstein–Nikolskii inequalities for polynomials and entire functions of exponential type*, arXiv:2609.32072v1. Levin and Lubinsky prove the asymptotic formula for $q = \infty$; for finite q, Gorbachev reports only the required lower estimate, with no matching asymptotic equality.

Answer: Open. **Evidence:** Heuristic.

The periodization argument for the supremum norm does not give the finite-q upper bound. The checked sources identify no method that closes this gap.

5.2. **Exponential improvement of the multidimensional bounds**

Can the upper and lower bounds in (32) for the multidimensional constant L_1^d be improved exponentially in d?

Problem 6, Section 7, printed p. 36, of Sergei Gorbachev, *Sharp Bernstein–Nikolskii inequalities for polynomials and entire functions of exponential type*, arXiv:2609.32072v1. The target is to improve both bounds in (32) by exponential factors as the dimension grows.

Answer: Open. **Evidence:** Heuristic.

Dai, Gorbachev and Tikhonov give the current explicit estimates for the multidimensional constant, but the checked results do not establish exponential improvements in dimension.

5.3. **Alternation of extremal Taylor coefficients**

Prove that the signs of the nonzero Taylor coefficients of extremal functions in the problems $c_{n,p}^{(r)}$ and $L_p^{(r)}$ alternate.

Problem 8, Section 7, printed p. 37, of Sergei Gorbachev, *Sharp Bernstein–Nikolskii inequalities for polynomials and entire functions of exponential type*, arXiv:2609.32072v1. At the endpoint $p = \infty$, $r = 0$, the source does not specify a unique choice of extremizer.

Answer: Partial. **Evidence:** Sketch.

At $p = \infty$, $r = 0$, an explicit normalized extremizer has two consecutive nonzero Taylor coefficients with the same sign, so a universal assertion fails. The existence of a selected alternating extremizer at that endpoint and the conjecture for finite p remain open.

5.4. **The Bessel extremal problem below the known range**

Solve problem (55) for $-1/2 < \alpha < -0.272$.

Problem 9, Section 7, printed p. 37, of Sergei Gorbachev, *Sharp Bernstein–Nikolskii inequalities for polynomials and entire functions of exponential type*, arXiv:2609.32072v1. Problem (55) is the Bessel-function extremal formula for the multidimensional constant in the survey's notation.

Answer: Partial. **Evidence:** Sketch.

For every $\alpha > -1/2$, integration against $t^{2\alpha+1}$ up to the first positive zero gives the lower bound $I_\alpha \geq a_0^*$. Equality is equivalent to a uniform tail bound for the Bessel extremizer, which remains open on the stated interval.

5.5. **Uniqueness of the radial extremal among all functions**

Prove that the radial extremal function in the problem L_1^d is unique not only among radial functions.

Problem 7, printed p. 37, of Sergei Gorbachev, *Sharp Bernstein–Nikolskii inequalities for polynomials and entire functions of exponential type*. Reconstructed question: is the normalized radial minimizer for L_1^d unique in the full class of integrable entire functions of spherical exponential type at most one on $\mathbb{R}^d$?

Answer: Answered: yes. **Evidence:** Sketch.

The full written proof uses raywise evenization to identify the even part with the radial minimizer, then a bounded-type division lemma and the simple real zeros of its profile to eliminate every odd perturbation.

Answers, "Uniqueness of the radial L_1^d extremal", p. 8.

5.6. **Degree monotonicity of sharp Bernstein–Nikolskii constants**

The sequences $A_{p,q,n}^{(s)}$ and $A_{p,q,n,\mathbb{R}}^{(s)}$ are decreasing in $n \in \mathbb{N}$.

Conjecture 1.3, printed p. 452, of Michael Ganzburg and Sergey Tikhonov, *On sharp constants in Bernstein–Nikolskii-type inequalities*, Constructive Approximation 45 (2017), 449–466. Here $s \in \mathbb{Z}_{\geq 0}$, $0 < p < q \leq \infty$, and $A_{p,q,n}^{(s)}$ is the normalized best constant for trigonometric polynomials of degree at most n on $[-\pi, \pi]$; the $\mathbb{R}$ subscript restricts to real-valued polynomials.

Answer: Partial. **Evidence:** Sketch.

The known $p = 2, q = \infty$ endpoint decreases by the cited Bennett–Jameson result. Our written estimates prove adjacent decrease in an explicit large-q range for $p = 2$, but leave general p and fixed finite q open.

5.7. **Equality of real and complex polynomial constants**

The questions as to whether $A_{p,q,n}^{(s)} = A_{p,q,n,\mathbb{R}}^{(s)}$ and $D_{p,q}^{(s)} = D_{p,q,\mathbb{R}}^{(s)}$ still remain open. However, these equalities are valid for $q = \infty$.

The paragraph "Complex and Real Constants," printed p. 451, of Ganzburg and Tikhonov's *On sharp constants in Bernstein–Nikolskii-type inequalities* raises these equalities. The question here is the polynomial equality for $0 < p < q < \infty$, all derivative orders $s \geq 0$ and degrees $n \geq 1$.

Answer: Answered: yes. **Evidence:** Sketch.

The phase-averaging argument in the public note proves equality for the polynomial constants throughout this range. The proof.

5.8. **Equality of real and complex entire-function constants**

The questions as to whether $A_{p,q,n}^{(s)} = A_{p,q,n,\mathbb{R}}^{(s)}$ and $D_{p,q}^{(s)} = D_{p,q,\mathbb{R}}^{(s)}$ still remain open. However, these equalities are valid for $q = \infty$.

The same paragraph, printed p. 451, raises the entire-function equality. For $0 < p < q < \infty$ and integer $s \geq 0$, $D_{p,q}^{(s)}$ is the sharp derivative ratio over entire functions of exponential type at most one with the indicated L^p and L^q norms; the real subscript restricts to real-valued functions.

Answer: Answered: yes. **Evidence:** Sketch.

The public note proves equality of the real and complex entire-function constants for every stated parameter. The proof.

5.9. **Finite-q limit of the polynomial constants**

> The limit $\lim_{n\to\infty} A^{(s)}_{p,q,n}$ exists and $D^{(s)}_{p,q} = \lim_{n\to\infty} A^{(s)}_{p,q,n}$ for $s \in \mathbb{Z}_+$ and $0 < p < q < \infty$. The real version of Conjecture 1.10 is believed to be true as well.

Conjecture 1.10 and the following paragraph, printed p. 454, of Ganzburg and Tikhonov's *On sharp constants in Bernstein–Nikolskii-type inequalities*.
Answer: Partial. **Evidence:** Sketch.

The source proves a lower-semicontinuity bound and the $q = \infty$ limit. A public note states the finite-q identity for real and complex constants, but that argument remains unchecked. The finite-q identity is therefore not treated here as an established answer.

5.10. **Four-digit accuracy of the monotone-delta constant**

> The upper bound is sharp "at least up to the first four significant digits."

The paragraph after Theorem 1, Section 1.2, p. 2, of Andrés Chirre, Dimitar K. Dimitrov, Emily Quesada-Herrera and Mateus Sousa, *An extremal problem and inequalities for entire functions of exponential type*, Proc. Amer. Math. Soc. 152 (2024). In the paper's notation, the question is whether $A_1 = \inf_f \|f\|_1/f(0)$, over nonnegative radially decreasing real entire functions of type at most 2π, agrees with the exhibited upper bound 1.27714 through the four significant digits requested by the authors.
Answer: Answered: yes. **Evidence:** Sketch.

A finite-section Gram estimate improves the lower bound to 1.2771345, while the published certified upper bound is 1.277135042105; together they determine the requested digits, without determining the exact constant.
Answers, "Four-digit accuracy of the monotone-delta constant", p. 9.

5.11. **Tangential Bernstein inequality on algebraic manifolds**

> "Prove that there is a constant C such that for any polynomial p"

Problem-session item 7, p. 389, in *Dirichlet Series and Function Theory in Polydiscs*, Oberwolfach Report 06/2014; proposed by Joaquim Ortega-Cerdà. The reconstructed question is whether, for every smooth compact algebraic manifold $M \subset \mathbb{R}^n$, there is C such that $\int_M |\nabla_t p|\, \mathrm{d}m \le C \deg(p) \int_M |p|\, \mathrm{d}m$ for all polynomials p, where ∇_t denotes the tangential gradient.
Answer: Answered: yes. **Evidence:** Published.

Berman and Ortega-Cerdà prove the L^q tangential-gradient Bernstein inequality for smooth compact real algebraic varieties; the $q = 1$ case is Theorem 9 of their 2018 paper.

6. Nehari theorems and Hankel forms

6.1. **A bounded symbol for the multiplicative Hilbert functional**

> Is the symbol $\varphi_0(s) = 1 + \sum_{n=2}^{\infty} (\log n)^{-1} n^{-1/2-s}$ the Riesz projection of a function in $L^\infty(\mathbb{T}^\infty)$?

Saksman and Seip, *Some open questions in analysis for Dirichlet series*, Problem 3.2, p. 4.
Answer: Answered: yes. **Evidence:** arXiv.

Perfekt's Theorem 1.1 and Corollary 1.2 prove boundedness of the coefficient functional and hence existence of a bounded symbol. This relies on his latest unrefereed preprint.
Conditional on: Karl-Mikael Perfekt, arXiv:2609.03855, latest version.

6.2. **Local L^q embeddings of the H^1 dual**

> It is natural to ask if the situation is even worse, namely that $(H^1)^*$ fails to be contained in $H^q_i(\mathbb{C}_{1/2})$ for any $q > 2$.

Konyagin, Queffélec, Saksman, and Seip, *Riesz projection and bounded mean oscillation for Dirichlet series*, §3.1, p. 9.
Answer: Partial. **Evidence:** Sketch.

Konyagin, Queffélec, Saksman, and Seip prove failure for $q > 4$; our argument extends this to $q > 7/2$. Whether failure holds throughout $2 < q \leq 7/2$ remains open.

7. Composition operators

7.1. **The endpoint for a boundary circle restriction**

Source passage. For the symbol $\varphi(s) = 3/2 - 2^{-s}$, Bayart and Brevig identify boundedness of C_φ with boundedness of the integration functional (34), then state: "Theorem 25. The functional L defined by (34) is bounded on H^p for any $p > 1$." The inferred endpoint question is whether C_φ is bounded on H^1; their result already covers every $p > 1$. See Theorem 25 and the discussion preceding it in *Composition operators and embedding theorems for some function spaces of Dirichlet series*.

Answer: Partial. **Evidence:** Published.

Bayart and Brevig prove boundedness for every $p > 1$ in Theorem 25; the original $1 \leq p < 2$ question therefore reduces to the endpoint $p = 1$. No endpoint result is given there.

Part 2. Where Nehari's theorem survives: structured Helson matrices

8. Embeddings and boundary values

8.1. **Carleson classes on the infinite polydisc**

> Theorem 22 invites the following question. Question. If μ defines an H^p-Carleson measure on $\mathbb{D}^\infty$ for some $0 < p < \infty$, does it also define an H^q-Carleson measure for every $p < q < \infty$?

Bayart, Brevig, Haimi, Ortega-Cerdà, and Perfekt, *Contractive inequalities for Bergman spaces and multiplicative Hankel forms*, §5, question after Theorem 22, p. 26.

Answer: Open. **Evidence:** Heuristic.

Bayart, Brevig, Haimi, Ortega-Cerdà, and Perfekt prove special implications and counterexamples in restricted settings, but no general passage from an arbitrary H^p-Carleson measure to all larger exponents.

9. Nehari theorems and Hankel forms

9.1. **Intrinsic duals of Dirichlet Hardy spaces**

> Describe $(\mathscr{H}^p)^*$ for p other than 2.

Saksman and Seip, *Some open questions in analysis for Dirichlet series*, Problem 2.8; the preceding paragraph includes $0 < p < 1$.

Answer: Partial. **Evidence:** Published.

The $p = 2$ dual is explicit, and for $p \geq 1$ a quotient description follows from general duality. An intrinsic description for other exponents, including the quasi-Banach range, remains open.

9.2. **A Nehari counterpart on the infinite torus**

> What is the counterpart to Nehari's theorem on $\mathbb{T}^\infty$? In particular, what can be said about the Riesz projection of $L^\infty(\mathbb{T}^\infty)$ and other BMO-type spaces on $\mathbb{T}^\infty$?

Saksman and Seip, *Some open questions in analysis for Dirichlet series*, Problem 3.1.

Answer: Partial. **Evidence:** Published.

Konyagin, Queffélec, Saksman, and Seip embed half-plane BMOA in the dual and show that the dual is not contained in any H^q with $q > 2$. A useful full symbol-space characterization is still missing.

9.3. **Positive Nehari theorem for Helson matrices**

> In general there are Carleson measures for $H^2(\mathbb{T}^\infty)$ which are not Carleson measures for $H^1(\mathbb{T}^\infty)$, but we do not know if these can be chosen to be concentrated on $(-1,1)^\infty$.

Perfekt and Pushnitski, *On Helson matrices: moment problems, non-negativity, boundedness, and finite rank*, question after Remark 4.11, p. 15. Their equivalent bounded-symbol formulation is also answered by the construction.

Answer: Answered: yes. **Evidence:** Sketch.

A finite real-cube measure has bounded H^2 embedding and unbounded absolute H^1 embedding. A positive polynomial test detects the strict two-atom gain, and disjoint tensor blocks amplify it.

Answers, "Positive Nehari theorem for Helson matrices", p. 10.

The proof

9.4. **Carleson moment representations of bounded Helson matrices**

> We do not know the answer to the following. Question 4.6. Let $M(\alpha)$ be bounded on $\ell^2(\mathbb{N})$. Does there exist a Carleson measure μ for $H^2(\mathbb{T}^\infty)$ such that α is represented as the moment sequence of μ, as in (4.4)?

Perfekt and Pushnitski, *On Helson matrices: moment problems, non-negativity, boundedness, and finite rank*, Question 4.6, p. 12.

Answer: Partial. **Evidence:** Sketch.

Finite-rank cases and several weighted ℓ^2 coefficient classes admit such representations. The general question allows complex measures on the open product disc and remains unresolved.

9.5. **Necessity of the apicality criteria**

> It is natural to ask which, if any, of these two conditions may also be necessary for φ not to be saturated.

Brevig and Seip, *Maximal norm Hankel operators*, §3, after Problem 1. The conditions are apicality of $|\varphi|$ at some level and nonempty interior of the low-modulus set.

Answer: Answered: no. **Evidence:** Sketch.

Neither condition is necessary: separate constructions show that neither is required for nonsaturation. A full written proof appears in the answers entry.

Answers, "Are the apicality criteria necessary for nonsaturation?", p. 10.

9.6. **Classifying saturated outer functions**

> Describe the outer functions in H^∞ that are saturated.

Brevig and Seip, *Maximal norm Hankel operators*, Problem 1. Saturation means equality in their equation (4), $\|H_\varphi\| = \|\varphi\|_\infty$.

Answer: Partial. **Evidence:** Published.

Their Theorems 4–6 give criteria and substantial classes, including apical weights, but not a complete description. The necessity question above is separate from the original classification problem.

9.7. **Power invariance of apical weights**

> Is it true that w is an apical Helson–Szegő weight if and only if w^κ is an apical Helson–Szegő weight for all positive numbers κ? In other words, we ask if (7) could be replaced simply by (27) $\operatorname{ess\,sup}_{L^+_w(\delta)} |\widetilde{v}| < \infty$.

Brevig and Seip, *Maximal norm Hankel operators*, Problem 2 and condition (27).

Answer: Answered: yes. **Evidence:** Sketch.

The finite localized conjugate bound can be reduced below any prescribed threshold by an analytic truncation. Brevig and Seip's power criterion then applies to every positive power.

Answers, "Power invariance of apical weights", p. 11.

The proof

9.8. **Norm-attaining vectors in complete Pick spaces**

It would be interesting to know what can be said of the norm attaining vectors and Hilbert points in this context.

Bampouras and Brevig, *Norm attaining vectors and Hilbert points*, paragraph after Theorem 4, p. 4. The scoped question is whether every $\mathcal{H}$-extremal vector is norm attaining.

Answer: Answered: yes in separable normalized complete Pick spaces. **Evidence:** Sketch.

For normalized separable complete Pick spaces, norm attainment is equivalent to the $\mathcal{H}$-extremal identity. In H^2, this says precisely that the boundary modulus is constant; the proof is in the answers entry.

Answers, "Norm-attaining vectors in complete Pick spaces", p. 11.

9.9. **A product Hardy-space Nehari theorem**

Does every bounded small Hankel operator on $H^2(\mathbb{C}_+^2)$ have a symbol in product BMO?

Bampouras and Perfekt, *Recovering Product BMO from Schatten Hankel Operators*, Introduction, equation (1).

Answer: Open. **Evidence:** Heuristic.

Bampouras and Perfekt prove a product-BMO conclusion for Schatten-class operators with finite exponent; their result does not cover every bounded operator. The endpoint question remains unresolved.

Conditional on: Bampouras and Perfekt's unrefereed public preprint.

9.10. **An intrinsic description of the analytic H^1 dual**

Give a concrete intrinsic description of all analytic symbols in $(\mathscr{H}^1)^*$.

Søvik, the book's open question in the H^1 discussion.

Answer: Partial. **Evidence:** Sketch.

BMOA symbols and conditional-square subclasses provide descriptions in special cases, while Hahn–Banach only gives a nonintrinsic quotient representation. A general concrete characterization remains open.

9.11. **Cancellation in signed curve symbols**

Wanted: an example where the signed matrix is bounded by cancellation but has no bounded symbol, or a proof that none exists for curve measures.

Søvik, the cancellation question in the multiplicative-Hilbert discussion. We address the finite real signed-measure case and show cancellation can bound the full matrix even when each Jordan-part matrix is unbounded; this does not produce a no-symbol example.

Answer: Partial. **Evidence:** Sketch.

An explicit finite signed measure has a Hilbert–Schmidt moment matrix and a bounded symbol even though its positive and negative parts separately have unbounded moment functionals. This resolves that cancellation example, not the general characterization.

9.12. **A countable signed curve test for positive Nehari**

Wanted: an example where the signed matrix is bounded by cancellation but has no bounded symbol, or a proof that none exists for curve measures.

Søvik, the cancellation question in the multiplicative-Hilbert discussion. The present question narrows this target to countably many sign-rotated pieces of the prime curve and asks whether an H^2 bound can coexist with an unbounded absolute H^1 embedding.

Answer: Answered: yes. **Evidence:** Sketch.

There is a countably atomic positive measure on sign-rotated prime-curve points with bounded H^2 embedding constant and infinite absolute H^1 embedding constant. A construction and proof are in the public note.

Answers, "A countable signed curve test for positive Nehari", p. 12.

The proof

9.13. **Products of Hankel operators in Schatten classes**

An interesting and important problem is to describe functions φ and ψ in L^2 for which $H^*_\varphi H_\psi \in S_p$; this problem is still open.

Vladimir V. Peller, *Hankel Operators and Their Applications*, Chapter 6, Concluding Remarks, printed p. 301. Our scoped question concerns bounded extension of the cross form on analytic polynomials.

Answer: Partial. **Evidence:** Sketch.

The sheet gives a necessary and sufficient criterion in terms of a bounded cross form for a substantial class, but does not settle all pairs or all possible composition domains. A full joint symbol characterization remains open.

9.14. **Recovering positive Fourier coefficient powers**

The very interesting problem of whether (12.1) holds for other values of $s \geq 1/2$ is still open.

Vladimir V. Peller, *Hankel Operators and Their Applications*, Chapter 7, §12, after Corollary 12.3, printed p. 373. Formula (12.1) asks whether finiteness of the negative-index weighted Fourier sum forces finiteness of the corresponding positive-index sum.

Answer: Partial. **Evidence:** Published.

The source proves the implication for positive integer s by convolution powers. Other exponents, including the endpoint $s = 1/2$, remain undetermined.

9.15. **Compact and noncompact inverse Hankel choices**

It is unknown if it is possible to choose $q \notin \operatorname{Range} R$ so that $I - Q$ is compact. It is also unclear whether it is possible to find q so that $I - Q$ is noncompact.

Megretskii, Peller, and Treil, *The Inverse Spectral Problem for Self-Adjoint Hankel Operators*, §III.7, p. 295; repeated by Peller, *Hankel Operators and Their Applications*, Chapter 12, §13, p. 537. The question is for each admissible operator in the stated construction.

Answer: Answered: both choices exist. **Evidence:** Sketch.

Rescaling the spectral bands of an admissible vector yields a compact choice and, separately, a noncompact one while preserving the construction's hypotheses. The full proof is in the answers entry.

Answers, "Compact and noncompact inverse Hankel choices", p. 13.

9.16. **Simon's Schatten majorization property**

Simon [2] conjectured that B_p has the majorization property for $p \in [2, \infty)$.

Peller, *Hankel Operators and Their Applications*, Chapter 6, Concluding Remarks, printed p. 300. The property is the implication from pointwise domination $|Tf| \leq R|f|$ to $T \in S_p$ whenever $R \in S_p$.

Answer: Answered: no for non-even finite p. **Evidence:** Published.

Peller's counterexamples disprove the conjecture for every finite $p \geq 2$ that is not an even integer; the even-integer cases hold.

9.17. **Generation of Douglas algebras**

Is every closed subalgebra of L^∞ containing H^∞ generated by H^∞ and the complex conjugates of inner functions?

Douglas's conjecture, stated by Donald Sarason in *Functions of vanishing mean oscillation*, §3, after Corollary 2, printed p. 401.

Answer: Answered: yes. **Evidence:** Published.

Chang's and Marshall's theorems on the structure of Douglas algebras establish the asserted generation.

9.18. **Continuity of the modulus of a Poisson extension**

Let f be a function in L^∞ with the property that $|f|$ is continuous in $\overline{D}$. Then f is in QC.

Donald Sarason, *Functions of vanishing mean oscillation*, Theorem 3, §4, printed p. 398. This gives the affirmative answer to the implicit question whether the modulus of the Poisson extension of f extending continuously to the closed disc forces $f \in QC = VMO \cap L^\infty$.
Answer: Answered: yes. **Evidence:** Published.

Sarason's Theorem 3 proves that this continuity forces f into QC.

9.19. **A converse Jensen-measure criterion**

An open problem is to obtain a converse to Lemma 5.1, characterizing $\sup_{\tau \in J_\phi} \|\tau_a\|$ in terms of A-subharmonic functions.

Cole and Gamelin, *Representing Measures and Hardy Spaces for the Infinite Polydisk Algebra*, §5, after Lemma 5.1, printed p. 119. Here J_ϕ is the set of Jensen measures and τ_a is the part absolutely continuous with respect to the fixed probability measure.
Answer: Answered: yes. **Evidence:** Sketch.

A real Hahn–Banach separation identifies the envelope with the supremum over Jensen measures; truncating their densities and applying minimax gives the sharp mass criterion. The full proof is in the answers entry.
Answers, "A converse Jensen-measure criterion", p. 15.

9.20. **Weak factorization on the unit ball**

Is every $f \in H^1(\mathbb{B}_n)$, $n > 1$, a product gh with $g, h \in H^2(\mathbb{B}_n)$?

Hayman and Lingham, *Research Problems in Function Theory*, 50th Anniversary Edition, Update 8.16, p. 193; the update attributes the ball problem to Rudin.
Answer: Answered: no. **Evidence:** Published.

Gowda's theorem shows that for $n > 1$ the products $H^2(\mathbb{B}_n)H^2(\mathbb{B}_n)$ form a first-category subset of $H^1(\mathbb{B}_n)$.

9.21. **A BMO route to Lindelöf for Dirichlet series**

Is there a BMO-type space of Dirichlet series between H^∞ and the symbols of bounded Hankel forms, such that every member has finite vertical p-means and $\zeta_1(\sigma + \cdot)$ belongs to it for every $\sigma > 1/2$?

McCarthy, *The Lindelöf Hypothesis and BMO*, Oberwolfach Report 06/2014, Problem session, item 2, pp. 384–385.
Answer: Partial. **Evidence:** Sketch.

McCarthy explains that neither endpoint candidate, $BMO_1 = (H^1)^*$ nor BMO_2, has the required all-moments property. Our tests rule out natural rotation-invariant BMO candidates; existence of a different space is unresolved.

9.22. **A Nehari theorem for multiplicative Hankel forms**

Does every bounded multiplicative Hankel form have a bounded symbol ψ on the polytorus $\mathbb{T}^\infty$?

Brevig and Perfekt, *Failure of Nehari's theorem for multiplicative Hankel forms in Schatten classes*, Introduction, p. 1, quoting Helson's question from *Hankel forms and sums of random variables.*
Answer: Answered: no. **Evidence:** Published.

Ortega-Cerdà and Seip construct bounded forms for which the least finite-dimensional symbol norms grow without bound, ruling out a bounded symbol on the infinite torus.

9.23. **Contractivity of diagonal restriction**

> Numerical estimation of the norms, with up to fifth degree polynomials for $g(w)$ and $(D_d f)(w)$, suggests the following conjectures: $\|D_d\|_p = 1$, $p \geq 2$, and $\|E_d\|_p > 1$, $p \neq 2$.

Birk D. Ramberg, *Nehari's theorem for multiplicative Hankel matrices*, numerical conjectures before Theorem 2, printed p. 5.

Answer: Partial. **Evidence:** arXiv.

The public NTNU institutional report proves contractivity for $p = 2$ and all positive even integers. It remains a public institutional report, not an arXiv posting; the general range $p \geq 2$ is open.

Conditional on: Birk D. Ramberg's unrefereed NTNU institutional report.

9.24. **Strict noncontractivity of the diagonal right inverse**

> Numerical estimation of the norms, with up to fifth degree polynomials for $g(w)$ and $(D_d f)(w)$, suggests the following conjectures: $\|D_d\|_p = 1$, $p \geq 2$, and $\|E_d\|_p > 1$, $p \neq 2$.

Birk D. Ramberg, *Nehari's theorem for multiplicative Hankel matrices*, numerical conjectures before Theorem 2, printed p. 5.

Answer: Answered: yes. **Evidence:** arXiv.

The public NTNU institutional report proves strict noncontractivity for every $p \neq 2$. It is an institutional report, not an arXiv posting.

Conditional on: Birk D. Ramberg's unrefereed NTNU institutional report.

9.25. **A bounded symbol for Hilbert–Schmidt forms**

> Helson proved that any Hankel form on $H^2 \times H^2$ which is of Hilbert–Schmidt class S_2 is induced by a bounded symbol on the infinite polytorus $\mathbb{T}^\infty$.

Brevig and Perfekt, *Weak product spaces of Dirichlet series*, §2. They identify the corresponding question as Helson's Question 3 in *Dirichlet Series*, p. 54; the exact wording of that question was not available for verification. The inferred target is whether every Hilbert–Schmidt multiplicative Hankel form has a bounded symbol.

Answer: Answered: yes. **Evidence:** Published.

Helson's §2 theorem proves the bounded-symbol assertion for every Hilbert–Schmidt multiplicative Hankel form. The exact wording in the original book was unavailable locally; the source and location are verified through Helson's citation and answer.

10. Hilbert points

10.1. **Equal-modulus coefficients of Hilbert points**

> Suppose that $1 \leq p < 2$. If $\varphi(z) = \sum_{j=1}^{d} c_j z_j$ is a Hilbert point in $H^p(\mathbb{T}^d)$, then the nonzero coefficients of φ all have the same modulus.

Brevig, Ortega-Cerdà, and Seip, *Hilbert points in Hardy spaces*, Conjecture 3.1. The source poses this question for $d \geq 3$.

Answer: Answered: no. **Evidence:** Sketch.

A three-variable example violates the equal-modulus conclusion. The public remarks give a full proof of the counterexample. The counterexample

10.2. **Unbounded Hilbert points below and above two**

> We do not know, for instance, whether there exists an unbounded Hilbert point in $H^p(\mathbb{T}^d)$ for some $p \neq 2$.

Brevig, Ortega-Cerdà, and Seip, *Hilbert points in Hardy spaces*, §4, paragraph following Question 4.1, p. 22. The curated target asks for a finite dimension and exponent with such an example.

Answer: Partial. **Evidence:** Sketch.

Finite-dimensional tests and a bootstrap rule out several candidate ranges and structures. They do not decide whether an unbounded Hilbert point exists in general.

10.3. **A cubic Hilbert-point exponent question**

Numerical evidence suggests that this φ is a Hilbert point in $H^p(\mathbb{T}^3)$ only when $p = 2, 4$. Unfortunately, we are only able to verify analytically that there is possibly a finite number of p in $[1, 4) \setminus \{2\}$ for which φ is a Hilbert point in $H^p(\mathbb{T}^3)$.

Brevig, Ortega-Cerdà, and Seip, *Hilbert points in Hardy spaces*, discussion following Question 4.1, p. 19. The curated target is whether the displayed cubic is a Hilbert point at any exponent in the source's remaining range.
Answer: Partial. **Evidence:** Sketch.

Our argument excludes the displayed cubic for exponents sufficiently close to 2, with the threshold depending on dimension. The source's broader remaining exponent range is unresolved.

10.4. **Dynamics of the nonlinear Hilbert-point projection**

Suppose that $\varphi_0(z) = \sum_{j=1}^{d} c_j z_j$ is in the unit sphere of $H^2_1(\mathbb{T}^d)$ and that $c_1 > c_j \geq 0$ for every $j = 2, \ldots, d$. Is it true that $\lim_{n\to\infty}(P_p^n \varphi_0)(z) = z_1$ whenever $1 \leq p < 2$?

Brevig, Ortega-Cerdà, and Seip, *Hilbert points in Hardy spaces*, Question 4.1, p. 15.
Answer: Partial. **Evidence:** Sketch.

Our argument proves convergence for p sufficiently near 2, with a threshold depending on d. The full range $1 \leq p < 2$ in the source question remains unresolved.

11. Fock-space spectral optimization

11.1. **Nearly invariant subspaces of Fock-type spaces**

Is it true that any nearly invariant subspace without common zeros in a Fock-type space

Aleman, Baranov, Belov, and Hedenmalm, *Backward shift and nearly invariant subspaces of Fock-type spaces*, final problem in the Introduction, p. 6. The complete question asks whether each such subspace of $\mathcal{F}^p_\alpha$ is one-generated, of the form $\mathcal{M}_G$; the paper defines these spaces with radial weights.
Answer: Partial. **Evidence:** arXiv.

The source proves ordering results in slow-growth regimes. The general radial Fock-type case remains without a verified answer in the checked literature.

11.2. **Uniqueness of discs for Ky Fan sums**

Which sets maximize the sum

Nicola, Riccardi, and Tilli, *The isoperimetric inequality for partial sums of Toeplitz eigenvalues in the Fock space*, Introduction. They ask which sets of prescribed area maximize the Ky Fan sum $\sum_{n=1}^{N} \lambda_n(T_\Omega)$. Abreu's later Theorem 1 proves disc optimality, while Remark 1 says that uniqueness up to translations and null sets was left open.
Answer: Partial. **Evidence:** arXiv.

Conditional on: arXiv:2607.28376v1.

The disc is a maximizer for every Ky Fan sum, according to an unrefereed preprint. Whether every maximizer is a translate of a disc, up to null sets, remains open.

11.3. **Maximizing a single higher Fock Toeplitz eigenvalue**

which sets maximize the second eigenvalue

Nicola, Riccardi, and Tilli, *The isoperimetric inequality for partial sums of Toeplitz eigenvalues in the Fock space*, Introduction. The source asks which sets $\Omega \subset \mathbb{C}$ of prescribed area maximize $\lambda_2(T_\Omega)$; it also raises the analogous question for a single λ_k, $k \geq 2$.
Answer: Partial. **Evidence:** arXiv.

Conditional on: arXiv:2503.07069.

For radial sets, an annulus is optimal for λ_2, but nonradial sets can improve on that radial optimizer. A global characterization for λ_2, and the analogous questions for higher single eigenvalues, remain open.

12. Hankel forms and Schatten classes

12.1. **Explicit diagonalization of a multiplicative Hilbert matrix**

Despite the similarity with Theorem A, we do not have an explicit diagonalisation of either $M(g_a)$ or $M_2(g_0)$; to find one is an interesting open problem.

Perfekt and Pushnitski, *Spectra of the Hilbert matrix*, Introduction, paragraph after Theorem 1.2. The known spectral description is abstract and does not give an explicit diagonalizing transform.
Answer: Open. **Evidence:** Heuristic.

The spectrum is known to be purely absolutely continuous on $[0, \pi]$, but no explicit kernel for the requested transform was found in the checked sources. The shifted and unshifted operators are both part of the question.

12.2. **Eigenvalues above the critical band for sparse prime sets**

eigenvalues above Λ_α

Ma, Lu, and Zu, *On the Spectra of Multiplicative Hilbert Operator on Bergman Spaces of Dirichlet Series*, Theorem 1.1(5), p. 3. The inferred extension asks whether a critical Hardy matrix on a regular sparse-prime semigroup can have eigenvalues above its essential band or embedded in it. This is our question, motivated by the cited Bergman theorem, not a question stated by the authors.
Answer: Answered: depends on the prime set. **Evidence:** Sketch.

Finite deletions can force arbitrarily many eigenvalues above the band, whereas for primes congruent to 1 modulo 4 the norm equals the band edge and no above-band eigenvalue occurs. Embedded eigenvalues are excluded under an explicit C^2 regularity hypothesis. The complete argument is in the book.
Answers, "Sparse-prime eigenvalues above the critical band", p. 15.

Book draft, eigenvalues of critical prime-set matrices section

12.3. **The unimodality of a multiplicative Hilbert norm**

Does there exist a unique $\alpha_{\min} > 1$ such that $\|T_\alpha\|$ is strictly decreasing on $(1, \alpha_{\min})$ and strictly increasing on $(\alpha_{\min}, \infty)$?

Ma, Lu, and Zu, *On the Spectra of Multiplicative Hilbert Operator on Bergman Spaces of Dirichlet Series*, §6.2, Question 1, p. 73 (version 3).
Answer: Answered: yes. **Evidence:** Sketch.
Conditional on: arXiv:2608.22882v3.

An interval-certified argument gives a unique minimum with $\alpha_{\min} \in (8/5, 3)$. The proof uses the source's kernel model, essential-spectrum description, and convexity theorem.
Answers, "Unimodality of a multiplicative Hilbert norm", p. 16.

12.4. **An explicit norm formula for the multiplicative Hilbert operator**

Can $\|T_\alpha\|$ be computed explicitly for $\alpha > \alpha_*$?

Ma, Lu, and Zu, *On the spectra of multiplicative Hilbert operator on Bergman spaces of Dirichlet series*, §6.2, Question 2, p. 74 (version 3, 9 October 2026).
Answer: Open. **Evidence:** arXiv.

The paper shows that this norm is the largest eigenvalue above the threshold, but gives no explicit formula for that eigenvalue.

12.5. **An explicit diagonalization below the spectral threshold**

For $1 < \alpha \le \alpha_*$, can one construct explicitly a unitary operator $U_\alpha : \mathcal{A}^2_{\alpha,0} \longrightarrow L^2((0,\Lambda_\alpha), \mathrm{d}\lambda)$ such that $(U_\alpha T_\alpha U_\alpha^{-1} g)(\lambda) = \lambda g(\lambda)$?

Ma, Lu, and Zu, *On the spectra of multiplicative Hilbert operator on Bergman spaces of Dirichlet series*, §6.3, Question 3, p. 74 (version 3, 9 October 2026).
Answer: Open. **Evidence:** arXiv.

The paper identifies the spectrum abstractly below α_*, but does not construct an explicit unitary map to multiplication on $L^2((0,\Lambda_\alpha))$.

12.6. **Closability and maximal realization in infinitely many variables**

Under the Carleman condition, closability of h_0 is equivalent to

Ma, Lu, and Zu, *Unbounded Helson forms with a moment representation*, Theorems 1.1–1.2. The inferred extension asks whether the closability criterion $\sqrt{n}\,\alpha(n) \to 0$ and the maximal-realization conclusion extend to product moment measures on $(-1,1)^\infty$. This is our question, not one stated by the authors.
Answer: Partial. **Evidence:** Sketch.

For independent coordinates, a parity-block calculation gives an exact closability criterion and product norm. In a natural class the maximal map is closed, but equality of its domain with the polynomial graph closure remains open; a support-only criterion is insufficient.

12.7. **Spectral information from unbounded Helson block profiles**

Moreover, H_μ is trace class if and only if $\sum_{n\ge1} \alpha(n^2) < \infty$.

Ma, Lu, and Zu, *Unbounded Helson forms with a moment representation*, §3. The inferred question is whether spectral type and compact-resolvent behavior can be classified from block profiles of the representing measure. This extension is not stated as a question in the source.
Answer: Partial. **Evidence:** Sketch.

Separated-atom criteria give discrete eigenvalue bounds, but block masses alone do not characterize compact resolvent, even within atomic measures. A useful criterion must retain within-block separation or Gram information.

12.8. **The critical essential spectrum of sparse-prime matrices**

For a set R of primes of density θ, the critical matrix on R-smooth numbers has essential spectrum $[0, c_R\,\Gamma(\theta/2)^2/\Gamma(\theta)]$, where $c_R = \lim_{x\downarrow0} x^\theta \zeta_R(1+x)$.

Søvik, book draft, Chapter 5, question on the critical sparse-prime spectrum.
Answer: Answered: yes. **Evidence:** Sketch.

The critical Mellin comparison identifies the essential spectrum with the stated interval under the actual limit hypothesis. The book draft contains a full written proof and records how oscillating profiles change the answer. Book draft, the companion-kernel theorem and translated-kernel limits

12.9. **A universal Mellin constant for positive Helson matrices**

Let the companion kernel of a positive Helson matrix grow like $c\,x^{-\kappa}$ at the boundary, with the fixed critical weight $u^{\kappa-1}\,\mathrm{d}u$. Is its critical essential spectrum always $[0, c\,\Gamma(\kappa/2)^2/\Gamma(\kappa)]$?

Søvik, book draft, Chapter 5, problem on the universal positive-Helson constant.
Answer: Answered: yes under a limit; no under mere comparability. **Evidence:** Sketch.

For the literal asymptotic $x^{\kappa}h(x) \to c$ with the critical weight, the interval formula holds. Comparability alone does not suffice: oscillating profiles can open a gap; a broader limit-operator criterion is in the book.
Answers, "A universal Mellin constant for positive Helson matrices", p. 17.

Book draft, the companion-kernel theorem and translated-kernel limits

12.10. **The exponent in bounded-rank Nehari growth**

The gap between the exponents $0.174251\ldots$ and $1/2$ remains open.

Søvik, bounded-rank Nehari question, final paragraph. The underlying question is to determine the sharp growth of $C(r)$, the least bounded-symbol norm divided by the operator norm among Hankel forms of rank at most r. The known bounds are $r^{\log_2(2/\sqrt{\pi})} \lesssim C(r) \leq \sqrt{r}$.
Answer: Partial. **Evidence:** Sketch.

Rank-tracked constructions give the lower exponent $\log_2(2/\sqrt{\pi}) \approx 0.174$, and a general rank bound gives exponent $1/2$. The family estimates do not close this gap.

12.11. **A Besov criterion for curve-measure Schatten classes**

For noninteger p, converting the corresponding spectral condition into a one-dimensional Besov criterion on F remains open.

Søvik, question following the integer trace formulas for curve-measure Helson matrices. Let ν be a finite positive Borel measure on $(0,\infty)$ and put $F(t) = \nu((0,t])$ and $M_\nu = (\int_0^\infty (mn)^{-1/2-u}\,\mathrm{d}\nu(u))_{m,n\geq 1}$. We take this as asking whether, for noninteger $p > 0$, membership in S_p can be characterized by a one-dimensional Besov condition on F.
Answer: Answered: yes, at an endpoint. **Evidence:** Sketch.

The criterion is the endpoint-localized first-difference condition $F \in B^1_{r,p}(0)$ for every $0 < r \leq \infty$; no fixed global classical Besov space on $(0,1)$ characterizes the same class. The proof follows from the dyadic Schatten criterion and a comparison with $F(t)/t$.
Answers, "A Besov criterion for curve-measure Schatten classes", p. 17.

Book draft, "Distribution functions and endpoint smoothness" section

13. Coefficient inequalities

13.1. **Real-power divisor reverse estimate**

"It seems reasonable to conjecture that (20) should hold whenever j is a real number larger than 1."

Source: Bondarenko, Heap, and Seip, "An Inequality of Hardy–Littlewood Type for Dirichlet Polynomials," arXiv:1405.6516v3, concluding remarks, p. 14, following equation (20). In (20), $q = 2^j$ and j is a positive integer; the question here is the stated real-j extension with

$$\left(\sum_{n=1}^{N} |a_n|^2 d(n)^{j-1}\right)^{1/2} \geq \left\|\sum_{n=1}^{N} a_n n^{-s}\right\|_{\mathcal{H}^{2j}}.$$

Answer: Answered: yes. **Evidence:** Published.

Llinares, "Contractive Inequalities between Dirichlet and Hardy Spaces," Revista Matemática Iberoamericana 40 (2024), Theorem 3.1 and Corollary 3.2, proves the reciprocal generalized-divisor weight estimate. Put $p = 2^j$, $b = j - 1$, and $\alpha = 2/p = 2^{-b}$. The prime-power comparison $1/d_\alpha(n) \leq d(n)^b$ gives the requested estimate for every real $j > 1$.

14. Contractivity and hypercontractivity

14.1. **Contractive Bergman inclusions $A^p_\alpha \subset A^q_\beta$**

"For any $0 < p \leq q$ and $\alpha, \beta \geq 1$ such that $q/\beta \leq p/\alpha$, does the contractive inequality $\|f\|_{A^q_\beta(\mathbb{D})} \leq \|f\|_{A^p_\alpha(\mathbb{D})}$ hold?"

The source uses the probability measure $(\alpha-1)(1-|z|^2)^{\alpha-2}\,\mathrm{d}m$ for $\alpha>1$ and sets $A^p_1=H^p$. Source: Bayart, Brevig, Haimi, Ortega-Cerdà, and Perfekt, "Contractive Inequalities for Bergman Spaces," arXiv:1701.06897v2, §2.2, pp. 9–10.
Answer: Answered: yes. **Evidence:** Published.

Llinares, "Contractive Inequalities between Weighted Bergman Spaces," *Revista Matemática Iberoamericana* 40 (2024), Corollary 1.1, characterizes contractive inclusions of normalized weighted Bergman spaces. Its parameter criterion includes this disc range.

14.2. **Sharp Hardy to weighted Bergman inequality**

"The inequality (3) holds for every $0<p\le 2$."

Here equation (3) is $\left(\sum_{n=0}^{\infty}|a_n|^2/c_{2/p}(n)\right)^{1/2}\le\|f\|_{H^p}$, and $c_\alpha(n)$ is defined by $(1-z)^{-\alpha}=\sum_{n=0}^{\infty}c_\alpha(n)z^n$. Source: Brevig, Ortega-Cerdà, Seip, and Zhao, "Contractive Inequalities for Bergman Spaces," arXiv:1706.00738v2, Conjecture 1, equation (3), p. 1.
Answer: Answered: yes. **Evidence:** Published.

Kulikov, *Geometric and Functional Analysis* 32 (2022), Theorem 1.1 and Corollary 1.3, proves the sharp Hardy-to-weighted-Bergman coefficient inequality for $0<p\le 2$, answering this conjecture affirmatively.

Part 3. Where Nehari's theorem fails: Schatten classes

15. Paley–Wiener spaces

15.1. **Bounded symbols for Schatten-class Hankel operators**

"whether the conclusion of the Nehari theorem holds when $H_\varphi\in S_p$, $1<p<\infty$"

The question in §4, printed p. 20, of Konstantinos Bampouras and Karl-Mikael Perfekt, *Besov spaces and Schatten class Hankel operators for Hardy and Paley–Wiener spaces in higher dimensions*, arXiv:2409.04184v6. For a bounded convex domain Ω, the reconstructed target is to classify the pairs (Ω,p) for which every S_p Hankel operator on $PW(\Omega)$ has a bounded symbol.
Answer: Partial. **Evidence:** Sketch.

Published results settle trace-class symbols and the bounded-operator case for simple convex polytopes. A recent arXiv paper claims the Hilbert–Schmidt case for convex domains without affine lines, while curved domains with a positive-curvature patch fail for $p>4$. The general curved-domain range $2<p\le 4$ remains open.

15.2. **Bounded symbols for Hankel operators on the disc**

Does every bounded Hankel operator on $PW(\mathbb{D})$ have a bounded symbol?

The first question in the introduction, printed p. 2, of Ole Fredrik Brevig and Karl-Mikael Perfekt, *The Nehari problem for the Paley–Wiener space of a disc*, J. Geom. Anal. 33 (2023), Article 16.
Answer: Answered: no. **Evidence:** Published.

Their Theorem 1 constructs a bounded Hankel operator on the disc Paley–Wiener space that has no bounded symbol. The Hilbert–Schmidt case is a separate question.

15.3. **Hilbert–Schmidt symbols for Hankel operators on the disc**

Does every Hankel operator on $PW(\mathbb{D})$ belonging to S_2 have a bounded symbol?

The second question in the introduction, printed p. 2, of Brevig and Perfekt, *The Nehari problem for the Paley–Wiener space of a disc*.
Answer: Answered: yes. **Evidence:** arXiv.

Bampouras and Perfekt's Corollary 1.8 proves the result for every open ball in $\mathbb{R}^n$, $n\ge 2$, including the disc. The proof is in their latest arXiv preprint, arXiv:2608.19806v1.

16. Hankel forms and Schatten classes

16.1. **AAK theory beyond the circle**

To extend the AAK theorem beyond its classical $H(T)$-setting (see e.g. [N], App. 4).

Janson, Peetre, and Rochberg, *Hankel forms and the Fock space*, Notes and Comments §4.1, p. 137.

Answer: Partial. **Evidence:** Published.

Extensions exist under shift hypotheses, while best rank-one approximation can fail to attain the classical singular-value bound in the Dirichlet space. The broad extension program across their analytic-space framework is not settled.

16.2. **Hankel Schatten theory below one**

It would be interesting to have any general results for $0 < p < 1$ too.

Janson, Peetre, and Rochberg, *Hankel forms and the Fock space*, Notes and Comments §4.2, p. 137.

Answer: Partial. **Evidence:** Published.

Wallstén's Fock-space results give a substantial special case, and an annulus criterion covers a restricted subunit range. A general theory for the analytic spaces and hypotheses of the paper remains open.

16.3. **A Bergman analogue of the metaplectic group**

Is there an analogue of the metaplectic group in the case of Bergman space (cf. [P6])?

Janson, Peetre, and Rochberg, *Hankel forms and the Fock space*, Appendix II, Problem 4.4, p. 137.

Answer: Partial. **Evidence:** Published.

Weighted Bergman realizations of the one-mode metaplectic representations and projective actions on Bergman spaces are known. The original question specifies neither the domain nor the covariance or symbol-space structure required of an analogue.

16.4. **Hankel forms on infinite-dimensional Fock space**

What happens on the infinite-dimensional Fock space for $p \neq 2, 4$ (in particular for $p = \infty$)?

Janson, Peetre, and Rochberg, *Hankel forms and the Fock space*, §7, Problem after Theorem 7.8, p. 103.

Answer: Open. **Evidence:** Heuristic.

The source derives formal identities at $p = 2$ and $p = 4$ while explicitly setting aside definition issues. The bounded search found no general classification for other exponents in the commutative infinite-dimensional Fock realization.

16.5. **Same-rank approximation by point evaluations**

We conjecture that it is possible to take N_m as the rank of H, i.e. that the set of continuous Hankel forms of rank $\leq r$ coincides with the closure (in any reasonable topology) of the set $\{(f, g) \mapsto \sum_{j=1}^{r} c_j f(z_j) g(z_j) : c_1, \ldots, c_r \in \mathbb{C}, z_1, \ldots, z_r \in \Omega\}$. (For $n = 1$, this follows easily from (14.3), but we have been unable to find a proof in higher dimension.)

Janson, Peetre, and Rochberg, *Hankel forms and the Fock space*, Remark 14.2, p. 126. In context, the conjecture includes the Fock-space case and claims the corresponding approximation for all finite-rank Hankel forms.

Answer: Answered: no. **Evidence:** Sketch.

A continuous rank-fourteen Hankel form on a six-dimensional Fock space has a neighborhood, detected by finitely many polynomial moments, containing no sum of at most fourteen evaluation forms. The one-variable assertion is not affected.
Answers, "Rank-fourteen obstruction to same-rank evaluation approximation", p. 20.

16.6. **Hankel forms concentrated on analytic subvarieties**

What can be said about the boundedness or smoothness of such forms?

Janson, Peetre, and Rochberg, *Hankel forms and the Fock space*, Remark 14.3, p. 126.
Answer: Partial. **Evidence:** Sketch.

Examples on a linear subvariety in Gaussian Fock space and one criterion in $\mathbb{C}^2$ address special cases. The source asks for a general description of the kernel ideal, the corresponding variety, and regularity, which remains open.

16.7. **Schatten classes without bounded symbols**

For every $p > 2$ there exists a multiplicative Hankel form H_φ in $S_p(\mathscr{H}^2)$ without a bounded symbol.

Brevig and Perfekt, *Weak product spaces of Dirichlet series* (2016), Conjecture 1 following Theorem 6, p. 10. The question concerns the full $\mathscr{H}^2$ space, with the constant row and column included.
Answer: Partial. **Evidence:** arXiv.
Conditional on: arXiv:2610.06728v1.

Brevig and Perfekt construct examples for $p > 5.7388\ldots$; Bampouras's unrefereed Theorem 1 extends failure to every $p > 4$. The range $2 < p \leq 4$ remains open.

16.8. **Existence of a trace limit for multiplicative Toeplitz matrices**

Let $\sigma_N = \{1, \ldots, N\}$; does the limit in (3.4) exist?

Nikolski and Pushnitski, *Szegő-type limit theorems for " multiplicative Toeplitz" operators and non-Følner approximations*, §3.8(a), p. 16. Equation (3.4) is the normalized trace $N^{-1} \operatorname{Tr} f(T_{\sigma_N}(\varphi))$ for bounded real φ and continuous f.
Answer: Answered: yes. **Evidence:** Published.

Guo, Li, and Zhou prove existence for these sections in Theorem 1.11 of their 2024 paper.

16.9. **A closed expression for the multiplicative trace limit**

Can it be expressed in a closed form in terms of f and φ?

Nikolski and Pushnitski, *Szegő-type limit theorems for " multiplicative Toeplitz" operators and non-Følner approximations*, §3.8(a), p. 16. The requested expression is for $N^{-1} \operatorname{Tr} f(T_{\sigma_N}(\varphi))$, with bounded real φ and continuous f.
Answer: Partial. **Evidence:** Sketch.

Our construction represents the limit as an expectation of a vector state of a random compression. Whether the question requires an elementary scalar formula remains open.

16.10. **The determinant formula for finite multiplicative measures**

Let $\varphi \geq 0$, $\varphi \in L^1(\mathbb{T}^\infty)$. Does the relation (3.9) necessarily hold in this generality? Furthermore, does it hold (similarly to the additive case) for multiplicative Toeplitz matrices of the form $\{\widehat{\mu}(j/k)\}_{j,k=1}^{\infty}$, where μ is a finite measure on $\mathbb{T}^\infty$ with $d\mu = \varphi\, dm + \mu_{\mathrm{sing}}$?

Nikolski and Pushnitski, *Szegő-type limit theorems for " multiplicative Toeplitz" operators and non-Følner approximations*, §3.8(b), p. 16. Relation (3.9) concerns determinants along multiplicative Følner sections.
Answer: Answered: yes. **Evidence:** arXiv.
Conditional on: arXiv:2412.13751v4.

Guo, Li, and Zhou establish (3.9) for nonnegative L^1 symbols; Austin's unrefereed Theorem A extends it to finite positive measures, including a singular part.

16.11. **Operator ideals with the Berger–Coburn phenomenon**

Are there nontrivial operator ideals in $B(H^2(\mathbb{T}), L^2(\mathbb{T}))$ or in $B(A^2(\mathbb{B}_n), L^2(\mathbb{B}_n))$ for which the Berger-Coburn phenomenon holds?

Hu and Virtanen, *On the Berger-Coburn phenomenon*, Open Problem 1, p. 4.
Answer: Answered: no. **Evidence:** Sketch.

Any ideal with the phenomenon must contain every bounded operator, so the only such ideal is the whole operator space. The proof is in our public note.
Answers, "Operator ideals with the Berger–Coburn phenomenon", p. 21.

Three open problems of Hu and Virtanen

16.12. **Schatten exponents for the Berger–Coburn phenomenon on Fock spaces**

Let F^2_ϕ be a doubling Fock space. For which values of p does the Berger-Coburn phenomenon hold for $S_p(F^2_\phi, L^2_\phi)$?

Hu and Virtanen, *On the Berger-Coburn phenomenon*, Open Problem 2, p. 9.
Answer: Answered: $p = 2$ on every space; some spaces for each $p > 1$. **Evidence:** Sketch.

For $0 < p \leq 1$ it fails on every doubling Fock space; for $1 < p < 2$ and $p > 2$ there are counterexample weights, while constant-Laplacian weights retain the phenomenon for all $1 < p < \infty$. The proof is in our public note.
Answers, "Schatten exponents for the Berger–Coburn phenomenon on Fock spaces", p. 21.

Three open problems of Hu and Virtanen

16.13. **Compactness and conjugate Fock Hankel operators**

Let F^2_ϕ be a doubling Fock space and $f \in L^\infty$. Is it true that if $H_f : F^2_\phi \to L^2_\phi$ is compact, then $H_{\bar{f}}$ is also compact?

Hu and Virtanen, *On the Berger-Coburn phenomenon*, Open Problem 3, p. 9.
Answer: Answered: yes. **Evidence:** Sketch.

Compactness forces the local distance from f to holomorphic functions to vanish at infinity. Rescaling by the doubling radius and applying Liouville's theorem yields vanishing mean oscillation, which gives compactness of the conjugate Hankel operator.
Answers, "Compactness and conjugate Fock Hankel operators", p. 22.

Three open problems of Hu and Virtanen

16.14. **Unbounded symbols and the Berger–Coburn phenomenon**

Characterize unbounded symbols f for which BCP holds for compact operators or for Schatten class operators H_f on Fock spaces.

Hu and Virtanen, *On the Berger-Coburn phenomenon*, Open Problem 4, p. 11.
Answer: Open. **Evidence:** Heuristic.

The source gives examples on both sides, but no characterization. The compact bounded-symbol result and the Schatten results for doubling spaces do not settle this unbounded-symbol problem.

16.15. **Logarithmic-BMO symbols on weighted Bergman A^1**

For $1 < p < \infty$, the problem whether $H^\omega_f : A^1_\omega \to L^1_\omega$ is bounded

Duan, Guo, Hong, and Yan, *Toeplitz and Hankel operators on Bergman spaces with doubling weights*, final paragraph before the references. The source asks whether this holds for every $f \in \mathrm{BMO}^p_{\omega,\log}$, even in the classical Bergman space; its hypotheses use radial two-sided doubling weights $\omega \in \mathcal{D}$.
Answer: Answered: yes. **Evidence:** Sketch.

For every such weight, the logarithmic local-oscillation condition implies the $A^1_\omega \to L^1_\omega$ Hankel bound. The full proof is in our public note.
Answers, "Hankel operators on weighted Bergman A^1", p. 22.

Hankel operators with logarithmic-BMO symbols

16.16. **Why the threshold $p = 4$ appears in two settings**

It would be interesting to understand whether the appearance of the same threshold in these two settings reflects some connection between them. In particular, the endpoint case $p = 4$ seems especially interesting, since its Schatten norm can be explicitly computed.

Bampouras, *Nehari's theorem for Schatten classes of Hankel forms on the polydisc and Paley-Wiener spaces*, arXiv:2610.06728v1, end of §1. In particular, the authors single out the computable S_4 endpoint; the question here is whether the matching threshold has a structural explanation.

Answer: Answered: the mechanisms differ. **Evidence:** Sketch.

On the torus the S_4 identity has a positive constant-term contribution, which yields the L^4 bound; in the Paley-Wiener setting the same bound fails. The planar cap-counting method reaches the same threshold through packing and square-root cancellation. The endpoint Nehari question itself remains open.

Answers, "Why the threshold $p = 4$ appears in two settings", p. 23.

16.17. **The multiplicative S_4 endpoint**

Every multiplicative Hankel form satisfies $|\langle f, \varphi\rangle| \le \|f\|_{L^1}\|H_\varphi\|_{S_4}$.

Søvik, book draft, Chapter 8, B2. The open question asks whether this proposed endpoint estimate holds on the full infinite torus. The estimate is equivalent to asking whether the multiplicative Schatten threshold is exactly 4; the constant one is forced by the known sharp normalization.

Answer: Partial. **Evidence:** Sketch.

The estimate holds for several structured families, including all affine symbols and reviewed constant-plus-quadratic families with one coordinate group of size at most two. General symbols remain open.

16.18. **A uniform Gaussian gap above** 2

Is there a dimension- and degree-free interval above 2 with no strict Gaussian blocks?

Søvik, book draft, Chapter 8, further questions near $p = 2$. A strict block means a symbol whose Gaussian dual norm exceeds the Schatten p-norm of its Hankel matrix.

Answer: Partial. **Evidence:** Sketch.

The book draft contains full written arguments excluding strict blocks at fixed dimension and degree and at bounded rank, and treats several structured families. No dimension- and degree-uniform gap is known; coherent-symbol limits still approach exponent 2 without settling the question. Book draft, "Near $p = 2$" section

16.19. **Finite-rank forms on a punctured spectral-algebra surface**

For the one complex dimensional Riemann surface of Example 3.4 it seems reasonable to conjecture that every finite rank symbol is of the form $b(\cdot) = \sum_{k=1}^{N} a_k Z^{n_k} K_{z_k}(\cdot)$ for some constants a_k, z_k, n_k.

Deck and Gross, *Hankel forms on complex manifolds*, Remark 6.10, p. 94, for the one-complex-dimensional Riemann surface in Example 3.4.

Answer: Answered: no. **Evidence:** Sketch.

On the punctured surface the constant symbol gives a bounded rank-one Hankel form, but it cannot be a finite sum of the proposed point-jet symbols: the spectral algebra contains only nonnegative powers, and the missing branch point prevents a finite jet representation. This counterexample addresses the literal example, not the broader manifold program.

Answers, "A finite-rank form missed by point jets on a punctured surface", p. 25.

16.20. **Real interpolation of Gaussian Fock scales**

It is an interesting (open) question to determine the spaces which arise by real interpolation from this scale.

Janson, Peetre, and Rochberg, *Hankel forms and the Fock space*, Introduction, p. 70; for fixed p this is related to spectral analysis of the dilation operator.

Answer: Partial. **Evidence:** Published.

For fixed interior p, dilation-resolvent results identify the couple, and a uniform estimate proposed in Example 0.2 is false. A classification across mixed exponents and Gaussian scales remains open.

16.21. **Higher-weight Hankel forms in several variables**

To extend the theory of higher weight Hankel forms in [JP2] to more general cases, for instance the unit ball in $\mathbb{C}^d$. This is basically a question of Invariant Theory.

Janson, Peetre, and Rochberg, *Hankel forms and the Fock space*, Notes and Comments §4.3, p. 137.

Answer: Partial. **Evidence:** Published.

Invariant-theory decompositions are known on balls and bounded symmetric domains, and Schatten criteria are available on the ball for $p \geq 1$. The full analytic theory on more general domains, including Cartan domains, remains only partly developed.

16.22. **The Wilf eigenvalue-counting asymptotic**

We have the following corollary, which was conjectured by Wilf. For the Hilbert matrix $((j+k+\alpha)^{-1})$, $j,k = 0,1,\ldots,N-1$, with $\alpha \neq 0,-1,-2,\ldots$, the number of eigenvalues which exceed πa $(0 < a < 1)$ is asymptotically $2/\pi \log N \operatorname{sech}^{-1} a$.

Widom, *Hankel matrices*, Corollary on printed pp. 30–31. The question attributed there to Wilf is whether this counting asymptotic holds for each fixed admissible α and $0 < a < 1$.

Answer: Answered: the stated asymptotic. **Evidence:** Published.

Widom attributes the formula to Wilf and proves it as a corollary.

16.23. **The Paley-Wiener L^1–S_4 endpoint**

Let Ω be bounded, convex, with C^2 boundary of positive curvature. Is there C_Ω with $|\langle h, \varphi\rangle| \leq C_\Omega \|h\|_1 \|H_\varphi\|_{S_4}$ for all Schwartz h, φ with Fourier support in $2\overline{\Omega}$?

Søvik, Paley-Wiener boundary-cap questions, endpoint question after the S_p-counting discussion.

Answer: Partial. **Evidence:** arXiv.

Conditional on: arXiv:2504.05986v3.

The S_2 estimate is known and curved-patch constructions give failure above $p = 4$. Reviewed packet estimates handle restricted families, but the full L^1–S_4 inequality is open.

16.24. **Exponential growth on product discs**

Is $\sup Q_p > 1$ on the disc for some $p \leq 4$?

Søvik, Paley-Wiener boundary-cap questions, the disc-block question. The source says a positive answer would tensorize to exponential S_p-Nehari constant growth on products $\mathbb{D}^m$; the relevant range is $2 < p \leq 4$.

Answer: Answered: yes. **Evidence:** Sketch.

For every $p \geq 2$ there are disc-band-limited functions with $|\langle h, \varphi\rangle| > \|h\|_1 \|H_\varphi\|_{S_p}$; tensorization therefore gives exponential growth in the product dimension. Book draft, "The product family is sharp at 4" section

16.25. **Dimension-uniform Nehari constants on finite tori**

Whether $C_p(d)$ remains bounded in d for any or all $p \leq 4$ is not decided by this argument.

Søvik, *Dimension growth of the Schatten-to-symbol constant*, final paragraph. Here $C_p(d)$ is normalized by the analytic L^1 norm and the full small Hankel matrix. The question is whether these constants remain uniformly bounded for any or all $p \leq 4$.

Answer: Partial. **Evidence:** Sketch.

Helson's theorem gives the bound for $p \leq 2$, while an unrefereed construction gives failure for $p > 4$. The interval $2 < p \leq 4$, including the endpoint, is unresolved; finite-dimensional restricted families do not settle the uniform constant.

16.26. **A Gaussian analytic L^1–L^4 inequality**

Is it true that $|\int_{\mathbb{C}^d} f\overline{\varphi}\, \mathrm{d}\mu_d| \leq \|f\|_{L^1(\mu_d)}\|\varphi\|_{L^4(\mu_d)}$ for all analytic polynomials f, φ on $\mathbb{C}^d$ and every d?

Søvik, Gaussian Hankel-block questions, the constant-one pairing question; μ_d is standard complex Gaussian probability measure.

Answer: Partial. **Evidence:** Sketch.

The inequality is proved for several substantial families, including all constant-plus-quadratic symbols of rank at most three, but arbitrary unequal higher-rank spectra and general analytic tests remain untreated.

Part 4. The borderline: the multiplicative Hilbert matrix

17. Paley–Wiener spaces

17.1. **Weak factorization on arithmetic semigroups**

Why does weak factorization fail on $\mathbb{T}^\infty$ but hold in Paley–Wiener? Bampouras derives weak factorization from his inequality. Helson's inequality holds on $\mathbb{T}^\infty$, yet weak factorization fails there. Isolate the extra input (a doubling-type property of ω?), and look for sub-semigroups of $\mathbb{N}$ where the divisor function satisfies it, so that weak factorization, and hence Nehari, holds.

Question 16, Topic 3, in Øistein Søvik's fourth-edition book questions. It is a program to identify the extra condition behind weak factorization and classify arithmetic subsemigroups where that condition recovers weak factorization and the Nehari theorem. The premise needs care: the cited Paley–Wiener inequality yields S_2-Nehari and W_2 orthogonal factorization, not full W_1 weak factorization.

Answer: Partial. **Evidence:** Sketch.

Our written argument proves failure of full W_1 factorization for every multiplicative submonoid containing infinitely many ordinary primes. Finite-prime monoids satisfy it, while non-simplicial finite-rank cases and general infinite-rank submonoids remain open.

18. Nehari theorems and Hankel forms

18.1. **Strict weak-product inclusion after differentiation**

The inclusion between the standard weak product and its skew counterpart is strict, $H^2 \odot H^2 \subsetneq \partial^{-1}H^2 \odot \partial H^2$.

Brevig and Perfekt, *Weak product spaces of Dirichlet series*, Conjecture 2, p. 14. Here $\partial(n^{-s}) = -(\log n)n^{-s}$ and the full Hardy space includes constants.

Answer: Open. **Evidence:** Heuristic.

Brevig and Perfekt establish the zero-constant-term comparison in their treatment of differentiated weak products. It leaves open whether constants make the full-space inclusion strict or whether equality holds.

18.2. **Fractional zeta primitives on $\mathscr{H}^p$**

For $0 < p \le 2$, does $\varphi_{1/p}$ defined in equation (19) induce a bounded linear functional on $\mathscr{H}^p$ exactly when $0 < p \le 1$?

Bondarenko, Brevig, Saksman, and Seip, *Linear space properties of H^p spaces of Dirichlet series*, §3, unnumbered conjecture.

Answer: Answered: yes, exactly for $0 < p \le 1$. **Evidence:** Sketch.

The completed proof establishes boundedness through $p = 1$ and failure above it. The full argument is in the public book draft. The proof

18.3. **An explicit bounded symbol for the multiplicative Hilbert matrix**

Can the Bessel operator V be dualized into an explicit bounded completion, i.e. an explicit symbol?

Søvik, the explicit-symbol question in the discussion of the multiplicative Hilbert matrix. The construction answers the symbol question by a different method.

Answer: Answered: yes. **Evidence:** Sketch.

A concrete localized sum of Poisson kernels gives a bounded symbol with norm at most $10^5 + 7$ and the prescribed coefficients. The construction is self-contained in the public book draft.

Answers, "An explicit bounded symbol for the multiplicative Hilbert matrix", p. 27.

The construction

18.4. **The sharp multiplicative Hilbert functional constant**

Is $C = \pi$, the norm of the multiplicative Hilbert matrix on $\mathscr{H}^2$?

Søvik, *Where Nehari Survives*, Chapter 9, the sharp-constant problem, p. 34.

Answer: Partial. **Evidence:** Sketch.

Søvik proves $\pi \le C \le 5.515$ in the cited chapter. Exact evaluation and a proof that $C > \pi$ remain open; numerical optimization alone does not establish a strict gap.

18.5. **The absolute real-curve Fejér–Riesz constant**

Is the sharp absolute real-prime-curve embedding constant equal to π?

Søvik, the question following the real-curve Fejér–Riesz estimates.

Answer: Partial. **Evidence:** Sketch.

Homogeneous and finite-degree families have constants strictly below π, but the global supremum is undetermined. The existing estimates do not give either a universal upper bound π or a strict lower bound above it.

18.6. **The growing prime-window limit**

Formulate the scaling limit of ℓ on functions of the primes in $(P, P^{1+\eta}]$ as $P \to \infty$, as a functional on a log-correlated Gaussian field, and compute its Nehari ratio. Is it 1, giving $C = \pi$? Or above 1 uniformly in the window, giving $C > \pi$ once windows are combined?

Søvik, *Where Nehari Survives*, Chapter 9, the prime-window question following the discussion of the real-curve constant.

Answer: Partial. **Evidence:** Sketch.

For each fixed window exponent the limit and its ratio are computed, and the ratio tends to $2/\sqrt{\pi}$ as the window narrows. The growing-window regime that could yield a log-correlated limit, and the resulting global constant, remain open.

19. Hilbert-type forms

19.1. **Large-parameter asymptotics for Hilbert-type forms**

What is the asymptotic decay of $\|B_\alpha\| - 1$ as $\alpha \to \infty$?

Brevig, *Sharp norm estimates for composition operators and Hilbert-type inequalities*, §5.2(b), p. 14. The source notes bounds of orders $4^{-\alpha}$ and $2^{-\alpha}$ and suggests the former as the correct one; the form has kernel $(mn)^{\alpha-1/2}/\max(m,n)^{2\alpha}$.

Answer: Answered: order $4^{-\alpha}$. **Evidence:** Sketch.

The Schur complement at the first coordinate reduces the excess norm to the squared norm of its off-diagonal row. The remaining block has norm bounded strictly below one, giving $\|B_\alpha\| - 1 \sim 4^{-\alpha}$.

20. Coefficient inequalities

20.1. **Divisor-power weighted small-p pseudomoments**

Determine the precise asymptotic growth of $\left\|\sum_{n=1}^{N}[d(n)]^{\gamma}n^{-1/2-s}\right\|_{H^p}$ when $N \to \infty$ for $p \le 1$.

Source: Saksman and Seip, "Some open questions in analysis for Dirichlet series," Problem 4.4, p. 6. The printed problem does not specify a range for γ; here $[d(n)]^{\gamma}$ means ordinary divisor powers, distinguished from generalized divisor coefficients.

Answer: Partial. **Evidence:** Sketch.

Gerspach–Lamzouri, *Quarterly Journal of Mathematics* 73 (2022), Theorem 1.2, gives the norm order for $\gamma < 0$; Gerspach, *International Mathematics Research Notices* (2022), Corollary 1, gives it for $\gamma = 0$. Our proof gives the matching order for $0 < \gamma < 1$ and $p_0 < p \le 1$, with upper bounds below p_0. The precise $\log\log N$ factors and the unrestricted positive-γ range remain open.

20.2. **Optimal power-weight coefficient contraction**

"The problem raised in the preceding paragraph is to find the largest exponent $\gamma = \gamma(p)$ for which the contractive inequality

$$\left(\sum_{n=0}^{\infty}|c_n|^2(n+1)^{\gamma}\right)^{1/2} \le \|f\|_{H^q(\mathbb{T})}$$

holds for every $f \in H^q(\mathbb{T})$, $0 < q < 2$."

Source: Bondarenko, Heap, and Seip, "An Inequality of Hardy–Littlewood Type for Dirichlet Polynomials," arXiv:1405.6516v3, concluding remarks, p. 14. We use the variable q throughout.

Answer: Answered: $\gamma(q) = \min\{\log_2(q/2), 1 - 2/q\}$. **Evidence:** Published.

Kulikov, Geometric and Functional Analysis 32 (2022), Section 5, equation (5.1), proves the binomial-weight coefficient inequality for every $0 < q < 2$. Comparing the binomial weights gives $\gamma(q) = \min\{\log_2(q/2), 1 - 2/q\}$. Linear perturbations and singular functions show sharpness. The value is $1 - 2/q$ for $0 < q \le 1$ and $\log_2(q/2)$ for $1 \le q < 2$.

20.3. **H¹ partial-sum norm growth**

Determine the precise asymptotic growth of the norm of the partial sum operator $S_N : \sum_{n=1}^{\infty} a_n n^{-s} \mapsto \sum_{n=1}^{N} a_n n^{-s}$ when $N \to \infty$ for $p = 1$ (or more generally, for $p \le 1$ or $p = \infty$).

Source: Saksman and Seip, "Some open questions in analysis for Dirichlet series," Problem 4.5, p. 6; the present answer addresses the $p = 1$ case.

Answer: Answered: $\asymp \log N/\log\log N$. **Evidence:** Sketch.

The published upper bound is $O(\log N/\log\log N)$; our entropy argument supplies a matching lower bound, so the H^1 order is $\log N/\log\log N$. The leading constant remains open.

Answers, "H¹ partial-sum norm growth", p. 28.

Public proof.

20.4. **Higher moments of homogeneous Dirichlet sums**

> Assume k is an integer larger than 1. For which m (depending on N) will the L^{2k} norms of m-homogeneous Dirichlet polynomials of length N be comparable to their L^2 norms?

Source: Saksman and Seip, "Some open questions in analysis for Dirichlet series," Problem 4.2, p. 5. The coefficient-one interpretation is a restricted interpretation of the printed question; the source itself says "m-homogeneous Dirichlet polynomials."
Answer: Partial. **Evidence:** Sketch.

Bondarenko and Seip, arXiv:1411.6388v2, Lemma 2 and equations (12) and (16), determine the sharp fourth-moment threshold for the canonical sums. Mahatab, arXiv:1710.08201v2, Theorem 1 (published in *International Journal of Number Theory* 16 (2020), 1295–1306), proves sixth-moment comparability when $m \ll (\log\log N)^{1/3}$. Our written proof characterizes comparability for every fixed $2k$ only in the upper-degree cone $m \geq \lfloor \log(N/2^m)/\log(3/2) \rfloor$; the sparse and intermediate thresholds remain open.

20.5. **Small-p critical-line polynomial norm**

> Determine the asymptotic behavior of $\left\| \sum_{n=1}^{N} n^{-1/2-s} \right\|_{H^p}$ when $N \to \infty$ for $0 < p \leq 1$.

Source: Saksman and Seip, "Some open questions in analysis for Dirichlet series," Problem 4.3, p. 5.
Answer: Answered: order determined. **Evidence:** Published.

Gerspach, International Mathematics Research Notices (2022), Corollary 1, determines the matching asymptotic order for every fixed positive p, hence throughout the source range $0 < p \leq 1$.

20.6. **Endpoint partial-sum operator norms**

> Determine the precise asymptotic growth of the norm of the partial sum operator $S_N : \sum_{n=1}^{\infty} a_n n^{-s} \mapsto \sum_{n=1}^{N} a_n n^{-s}$ when $N \to \infty$ for $p = 1$ (or more generally, for $p \leq 1$ or $p = \infty$).

This entry concerns $0 < p < 1$ and $p = \infty$; the $p = 1$ case is answered separately above. Source: Saksman and Seip, "Some open questions in analysis for Dirichlet series," Problem 4.5, p. 6.
Answer: Partial. **Evidence:** Sketch.

Our proof establishes the $p = 1$ order $\log N/\log\log N$; see the public proof. Bondarenko, Brevig, Saksman, and Seip, "Linear Space Properties of H^p Spaces of Dirichlet Series," *Transactions of the American Mathematical Society* 372 (2019), Theorems 5.2–5.3, give the published H^1 upper bound and the logarithmic scale for each fixed $0 < p < 1$, with unmatched leading constants; Chen and Fu, arXiv:2608.17989v1, Theorem 1.1, prove the $\mathcal{H}^\infty$ order $\log N$. The sharp fixed-p rate for $0 < p < 1$ remains open.

20.7. **Absolute-coefficient Hardy inequality**

> "The validity of the Hardy inequality, featuring absolute values on the coefficients, $|a_1| + \sum_{n\geq 2} \frac{|a_n|}{\sqrt{n}\log n} \lesssim \|f\|_{\mathcal{H}^1}$, remains an open problem."

Source: Perfekt, "A Fejér–Riesz inequality for Dirichlet series," arXiv:2609.03855v2, equation (7), p. 4.
Answer: Partial. **Evidence:** Sketch.

Perfekt proves the signed H^1 functional inequality, while the absolute-value inequality remains explicitly open. Our checked work gives restricted-support cases but no unrestricted conclusion.

20.8. **Bohr coefficient inequality at the critical endpoint**

> For every $a_n n^{-s} \in \mathcal{H}^\infty$ and every $0 \leq c < 1/\sqrt{2}$, $\sum_{n=1}^{\infty} |a_n| e^{c\sqrt{\log n \log\log n}}/\sqrt{n} < \infty$. Moreover, the constant $1/\sqrt{2}$ cannot be improved.

Source: Defant, García, Maestre, and Sevilla-Peris, "Dirichlet Series and Holomorphic Functions in High Dimensions," Corollary 9.2, p. 206. The endpoint question is inferred from the displayed strict range and the sharpness statement; the cited corollary does not decide $c = 1/\sqrt{2}$.

Answer: Partial. **Evidence:** Sketch.

The subcritical inequality holds for $c < 1/\sqrt{2}$ and fails for $c > 1/\sqrt{2}$. The endpoint remains open; the recorded second-order criteria leave an interval of possible weights.

20.9. **Sidon constant of Dirichlet polynomials in $\mathcal{H}^1$**

"For every $f \in H^1(\mathbb{T}^\infty)$, we have

$$\left(\sum_{\alpha \in \mathbb{N}_0^{(\mathbb{N})}} \left(\frac{|\widehat{f}(\alpha)|}{\sqrt{\alpha+1}} \right)^2 \right)^{1/2} \leq \int_{\mathbb{T}^\infty} |f(w)| \, \mathrm{d}w.$$

"

Here $\alpha + 1 = \prod_j (\alpha_j + 1)$ under the Bohr lift. The sharp order of $S_1(x) = \sup\{\sum_{n \leq x} |a_n| : \|\sum_{n \leq x} a_n n^{-s}\|_{\mathcal{H}^1} \leq 1\}$ is inferred from this Sidon-type inequality and its surrounding discussion; the cited theorem does not explicitly pose that growth question. Source: Defant, García, Maestre, and Sevilla-Peris, "Dirichlet Series and Holomorphic Functions in High Dimensions," Theorem 12.14, p. 299.

Answer: Partial. **Evidence:** Sketch.

The checked bounds are $\sqrt{x}\exp(c(\log\log x)^{1/3})$ below and $(1+o(1))\sqrt{x\log x}$ above. The exact order is unknown.

20.10. **Rectangular factorization in the disk algebra (Queffélec)**

Has any function $f \in A(\mathbb{D})$ a rectangular factorization $f = g * h$ with both g and h in $A(\mathbb{D})$?

Source: Hervé Queffélec, "Dirichlet Series and Function Theory in Polydiscs," Oberwolfach Report 06/2014, Problem Session, §3.2, Question 2, p. 386 (reported by O. F. Brevig).

Answer: Answered: no. **Evidence:** Sketch.

If $g, h \in A(\mathbb{D})$, then their Taylor coefficients lie in ℓ^2, so the coefficients of $g * h$ lie in ℓ^1 by Cauchy–Schwarz. Since $A(\mathbb{D})$ strictly contains the Wiener algebra, the proposed factorization fails for some $f \in A(\mathbb{D})$. The source's separate factorization question for $f \in W$ remains open.

21. Dirichlet series and frequencies

21.1. **The logarithm in partial-sum bounds**

Question. Defant and Schoolmann ask: "Determine optimal bounds for $\|S_N : \mathscr{D}_\infty((\log n)) \to \mathscr{D}_\infty((\log n))\|$." Problem 2.5, p. 8 of the arXiv and published surveys.

Answer: Partial. **Evidence:** arXiv.

Chen and Fu claim the sharp order $\log N$ in an arXiv preprint. The source asks for optimal bounds, and the exact leading constant remains open.

Part 5. Related function-theoretic questions

22. Extremal coefficient problems

22.1. **Suffridge singular-inner-factor conjecture**

"It is conjectured on [13, p. 187] that this singular inner function is trivial when $0 < p < 1$."

Source: Brevig and Saksman, "The Function and Operator Theory of the Bohnenblust–Hille Inequality," arXiv:1905.09547, §5.2, p. 12, reporting Suffridge's conjecture. In context, the function is the singular inner factor of an extremal for the nonvanishing-coefficient problem.

Answer: Answered: yes. **Evidence:** Sketch.

The public outer-function theorem rules out every nontrivial inner factor for extremals of nonzero bounded functionals, which implies Suffridge's singular-inner-factor assertion.

Public proof.

22.2. **Brevig–Saksman nonvanishing conjecture**

> "For $0 < p < 1$ any extremal f for $C(k, p)$ does not vanish in $\mathbb{D}$."

Source: Brevig and Saksman, "The Function and Operator Theory of the Bohnenblust–Hille Inequality," arXiv:1905.09547, §5.1, Conjecture 1, p. 12.

Answer: Answered: yes. **Evidence:** Sketch.

Our public theorem proves the stronger result that every extremal for a nonzero bounded functional on H^p, $0 < p < 1$, is outer, hence zero-free in the disc.

Public proof.

22.3. **Brevig–Saksman strict coefficient monotonicity**

> "For $0 < p < 1$ the sequence $C(k, p)$ is strictly increasing."

Source: Brevig and Saksman, "The Function and Operator Theory of the Bohnenblust–Hille Inequality," arXiv:1905.09547, §5.1, Conjecture 2, p. 12.

Answer: Answered: yes. **Evidence:** Sketch.

The same outer-function theorem implies strict increase of $C(k, p)$ in k for every $0 < p < 1$.

Public proof.

22.4. **BGI Question 1: at most one zero**

> Suppose that $0 < p < 1$. Is it true that the extremal for $\Phi_k(p, t)$ has at most one zero in $\mathbb{D}$?

Source: Ole Fredrik Brevig, Sigrid Grepstad, and Sarah May Instanes, "F. Wiener's trick and an extremal problem for H^p," §6, Question 1, p. 16. The paper defines $\Phi_k(p, t)$ in (1), with $k \geq 1$ and $0 \leq t \leq 1$.

Answer: Answered: yes. **Evidence:** Sketch.

Our proof establishes that every extremal has at most one zero in the disc, counted with multiplicity, for all $k \geq 1$, $0 < p < 1$, and $0 \leq t \leq 1$.

Public proof.

22.5. **BGI Question 2: terminal equality interval**

> Fix $k \geq 2$ and $0 < p < 1$. Is there some t_0 such that $\Phi_k(p, t) = \Phi_1(p, t)$ holds for every $t_0 \leq t \leq 1$?

Source: Ole Fredrik Brevig, Sigrid Grepstad, and Sarah May Instanes, "F. Wiener's trick and an extremal problem for H^p," §6, Question 2, p. 17.

Answer: Partial. **Evidence:** Sketch.

The authors give numerical and analytical evidence for $k = 2$, with one extremal case excluded only numerically. Our conditional proof establishes the exact terminal equality interval for $k \geq 3$; the $k = 2$ case remains open.

Conditional on: the canonical-factorization and positive-parameter extremal results stated in the answer entry.

Answers, "No disconnected equality points below terminal threshold", p. 30.

23. Hardy and Bergman extremals

23.1. **Characterizing zero sets of Bergman spaces**

> Characterise the zero sets for A^p, or at least find some non-trivial converse to (8.1).

Problem 8.6 and Update 8.6, printed p. 190, of Hayman and Lingham's *Research Problems in Function Theory, Fiftieth Anniversary Edition*. The question asks for a characterization,

or a nontrivial converse to Horowitz's growth bound, for zero sequences of the unweighted Bergman space A^p, $p > 0$.
Answer: Answered: yes. **Evidence:** Published.

Korhonen and Rättyä's weighted-Bergman zero-set criterion specializes to an exact characterization for A^p (Theorem 1 and Corollary 2, 2019).

23.2. **Separation of simplest-fraction sets in A^2_1**

> In particular, are pairwise distances between these sets bounded away from 0 in the space A^2_1?

Question 2, p. 9, of Evgeny Abakumov, Alexander Borichev and Konstantin Fedorovskiy, *Chui's conjecture in Bergman spaces*, Math. Ann. 379 (2021), 1507–1532. Here SF_n denotes the set of sums of n Cauchy kernels with poles on the unit circle, counted with multiplicity, and A^2_1 has its standard normalized weighted-area norm.
Answer: Answered: yes. **Evidence:** Sketch.

A Fourier-coefficient argument proves the uniform lower bound $\operatorname{dist}_{A^2_1}(SF_n, SF_{n+k}) \geq \pi/\sqrt{6}$ for all integers $n, k \geq 1$.
Answers, "Separation of simplest-fraction sets in A^2_1", p. 35.

23.3. **Power weights in the simplest-fraction problem**

> Does Theorem 1 hold for larger classes of g? For instance, for $g(t) = t^\alpha$, $\alpha > 1$?

Question 1, p. 9, of Evgeny Abakumov, Alexander Borichev and Konstantin Fedorovskiy, *Chui's conjecture in Bergman spaces*, Math. Ann. 379 (2021), 1507–1532. Theorem 1 concerns minimizers for the radial weight g in the simplest-fraction problem.
Answer: Partial. **Evidence:** arXiv.

Corbett's Theorem 8 proves equispacing for every power weight $g(t) = t^\alpha$, $\alpha > 0$. The question for general classes of g remains open.
Conditional on: Corbett's unrefereed arXiv preprint, Theorem 8.

23.4. **The conjectured distance formula for simplest-fraction sets**

> Our conjecture is that the answer is positive, and, moreover, if $\alpha > 0$, $n, k \geq 1$, then
>
> $$\operatorname{dist}_{A^2_\alpha}(SF_n, SF_{n+k}) = \|\Psi_k\|_\alpha.$$

The conjectured formula in Question 2, p. 9, of Evgeny Abakumov, Alexander Borichev and Konstantin Fedorovskiy, *Chui's conjecture in Bergman spaces*, Math. Ann. 379 (2021), 1507–1532. The separate question there about positive separation in A^2_1 is listed in the preceding entry.
Answer: Answered: no. **Evidence:** Sketch.

For every $\alpha > 2$, a second-variation calculation shows that one pole is not a nearest point to the two-pole set, contradicting the proposed formula. The exact distances for $0 < \alpha \leq 2$ remain undetermined.

23.5. **Two-factor division in weighted Bergman spaces**

> The norm computation "constitutes an interesting open problem in our opinion."

The final paragraph of Section 3.2, p. 6, of Adrián Llinares and Dragan Vukotić, *Extremal problems for vanishing functions in Bergman spaces* (arXiv:2012.05091v2). The inferred target is the sharp norm of $f \mapsto f/(\varphi_a\varphi_b)$ on the subspace of A^p_α functions vanishing at $a, b \in \mathbb{D}$, for $\alpha > -1$, $p > 0$, including repeated zeros and the dependence on the separation of the points.
Answer: Partial. **Evidence:** Sketch.

The authors solve the one-factor problem sharply and note that iterating it does not give a sharp two-factor bound. Our Hilbert-space calculation gives a local expansion as the two zeros coalesce, but the general norm remains open.

23.6. **Norms and extremals of the backward shift on H^p**

> "We hope this note may inspire further work for $p \neq 1, 2, \infty$, including the case $0 < p < 1$."

The last paragraph of the introduction, p. 2, of Ole Fredrik Brevig and Kristian Seip, *The norm of the backward shift on H^1 is $2/\sqrt{3}$*, Pure Appl. Funct. Anal. 9 (2024), no. 4, 991–994. The reconstructed question is to determine $\|B\|_{H^p \to H^p}$ and its extremals for the exponents not covered by the exact cases $p = 1, 2, 4, \infty$. Bampouras and Llinares give the $p = 4$ value and extremal family in *The norm of the backward shift on H^4 is $\sqrt[4]{\varphi}$*, Theorem 1.1.
Answer: Partial. **Evidence:** Sketch.

We prove quadratic-order growth of the norm near $p = 2$ and refute the inner-translate candidate at $p = 6$; the exact norm outside the known cases remains open.

23.7. **Best constant in a radial H^1 difference inequality**

> "It would be interesting to know what the best constant C in the estimate appearing Lemma 2 is."

The paragraph after Lemma 2, p. 2, of Ole Fredrik Brevig, *Two footnotes to the F. and M. Riesz theorem* (arXiv:2407.06843v2). The estimate in Lemma 2 is $\|f_r - f_\varrho\|_1 \leq C(\|f_\varrho\|_1^2 - \|f_r\|_1^2)^{1/2}$ for analytic f on the unit disc and $0 \leq r \leq \varrho < 1$.
Answer: Answered: $\sqrt{2}$. **Evidence:** Sketch.

The last Cauchy–Schwarz step in the published proof can be sharpened to give the sharp constant $\sqrt{2}$ for every fixed pair of radii; the lower bound is attained asymptotically by a first-order perturbation of a constant function.

23.8. **The inner-translate candidate for the backward-shift norm**

> the extremal functions are determined by inner functions with a prescribed value at the origin

This is an implicit question raised by the passage in Section 1, p. 2, of Konstantinos Bampouras and Adrián Llinares, *The norm of the backward shift on H^4 is $\sqrt[4]{\varphi}$* (arXiv:2605.10469v1). The inferred candidate asks whether, for $2 < p < \infty$, $\|B\|_{H^p \to H^p}$ equals the supremum over inner translates $I - c$ with $I(0) = a$, $0 < a < 1$, and $c \in \mathbb{C}$.
Answer: Answered: no. **Evidence:** Sketch.

At $p = 6$ the candidate supremum is attained inside its parameter range, but the full variational equation for an operator extremizer forces an impossible third-order pole cancellation. *Answers*, "The inner-translate candidate for the backward-shift norm", p. 36.

23.9. **Largest majorant-comparison radius at fixed constant term**

> It would be interesting to determine the largest r

The last paragraph, p. 3, of Ole Fredrik Brevig, *A bootstrap proof of Bohr's theorem*, arXiv: 2609.26850v1. Here S is the Schur class and $\varphi(z) = (w - z)/(1 - \overline{w}z)$ with $w = f(0)$; the question is to find the largest radius as a function of $0 < \varrho < 1$.
Answer: Partial. **Evidence:** Sketch.

The paper proves the lower bound $r(\varrho) \geq \max(\varrho, 1/3)$; our reviewed example gives $r(1/2) \leq 11/16$, but the exact threshold is open.

23.10. **Bombieri's Bohr radius for a prescribed constant term**

> "Find the best constant $B(\lambda)$"

Problem 5.18 and Update 5.18, printed p. 91, of W. K. Hayman and E. F. Lingham, *Research Problems in Function Theory, Fiftieth Anniversary Edition* (Springer, 2019). The source asks for the sharp radius $B(\lambda)$ such that the majorant $F(r) = \lambda + \sum_{n \geq 1} |a_n| r^n$ is at most one for

every analytic $f(z) = \lambda + \sum_{n\geq 1} a_n z^n$ on the disc. For the present range $0 < \lambda < 1/2$, the exact value remains unknown; the source's update reports no progress.
Answer: Partial. **Evidence:** Sketch.

Our proof sheet gives a sharp expansion at $\lambda = 0$ and improves the lower bound for small positive λ, but it does not determine the interior formula.

24. Contractivity and hypercontractivity

24.1. Defant–Pérez conjecture on hypercontractive constants for m-homogeneous polynomials

"Conjecture 5.1. For every $1 \leq p < q < \infty$ and $m \in \mathbb{N}$ we have that

$$H_m^{q,p} \leq m^{1/(2q)-1/(2p)} \left(\frac{q}{p}\right)^{m/2}.$$

"

Source: Andreas Defant and Alejandro Pérez, "Optimal Comparison of p-Norms of Dirichlet Polynomials," arXiv:1603.02128v1, §5, Conjecture 5.1, p. 10.
Answer: Partial. **Evidence:** Sketch.

Defant and Pérez, *Israel Journal of Mathematics* 221 (2017), Conjecture 5.1 and the paragraph following it, prove the bound when $p < q$ are powers of two. Our proof gives the exact constant for $p = 2\ell$ and $q = 2\ell s$ with integers $\ell \geq 1$, $s \geq 2$. The proposed bound for general real exponents remains open.

24.2. All-weight Bergman dilation hypercontractivity

"As in the previous subsection, we conjecture that Theorem 6 is true for all values of $\alpha \geq 1$."

In Theorem 6 the dilation $f(z) \mapsto f(rz)$ is conjectured to contract A^p_α into A^q_α precisely for $r \leq \sqrt{p/q}$, when $0 < p \leq q < \infty$. The source proves the criterion for $\alpha = (n+1)/2$, $n \in \mathbb{N}$.
Source: Bayart, Brevig, Haimi, Ortega-Cerdà, and Perfekt, "Contractive Inequalities for Bergman Spaces," arXiv:1701.06897v2, §2.3, p. 11.
Answer: Partial. **Evidence:** Sketch.

The sharp dilation criterion is proved for the stated discrete range of weights. Whether it holds for every $\alpha \geq 1$ remains open.

24.3. Hyperbolic superlevel estimate

"Suppose that $f \in H^2$. Then

$$\mu\big(\{z \in \mathbb{D} : |f(z)|^2(1-|z|^2) > \lambda \|f\|_{H^2}^2\}\big) \leq 1/\lambda - 1.$$

"

Here $0 < \lambda < 1$ and $\mathrm{d}\mu = \pi^{-1}(1-|z|^2)^{-2}\,\mathrm{d}x\,\mathrm{d}y$ is hyperbolic measure. Source: Brevig, Ortega-Cerdà, Seip, and Zhao, "Contractive Inequalities for Bergman Spaces," arXiv:1706.00738v2, Conjecture 2, p. 7.
Answer: Answered: yes. **Evidence:** Published.

Kulikov, *Geometric and Functional Analysis* 32 (2022), Theorem 3.1, proves the estimate for every norm-one $f \in H^2$, which gives the stated scale-invariant form.

24.4. Reverse coefficient contractivity

"The inequality

$$\|f\|_{H^q} \leq \left(\sum_{n=0}^{\infty} |a_n|^2 c_{q/2}(n)\right)^{1/2}$$

holds for every $2 \leq q < \infty$."

Source: Brevig, Ortega-Cerdà, Seip, and Zhao, "Contractive Inequalities for Bergman Spaces," arXiv:1706.00738v2, Conjecture 3, p. 9.
Answer: Answered: yes. **Evidence:** Published.

Llinares, *Revista Matemática Iberoamericana* 40 (2024), Theorem 2.1, proves $\|f\|_{H^q} \leq \|f\|_{D_{q/2}}$ for $q > 2$; at $q = 2$ the two norms agree. This is the stated coefficient inequality.

24.5. **One-dimensional critical Riesz exponent**

> "The Riesz projection P is a contraction from $L^r(\mathbb{T})$ to H^q with $q = 4(1 - 1/r)$ for $1 < r \leq \infty$. If f is in $L^1(\mathbb{T})$, then
>
> $$\exp\left(\int_0^{2\pi} \log|Pf(e^{i\theta})| \, \mathrm{d}\theta/(2\pi)\right) \leq \|f\|_{L^1}.$$
>
> "

Source: Brevig, Ortega-Cerdà, Seip, and Zhao, "Contractive Inequalities for Bergman Spaces," arXiv:1706.00738v2, Conjecture 4 and equation (19), p. 11.
Answer: Answered: yes. **Evidence:** arXiv.

Fu and Li, arXiv:2609.08603v1, Theorem 1.1, prove the exact critical exponent for every finite $1 < r < \infty$. The endpoint cases $r = 1$ and $r = \infty$ were settled in earlier published work; the current interior proof remains an arXiv result.

24.6. **Monotonicity of normalized Bergman power norms**

> "Suppose that f is an outer function with $\|f\|_{H^2} = 1$. Is it true that
>
> $$\frac{\partial}{\partial\alpha} U_f(\alpha) \leq 0$$
>
> for all $\alpha > 1$?"

Here $U_f(\alpha) = \int_{\mathbb{D}} |f|^{2\alpha}(\alpha - 1)(1 - |z|^2)^\alpha \, \mathrm{d}\mu$. Source: Brevig, Ortega-Cerdà, Seip, and Zhao, "Contractive Inequalities for Bergman Spaces," arXiv:1706.00738v2, Question 1, p. 6.
Answer: Answered: yes. **Evidence:** Published.

Kulikov, *Geometric and Functional Analysis* 32 (2022), Corollary 1.3, proves contractivity when $p/\alpha = q/\beta = 2$. This implies monotonicity of $U_f(\alpha)$ and answers the question.

24.7. **Radial level-set maximal integral**

> "Is it true that
>
> $$\int_0^{2\pi} \max\{|f(re^{i\theta})|^2 : |f(re^{i\theta})|^2(1 - r^2) = \lambda\} \, \frac{\mathrm{d}\theta}{2\pi} \leq \|f\|_{H^2}^2$$
>
> holds whenever $0 < \lambda < |f(0)|$?"

The source first normalizes $\|f\|_{H^2} = 1$ and, by Möbius invariance, assumes that $u(re^{i\theta}) = |f(re^{i\theta})|^2(1 - r^2)$ attains its maximum at the origin. Source: Brevig, Ortega-Cerdà, Seip, and Zhao, "Contractive Inequalities for Bergman Spaces," arXiv:1706.00738v2, Question 2, equation (16), p. 8.
Answer: Partial. **Evidence:** Sketch.

No complete resolution is recorded. Available weak-type and layer-cake estimates address related level-set quantities but leave the maximal radial integral open.

24.8. **Infinite-torus Riesz critical exponent**

> "Question. What is the largest $p = p_\infty$ such that the Riesz projection (1) is bounded from $L^\infty(\mathbb{T}^\infty)$ to $L^p(\mathbb{T}^\infty)$?"

Here (1) is the Riesz projection $Pf = \sum_{\alpha \in \mathbb{N}_0^{(\mathbb{N})}} c_\alpha z^\alpha$. Source: Ole Fredrik Brevig, "Linear Functions and Duality on the Infinite Polytorus," arXiv:1806.10849v2, Introduction, p. 1.
Answer: Answered: yes. **Evidence:** Published.

Konyagin, Queffélec, Saksman, and Seip, Studia Mathematica 262 (2022), Theorem 2.1 and Corollary 2.2, identify the largest exponent as $p_\infty = 2$ and prove unboundedness for every $p > 2$.

24.9. **Lieb–Solovej majorization conjecture for** $\mathrm{SU}(1,1)$

> "Conjecture 6.1 (Majorization of Φ_L). For all density matrices ρ on D_K the eigenvalues of $\Phi_L(\rho)$ will be majorized by the eigenvalues of $\Phi_L(|z\rangle_K\langle z|)$. "

Here D_K is the source's representation space and Φ_L is the covariant channel defined immediately before the conjecture. Source: Elliott H. Lieb and Jan Philip Solovej, "Wehrl-Type Coherent State Entropy Inequalities for $\mathrm{SU}(1,1)$ and its $ax+b$ Subgroup," arXiv: 1906.00223v2, §6, Conjecture 6.1, p. 12.

Answer: Partial. **Evidence:** Sketch.

Kulikov and Frank prove related coherent-state convex-functional and entropy inequalities, but not this channel eigenvalue majorization. Van Haastrecht, *Journal of Functional Analysis* 289 (2025), Proposition 4.4 and the discussion on p. 49, explicitly leaves lowest-component majorization open; the source conjecture remains unresolved.

24.10. **Outer-function Bergman norm monotonicity**

> Question 3. Suppose that f is an outer function. For $\alpha > 1$, denote
>
> $$N_f(\alpha) = \|f\|_{A^{2\alpha}_\alpha} = \left(\int_{\mathbb{D}} |f(z)|^{2\alpha} (\alpha-1)(1-|z|^2)^\alpha \, \mathrm{d}\mu(z) \right)^{1/(2\alpha)}.$$
>
> Is it true that
>
> $$\frac{\partial}{\partial \alpha} N_f(\alpha) \le 0$$
>
> for all $\alpha > 1$?

Here $\mathrm{d}\mu(z) = (1-|z|^2)^{-2} \, \mathrm{d}x \, \mathrm{d}y/\pi$. Source: Hui Dan, Kunyu Guo, and Yi Wang, "A Sharp Inequality of Hardy–Littlewood Type via Derivatives," arXiv:1908.01320v1, Question 3, p. 3.

Answer: Answered: yes. **Evidence:** Published.

The equal-ratio weighted Bergman inclusion of Kulikov, quoted as Theorem A by Llinares, arXiv:2209.14104v2, together with Llinares, Revista Matemática Iberoamericana 40 (2024), Corollary 1.1, gives $N_f(\beta) \le N_f(\alpha)$ for $\beta > \alpha > 1$. Differentiation under the integral then gives the source's derivative inequality.

24.11. **Kernel-matrix monotonicity conjecture and equivalent formulation**

> Conjecture 4. Suppose $\alpha > 0$ and $(\mathbf{c}, \mathbf{w}) \in \Lambda_\alpha$. Then $D_\alpha(\mathbf{c}, \mathbf{w})$ defined as in (3.2) and (3.4) is non-positive.

The source states that this is equivalent to the following reformulation:

> Conjecture 6. Suppose $\alpha > 0$, $\mathbf{w} \in (-1,1)^k$ and $\mathbf{f} \in \mathbb{H}^k$. Then $\widehat{D}_\alpha(\mathbf{f}, \mathbf{w}) \le 0$.

Here $\mathbb{H} = \{z \in \mathbb{C} : \operatorname{Re} z > 0\}$; Λ_α consists of real nodes $\mathbf{w} \in (-1,1)^k$ and coefficients $\mathbf{c}$ for which the samples of $\mathbf{f}_\alpha = \mathbf{c} W_\alpha$, with $W_\alpha = [(1-\overline{w_i} w_j)^{-\alpha}]$, lie in $\mathbb{H}$. Source: Hui Dan, Kunyu Guo, and Yi Wang, "A Sharp Inequality of Hardy–Littlewood Type via Derivatives," arXiv: 1908.01320v1, Conjecture 4, p. 19, and equivalent Conjecture 6, p. 20.

Answer: Partial. **Evidence:** Sketch.

Our construction gives a three-node rational example at $\alpha = 1$ with $D_1(\mathbf{c}, \mathbf{w}) > 268.1785$, contradicting both formulations. The counterexample has not received an independent review.

24.12. **Conditional norm monotonicity below the reproducing-kernel range**

> Question 5. Suppose Conjecture 4 holds. Does it imply that the norms $\|f^\alpha\|^{1/\alpha}_{A^{2\alpha}_\alpha}$ are non-increasing for $\alpha > 0$, for some set of functions f?

The question is conditional on the source's kernel-matrix Conjecture 4; the phrase "some set of functions" is left unspecified in the source. Source: Hui Dan, Kunyu Guo, and Yi Wang, "A Sharp Inequality of Hardy–Littlewood Type via Derivatives," arXiv:1908.01320v1, Question 5, p. 20.

Conditional on: Conjecture 4 of Dan, Guo, and Wang.

Answer: Open. **Evidence:** Heuristic.

The source's premise is contradicted by the three-node example above. No specific function class is given for the proposed below-one monotonicity consequence.

24.13. **Entropy inequality for arbitrary positive semidefinite matrices**

Question 7. Suppose $\mathbf{A} = [a_{ij}]$ is semi-positive definite and $\operatorname{Re} a_{ij} \geq 0$, for all $i, j = 1, \ldots, k$. Suppose $\mathbf{c} \in \mathbb{C}^k$ is such that $\mathbf{cA} = 0$. Do we have

$$\widetilde{D}(\mathbf{c}, \mathbf{A}) = -\sum_{i,j=1}^{k} c_i \overline{c_j} a_{ij} \operatorname{Log} a_{ij} \leq 0?$$

Source: Hui Dan, Kunyu Guo, and Yi Wang, "A Sharp Inequality of Hardy–Littlewood Type via Derivatives," arXiv:1908.01320v1, Question 7, p. 20. This is stated for arbitrary positive semidefinite matrices, not only kernel matrices.

Answer: Partial. **Evidence:** Sketch.

Our construction gives a four-node positive semidefinite counterexample. Its archived proof received an independent review, but the remaining matrix sizes are still open.

24.14. **Asymptotic norm of the Bloch-to-Bergman inclusion**

"Conjecture 3.9. For any $\alpha > -1$,

$$\lim_{p\to\infty} \frac{C_\alpha(p)}{p/(2e)} = \frac{1}{\sqrt{(\alpha+1)(\alpha+2)}}.$$

"

Here $C_\alpha(p)$ denotes the inclusion norm from the Bloch space with norm $|f(0)| + \sup_{z\in\mathbb{D}}(1 - |z|^2)|f'(z)|$ to the normalized weighted Bergman space A^p_α. Source: Adrián Llinares, "Norms of Inclusions between Some Spaces of Analytic Functions," arXiv:2107.11315v2, §3, Conjecture 3.9, p. 9.

Answer: Answered: no. **Evidence:** Sketch.

A uniform hyperbolic-average deficit gives constants $\eta < 1$ and $\varepsilon_0 > 0$ for which the limsup is below the conjectured constant whenever $-1 < \alpha < -1 + \varepsilon_0$. The actual limit and its existence remain undetermined.

Answers, "Asymptotic norm of the Bloch-to-Bergman inclusion", p. 37.

24.15. **Sharp weighted Bergman contraction**

"Conjecture 1.3 (Lieb, Solovej). Let $\alpha > -1$. If $p > 2$, then

$$\|f\|_{A^p_{(\alpha+2)p/2-2}} \leq \|f\|_{A^2_\alpha}, \qquad \forall f \in A^2_\alpha.$$

In addition, equality is possible if and only if f is a constant multiple of the Bergman kernel."

Source: Adrián Llinares, "On a Conjecture about Contractive Inequalities for Weighted Bergman Spaces," arXiv:2112.09962v1, Conjecture 1.3, p. 3.

Answer: Answered: yes. **Evidence:** Published.

Kulikov, Geometric and Functional Analysis 32 (2022), Theorem 1.1, proves the sharp weighted Bergman contraction for all weights above the Hardy endpoint and classifies equality by reproducing kernels, answering Conjecture 1.3.

24.16. **Higher-dimensional critical Riesz exponents**

"Critical exponent conjecture ($d > 1$).

$$p_d(q) = \begin{cases} a_d(q), & \dfrac{2d}{d+1} \leq q \leq \infty, \\ -1, & 1 \leq q < \dfrac{2d}{d+1}. \end{cases}$$

"

Here $a_d(q) = 2 + 2/(d + 2/(q-2))$, with endpoint values interpreted as limits. Source: Ole Fredrik Brevig, Adrián Llinares, and Kristian Seip, "Critical Exponents of the Riesz Projection," arXiv:2402.09787v1, pp. 4–5.

Answer: Partial. **Evidence:** Sketch.

The published paper establishes sharp bounds in selected dimensions and endpoint ranges. The full d-dimensional critical-exponent formula remains open.

24.17. **Equality cases at the critical Riesz exponent**

> "Conjecture. Suppose that $1 < q < 2$ or $2 < q \le \infty$. If there is a nontrivial function ψ in L^q such that
>
> $$\|P_+\psi\|_{p(q)} = \|\psi\|_q,$$
>
> then there is a constant $C \neq 0$ and an inner function I such that $\psi = CI$."

Here $p(q)$ is the critical exponent for the Riesz projection. Source: Ole Fredrik Brevig, Adrián Llinares, and Kristian Seip, "Critical Exponents of the Riesz Projection," arXiv:2402.09787v1, §3, p. 10.

Answer: Answered: yes. **Evidence:** Sketch.

Our proof establishes the equality cases for all finite $q \neq 2$, conditional on Fu–Li's strict subharmonic minorant; the $q = \infty$ case is Brevig–Llinares–Seip, Theorem 6.

Conditional on: Fu and Li, arXiv:2609.08603v1, Lemma 2.1.

Public proof.

24.18. **Sharp boundary growth of point evaluation**

> "This result does not only extend the already known lower estimates given in [2, p. 26] from the range $\alpha \in (-1, 0)$ to any value of $\alpha > -1$, but it also provides an affirmative answer to Question 2 from [8]."

The cited Question 2 asks for the order of the evaluation norm near the boundary. The inferred target is whether, for $1 \le p < \infty$ and $\alpha > -1$, $\|\delta_s\|_{(A^p_\alpha)^*} \asymp (\operatorname{Re} s/(\operatorname{Re} s - 1/2))^{(\alpha+2)/p}$ for $\operatorname{Re} s > 1/2$. Source: Carlos Gómez-Cabello, Pascal Lefèvre, and Hervé Queffélec, "Integration Type Operators and Point Evaluation on Weighted Bergman Spaces of Dirichlet Series," arXiv:2402.12523v2, p. 3, referring to Fu, Guo, and Yan, Question 2. The original question text was not locally available.

Answer: Answered: yes. **Evidence:** Published.

Gómez-Cabello, Lefèvre, and Queffélec, Proceedings of the American Mathematical Society 153 (2025), Theorem 5.2, proves the global point-evaluation estimate in equation (1.1), including the stated boundary growth up to constants.

24.19. **Tight Bergman inclusions on the unit ball**

> Conjecture 22(a). Suppose $0 < p, q < \infty$ and $\alpha, \beta > -1$. Then $A^p_\alpha \subset A^q_\beta$ is a tight fitting if and only if
>
> $$p < q \quad \text{and} \quad (n + 1 + \alpha)q = (n + 1 + \beta)p.$$
>
> Furthermore, $\|f\|_{A^p_\alpha} = \|f\|_{A^q_\beta}$ if and only if $f(z) = c/(1 - \langle z, a\rangle)^{2(n+1+\alpha)/p}$, where $c \in \mathbb{C}$ and $a \in \mathbb{B}^n$.

Source: Guanlong Bao, Pan Ma, Fugang Yan, and Kehe Zhu, "Embedding and Compact Embedding between Bergman and Hardy Spaces," arXiv:2502.08406v1, Conjecture 22(a), p. 19.

Answer: Answered: yes. **Evidence:** arXiv.

Kalaj, arXiv:2607.20055v3, Theorem 7.2 and Corollary 7.4, proves the norm-one contraction under $0 < p < q$, $n < \gamma < \delta$, and $p/\gamma = q/\delta$, with equality only for the stated kernel functions. The result is an arXiv claim.

24.20. **Tight Hardy-to-Bergman inclusions on the unit ball**

> Conjecture 22(b). Then $H^p \subset A^q_\alpha$ is a tight fitting if and only if
>
> $$p < q \quad \text{and} \quad nq = (n + 1 + \alpha)p.$$
>
> Furthermore, $\|f\|_{A^q_\alpha} = \|f\|_{H^p}$ if and only if $f(z) = c/(1 - \langle z, a\rangle)^{2n/p}$, where $c \in \mathbb{C}$ and $a \in \mathbb{B}^n$.

The hypotheses are $0 < p, q < \infty$, $\alpha > -1$, and $n \geq 1$, as in the preceding sentence of Conjecture 22. Source: Guanlong Bao, Pan Ma, Fugang Yan, and Kehe Zhu, "Embedding and Compact Embedding between Bergman and Hardy Spaces," arXiv:2502.08406v1, Conjecture 22(b), p. 19.
Answer: Answered: yes. **Evidence:** Sketch.
Conditional on: the complex-hyperbolic isoperimetric hypothesis stated in our proof.

Our proof establishes the full Hardy-to-Bergman statement conditional on the complex-hyperbolic isoperimetric hypothesis, with exact equality kernels. Kalaj's latest preprint claims the geometric input; the equality bridge remains conditional.
Answers, "Tight Hardy-to-Bergman inclusions on the unit ball", p. 39.

24.21. **Contractive projections on H^p for $0 < p < 1$**

> "It seems that the assumption $Pz^n \in H^\infty(\mathbb{T})$ in Theorem 3.8 arises from technical reasons. We conjecture that this assumption can be dropped. Specifically, we conjecture that if P is a contractive projection on $H^p(\mathbb{T})$ with $0 < p < 1$ satisfying $P1 = 1$, then it automatically maps bounded functions to bounded functions, and therefore $Pf = f(0)$ or P is the identity."

Source: Xiang Fu, Kunyu Guo, and Dilong Li, "Contractive Projections, Conditional Expectations, and Idempotent Coefficient Multipliers on H^p Spaces $(0 < p < 1)$," arXiv:2503.06615v2, paragraph following Theorem 3.8, pp. 15–16.
Answer: Answered: yes. **Evidence:** Sketch.
Conditional on: Fu, Guo, and Li, arXiv:2503.06615v2, Theorem 3.8.

Our argument proves the conjectured dichotomy conditional on Fu, Guo, and Li's Theorem 3.8. The reduction to the source theorem is written and independently checked.
Answers, "Contractive projections on H^p for $0 < p < 1$", p. 41.

24.22. **Linearity in the Dirichlet range**

> "Fix $0 < \alpha < 1$ and $p \neq 2$. Is A^p_α a linear space?"

Source: Brevig, Kulikov, Seip, and Zlotnikov, "Contractive Inequalities in the Dirichlet Range," arXiv:2510.14333v1, Problem 1, p. 4.
Answer: Partial. **Evidence:** arXiv.

For $0 < \alpha < 1$, $p > 0$, $p \neq 2$, and $\alpha + p > 1/2$, Dayan, Llinares, and Monsalve-López, arXiv:2604.21347, Theorem 1.3, give monomial counterexamples to linearity. The triangle $0 < p < 1/2$, $\alpha \leq 1/2 - p$ is not covered by that cited result.

24.23. **Monotonicity in p below the Hardy endpoint**

> "Fix $0 < \alpha < 1$ and let f be an analytic function in $\mathbb{D}$. Is the function $p \mapsto \|f\|_{A^p_\alpha}$ increasing?"

Source: Brevig, Kulikov, Seip, and Zlotnikov, "Contractive Inequalities in the Dirichlet Range," arXiv:2510.14333v1, Problem 2, p. 8.
Answer: Answered: no. **Evidence:** arXiv.

Dayan, Llinares, and Monsalve-López, arXiv:2604.21347, Introduction, give a monomial counterexample for every $0 < \alpha < 1$, so the proposed monotonicity fails throughout the stated range.

24.24. **Exact norm of a weighted Bergman multiplier**

> Question 1.6. For a multiplier g with $\|g\|_{B_{(\beta-\alpha)/2}} = 1$, what is its exact multiplier norm?

Here $\beta > \alpha > -1$, $M_g : A^2_\alpha \to A^2_\beta$ is multiplication by g, and $\|g\|_{B_\delta} = \sup_{z\in\mathbb{D}} |g(z)|(1 - |z|^2)^\delta$.
Source: Jianjun Jin, "On the Norms of the Multiplication Operators between Weighted Bergman Spaces," arXiv:2603.16170v9, Question 1.6, p. 3.
Answer: Partial. **Evidence:** Sketch.

Our calculation gives exact norms for normalized monomial multipliers and an explicit lower bound for the general supremum. The exact norm for an arbitrary analytic multiplier remains open.

24.25. Sharp supremum of weighted Bergman multiplier norms

Question 1.7. What is the exact value of

$$M(\alpha,\beta) := \sup_{\|g\|_{B_{(\beta-\alpha)/2}}=1} \|g\|_{M_{\alpha,\beta}}?$$

Here $\beta > \alpha > -1$ and $M_{\alpha,\beta}$ denotes the norm of the multiplication operator $M_g : A^2_\alpha \to A^2_\beta$. Source: Jianjun Jin, "On the Norms of the Multiplication Operators between Weighted Bergman Spaces," arXiv:2603.16170v9, Question 1.7, p. 3. The source computes the norm for $g_0(z) = (1-z^2)^{-(\beta-\alpha)/2}$ in Theorem 1.8; the separate Question 5.17 is outside this target.
Answer: Partial. **Evidence:** Sketch.

The monomial calculation supplies an explicit lower bound, but no interaction-sensitive upper bound determines the supremum.

24.26. Does every Banach space of cotype q have hypercontractive polynomial cotype q?

The question we now ask is whether or not every Banach space with cotype q has hypercontractive polynomial cotype q.

Source: Defant, García, Maestre, and Sevilla-Peris, "Dirichlet Series and Holomorphic Functions in High Dimensions," §25.4, p. 623. Hypercontractive polynomial cotype is defined immediately before this question by equation (25.13).
Answer: Answered: yes. **Evidence:** Published.

Carando, Marceca, and Sevilla-Peris, Transactions of the American Mathematical Society 373 (2020), Theorem 3.1 and Remark 3.3, prove that cotype q is equivalent to exponential-degree polynomial cotype q, including an L^1 norm on the right. This gives the affirmative answer for every Banach space of cotype q.

24.27. Haagerup sharp Steinhaus Khintchine constants

"Our main result is a step towards substantiating the Haagerup conjecture (cf. Pełczyński [8], Section 4)."

The inferred target asks whether the optimal lower Steinhaus constant is $A_p = \min\{\gamma_p, d_p\}$ for $0 < p < 2$, and the optimal upper constant is $B_p = \gamma_p$ for $p > 2$, where $\gamma_p = \Gamma((p+2)/2)^{1/p}$ and $d_p = \sqrt{2}(\Gamma((p+1)/2)/(\sqrt{\pi}\Gamma((p+2)/2)))^{1/p}$. Sawa's §4.3 defines these candidates; the added-in-proof result gives the lower bound with constant γ_p for $p_0 < p \le 2$, where $p_0 = 0.47562\ldots$, and the sharp upper bound γ_p for $p \ge 2$. Source: Jacek Sawa, "Some Remarks on the Khintchine Inequality for Complex Steinhaus Variables," §4.3, pp. 124–125 and Added in proof, p. 125 (scan pp. 10–11). The quoted sentence motivates but does not itself state the inferred sharp-constant question.
Answer: Partial. **Evidence:** arXiv.

Rapaport, Tkocz, and Wu, arXiv:2512.09077v2, Section 3, identify and repair the induction gap in König's lower-constant proof for $0 < p < 1$. This supplies a preprint result for part of Sawa's proposed constants; the full two-sided conjecture for all $0 < p < 2$ and $p > 2$ is not settled here.

24.28. Sphere orthogonal-polynomial hypercontractivity

Conjecture. If μ is the uniform measure on a sphere in $\mathbb{R}^d$, then $\|T_\omega\|_{p,q} = 1$ for $-\sqrt{(p-1)/(q-1)} \le \omega \le \sqrt{(p-1)/(q-1)}$.

Source: Svante Janson, "On hypercontractivity for multipliers on orthogonal polynomials," Section 6, Conjecture, pp. 107–108.
Answer: Answered: yes. **Evidence:** Published.

Beckner, Proceedings of the National Academy of Sciences 89 (1992), proves the sharp logarithmic Sobolev inequality on spheres; its hypercontractive consequence gives the Nelson range $|\omega| \le \sqrt{(p-1)/(q-1)}$ for the degree multiplier. This settles Janson's conjecture in the stated probability normalization.

24.29. **Gegenbauer probability-measure hypercontractivity**

Conjecture. The same is true for the measures $c_\alpha(1-x^2)^\alpha \, \mathrm{d}x$ on $(-1,1)$, $\alpha > -1$ (Gegenbauer polynomials).

Source: Svante Janson, "On hypercontractivity for multipliers on orthogonal polynomials," Section 6, Conjecture, p. 108.
Answer: Answered: no. **Evidence:** Sketch.

A degree-two polynomial and an exact rational moment computation disprove the universal conjecture for α sufficiently close to -1.
Answers, "Gegenbauer probability-measure hypercontractivity", p. 42.

24.30. **Measures with sharp orthogonal-polynomial hypercontractivity**

"The general and still unsolved problem is to decide when the operators are contractions and, in particular, for which measures this holds with Nelson's conditions on ω."

Source: Svante Janson, "On hypercontractivity for multipliers on orthogonal polynomials," Introduction, p. 97. Here the operators are Janson's orthogonal-polynomial degree multipliers.
Answer: Partial. **Evidence:** Sketch.

The source supplies sufficient conditions for several classes of measures and shows the Nelson condition is necessary. A general characterization of measures remains open.

25. Bohr radii and monomial convergence

25.1. **Second-order asymptotics of the polydisc Bohr radius**

$$\lim_{n\to\infty} \frac{K_n}{\sqrt{(\log n)/n}} = 1.$$

Source: Defant, García, Maestre, and Sevilla-Peris, "Dirichlet Series and Holomorphic Functions in High Dimensions," Theorem 10.23, p. 265. The next-order question is inferred from this leading asymptotic; the theorem does not state a second-order problem.
Answer: Partial. **Evidence:** Sketch.

The leading equivalent is known. A recent preprint gives an O(loglog n/log n) remainder and bounds the normalized liminf and limsup, but not the true second-order term.

25.2. **Critical monomial-convergence boundary on c_0**

For $c < 1$, $(c/\sqrt{n})_{n\in\mathbb{N}} \in \operatorname{mon} H^\infty(B_{c_0})$, while for $c > 1$ the corresponding sequence is not in $\operatorname{mon} H^\infty(B_{c_0})$. In the case $c > 1$, the sequence is understood as constantly equal to $1/c$ for $1 \le n \le c^2$ and $c/\sqrt{n}$ for $n > c^2$.

Source: Defant, García, Maestre, and Sevilla-Peris, "Dirichlet Series and Holomorphic Functions in High Dimensions," §10.6, formula (10.54), pp. 260–261. The endpoint question asks whether the critical $n^{-1/2}$ sequence, with its first coordinate clipped to lie in the open ball, belongs to the set. This is inferred from the strict threshold and the source's finite-coordinate observation; it is not the same as the displayed $c > 1$ sequence.
Answer: Partial. **Evidence:** Sketch.

The membership criterion is known when the normalized limsup is below 1 and fails above 1. The critical sequence remains unresolved.

25.3. **Exact asymptotics of the Bohr radius of ℓ_p^n balls**

"Whether or not the equality from (19.2) holds for $1 < p < 2$ is unknown."

Equation (19.2) in the source is the proposed limit

$$\lim_{n\to\infty} \frac{K(B_p^n)}{((\log n)/n)^{1-1/\min(p,2)}} = 1.$$

For $1 < p < 2$, this asks whether the normalized Bohr radius tends to 1 with exponent $1-1/p$. Source: Defant, García, Maestre, and Sevilla-Peris, "Dirichlet Series and Holomorphic Functions in High Dimensions," Chapter 19, after Theorem 19.1, equations (19.1)–(19.2), p. 487.

Answer: Answered: no. **Evidence:** Sketch.

Our public note proves a strict upper constant below 1 for each fixed $1 < p < 2$, disproving the proposed limit. It does not determine whether the normalized Bohr radius converges. *Answers*, "Exact asymptotics of the Bohr radius of ℓ_p^n balls", p. 42.

Public proof.

25.4. **Exact set of bounded holomorphic monomial convergence on ℓ_p**

For $1 < p < \infty$, define q by $1/q = 1/2+1/\max\{p,2\}$. Then $B_{\ell_p} \cap \ell_q \subset \operatorname{mon} H^\infty(B_{\ell_p})$. Conversely, if $z \in \operatorname{mon} H^\infty(B_{\ell_p})$, then $z_n^* \lesssim (\log n)^{1-1/\min\{p,2\}}/n^{1/2+1/\max\{p,2\}}$.

Source: Defant, García, Maestre, and Sevilla-Peris, "Dirichlet Series and Holomorphic Functions in High Dimensions," Theorem 20.23, p. 523. The exact-set target is inferred from the logarithmic gap between the theorem's sufficient and necessary conditions.

Answer: Partial. **Evidence:** Sketch.

The book gives strict inclusions and rearrangement bounds for the set of monomial convergence. Its exact description remains open.

25.5. **Exact monomial-convergence sets for homogeneous polynomials on ℓ_p**

For $1 < p < \infty$ and $m \geq 2$, $\ell_{2m/(m-1),\infty} \cdot \ell_p \subset \operatorname{mon} P_m(\ell_p) \subset \ell_{((m-1)/(2m)+1/p)^{-1},\infty}$ for $2 \leq p < \infty$, and $\ell_{mp,1} \subset \operatorname{mon} P_m(\ell_p) \subset \ell_{mp,\infty}$ for $1 < p \leq 2$.

Source: Defant, García, Maestre, and Sevilla-Peris, "Dirichlet Series and Holomorphic Functions in High Dimensions," Theorem 20.7, p. 510. Determining the exact universal convergence set between these inclusions is an inferred target; the theorem itself gives the stated bounds.

Answer: Partial. **Evidence:** Sketch.

Known inclusions characterize several large subsets of the universal convergence set, but the exact set for each degree and $1 < p < \infty$ is unknown.

25.6. **Exact values of multidimensional Bohr radii**

Bohr "gives the exact value of K_1 (actually the only exact value known so far)."

Source: Defant, García, Maestre, and Sevilla-Peris, "Dirichlet Series and Holomorphic Functions in High Dimensions," §8.1, p. 183. The question for K_n with $n \geq 2$ or for other Reinhardt domains is inferred from this statement, not explicitly posed in the quoted sentence.

Answer: Open. **Evidence:** Heuristic.

The exact multidimensional Bohr radius is unknown beyond the classical one-variable and special arithmetic-domain cases.

25.7. **Optimal growth of the complex multilinear Bohnenblust–Hille constants**

"The optimal exponent here so far seems not to be relevant in our context."

This sentence concerns the optimal growth exponent for the complex multilinear Bohnenblust–Hille constants B_m. The natural question of their optimal growth is inferred from this discussion; boundedness is stronger than merely asking whether the exponent is zero. Source: Defant, García, Maestre, and Sevilla-Peris, *Dirichlet Series and Holomorphic Functions in High Dimensions*, §6.7 Notes, p. 152.

Answer: Partial. **Evidence:** Published.

Bayart, Pellegrino, and Seoane-Sepúlveda, *Advances in Mathematics* 264 (2014), Proposition 3.1 and Corollary 3.2, prove $B_m \leq Cm^{(1-\gamma)/2}$. Their exact exponent $(1-\gamma)/2$ is

approximately 0.21139217; the textbook's displayed decimal 0.21392 is inconsistent. The optimal growth and dimension-free boundedness remain open.

25.8. **Growth of the complex polynomial Bohnenblust–Hille constants**

> "There are two ways to improve this constant: to find a better constant in the multilinear inequality and/or to use better polarization constants."

The passage concerns the polynomial Bohnenblust–Hille constants B_m^{pol}. The question of their optimal growth, in particular whether $\sup_m B_m^{\mathrm{pol}} < \infty$, is inferred from the discussion. Source: Defant, García, Maestre, and Sevilla-Peris, *Dirichlet Series and Holomorphic Functions in High Dimensions*, §6.7, p. 152. Theorem 8.19 and Theorem 10.22 give the cited subexponential controls and $\lim_m (B_m^{\mathrm{pol}})^{1/m} = 1$.
Answer: Partial. **Evidence:** Sketch.

The source's Theorems 8.19 and 10.22 give subexponential upper bounds and the exact root limit $\lim_m (B_m^{\mathrm{pol}})^{1/m} = 1$. These results do not determine whether the constants are bounded or their optimal subexponential growth.

25.9. **Lempert's problem on monomial convergence and ℓ_1**

> "Does a Banach sequence space X equal ℓ_1 whenever $rB_X \subset \operatorname{mon} H^\infty(B_X)$ for some $0 < r < 1$?"

Here X has a 1-unconditional basis and $\ell_1 \subset X \subset c_0$. Defant, García, Maestre, and Sevilla-Peris, Lemma 20.27, identify equivalent formulations using finite sections and the Bohr radii. Source: Lempert's question as quoted by Defant, García, Maestre, and Sevilla-Peris, "Dirichlet Series and Holomorphic Functions in High Dimensions," §20.5, pp. 526–527, Question 20.26 and Lemma 20.27.
Answer: Answered: no. **Evidence:** arXiv.

Wang, arXiv:2610.08873v1, Theorem 4.16 with Proposition 4.5 and Corollary 4.7, claims a complex Banach sequence space Y strictly between ℓ_1 and c_0 for which every Reinhardt domain has the asserted monomial convergence. This gives a negative answer conditional on the unrefereed proof.

25.10. **Polynomial Bohnenblust–Hille constants on the Boolean cube**

> "The aim is to show that $BH(d)$ is at most of the polynomial growth or is even bounded by some universal constant."

Here $BH(d)$ is the Bohnenblust–Hille constant for d-homogeneous polynomials on the Boolean cube. The inferred question is whether $BH(d)$ has polynomial growth in d, or is bounded independently of d; the original AimPL problem-list numbering is unavailable, so no Problems 3.1–3.2 wording is claimed. Source: AIM workshop report, "Hypercontractivity and Quantum Information Theory," Bohnenblust–Hille section, p. 2.
Answer: Answered: yes. **Evidence:** arXiv.

Ivanisvili, arXiv:2609.12427, and Slote, Tseng, and Volberg, arXiv:2609.21144, prove polynomial-in-degree Bohnenblust–Hille bounds on the Boolean cube for degree-at-most-d polynomials; restriction to homogeneous polynomials gives polynomial growth for $BH(d)$. The optimal power is unresolved.

25.11. **Asymptotics of the Bohr radius for degree-N polynomials (Khavinson, Fournier)**

> "What is the rate of convergence of R_N to $1/3$ as $N \to \infty$? In other words, what is the asymptotic of $R_N - 1/3$?"

Here R_N is the Bohr radius for polynomials of degree at most N bounded by one on $\mathbb{D}$. The source says Fournier suggested the asymptotic $R_N - 1/3 \sim \pi^2/(3N^2)$. Source: Dmitry Khavinson, after Guadarrama and Fournier, "Problems for Polynomials of One and Several Variables," Oberwolfach Report 06/2014, Problem Session, §3, item 3, p. 382.
Answer: Answered: yes. **Evidence:** Published.

Chu, Contemporary Mathematics 638 (2015), pages 39–43, proves $R_N = 1/3 + \pi^2/(3N^2) + o(N^{-2})$, confirming Fournier's proposed asymptotic.

25.12. **Are the perfect squares stationary? (Queffélec)**

Is the set S of perfect squares stationary?

Source: Hervé Queffélec, "Dirichlet Series and Function Theory in Polydiscs," Oberwolfach Report 06/2014, Problem Session, §3.1, Question 1, pp. 385–386 (reported by O. F. Brevig). Here stationary means that Pisier's randomized supremum norm is bounded by a constant times the ordinary supremum norm.

Answer: Answered: no. **Evidence:** Sketch.

Our explicit polynomials supported on odd squares have Pisier norm larger than their supremum norm by a factor of order $\sqrt{\log N}$, so the full square set is not stationary. *Answers*, "Are the perfect squares a stationary set?", p. 43.

26. Random multiplicative questions

26.1. **Farmer–Gonek–Hughes global zeta maximum**

"Conjecture A.

$$\max_{t\in[0,T]}\left|\zeta\left(\frac{1}{2}+it\right)\right| = \exp\left((1+o(1))\sqrt{\tfrac{1}{2}\log T\log\log T}\right).$$

"

Source: David W. Farmer, S. M. Gonek, and C. P. Hughes, "The Maximum Size of L-Functions," arXiv:math/0506218v3, Conjecture A, equation (1.3), p. 2; published in *Journal für die reine und angewandte Mathematik* 638 (2010), 215–236.

Answer: Partial. **Evidence:** Sketch.

No proof of the global maximum conjecture is known in the bounded search. Our conditional coefficient-spread result is an obstruction for one resonator family only.

26.2. **Sharp logarithmic exponent of Gál sums for 0¡alpha¡1/2**

"For every α, $0 < \alpha < 1/2$, there exist positive constants $a(\alpha)$ and $b(\alpha)$ such that

$$N^{1-2\alpha}(\log N)^{a(\alpha)} \le \Gamma_\alpha(N) \le N^{1-2\alpha}(\log N)^{b(\alpha)}$$

for sufficiently large N."

Here $\Gamma_\alpha(N) = \sup_{|M|=N} |M|^{-1}\sum_{m,n\in M}\big((m,n)/[m,n]\big)^\alpha$ is the normalized Gál sum over arbitrary finite sets of positive integers. The source does not ask for a single sharp logarithmic exponent; the target here is the inferred question whether $\Gamma_\alpha(N) = N^{1-2\alpha}(\log N)^{c(\alpha)+o(1)}$ for some $c(\alpha)$. Source: Andriy Bondarenko, Titus Hilberdink, and Kristian Seip, "Gál-Type GCD Sums beyond the Critical Line," arXiv:1512.03758v2, Theorem 1, p. 2.

Answer: Partial. **Evidence:** Sketch.

Bondarenko, Hilberdink, and Seip give matching polynomial-scale bounds with nonmatching logarithmic exponents. The exact exponent function remains open.

26.3. **Gál sums for 0¡alpha¡1/2 over square-free sets: can the log power be removed?**

> "One may notice that it would not be possible to construct a similar example if we required the set to consist only of square-free numbers. Hence it remains an open problem to prove the analogue of Theorem 1 in the square-free case. More specifically, we may ask whether the logarithmic power can be discarded in this case as well."

For $0 < \alpha < 1/2$, let $\Gamma_\alpha^{\mathrm{sf}}(N)$ be the normalized Gál maximum over square-free N-element sets. The question is whether the logarithmic factor in the unrestricted estimate can be removed, that is, whether $\Gamma_\alpha^{\mathrm{sf}}(N) \asymp N^{1-2\alpha}$. Source: Andriy Bondarenko, Titus Hilberdink, and Kristian Seip, "Gál-Type GCD Sums beyond the Critical Line," arXiv:1512.03758v2, §1, p. 3.

Answer: Partial. **Evidence:** Sketch.

The known unrestricted estimates include logarithmic factors, while the square-free lower-bound construction does not settle whether they can be removed.

26.4. **Exact a.s. growth of random multiplicative sums**

Determine the almost sure growth rate (in N) of the character sum $\sum_{n=1}^{N} \chi(n)$.

Source: Saksman and Seip, "Some open questions in analysis for Dirichlet series," Problem 4.6, p. 6. Immediately before the problem, $\chi(p)$ are defined as independent uniform random variables on $\mathbb{T}$ and extended multiplicatively.

Answer: Partial. **Evidence:** Sketch.

Harper, *International Mathematics Research Notices* (2023), Theorem 1, proves arbitrarily large fluctuations at the $1/4$ exponent, with any prescribed diverging loss. Durkan and Pearce-Crump, arXiv:2607.29429v2, Theorem 1 and Corollary 1.1, prove the matching upper exponent; Verreault, arXiv:2608.21354v2, Theorem 1.1, gives an upper bound with a $\log_3 x$ factor. Thus the exponent is determined, but the exact almost-sure order and normalized limsup remain open.

26.5. **GCD sums and Hardy operator theory**

Link the estimates for GCD sums to the function and operator theory of the spaces H^p.

Source: Saksman and Seip, "Some open questions in analysis for Dirichlet series," Problem 5.1, p. 7.

Answer: Partial. **Evidence:** Sketch.

Aistleitner, Berkes, and Seip identify GCD sums with Poisson integrals and give H^2 spectral consequences. Further operator-theoretic applications remain unspecified and open.

26.6. **GCD sums and zeta applications**

Develop further applications to and links with the Riemann zeta function.

Source: Saksman and Seip, "Some open questions in analysis for Dirichlet series," Problem 5.2, p. 7.

Answer: Partial. **Evidence:** Sketch.

Published work gives sharp Gál-sum estimates and zeta resonance applications; a separate square-free divisor-form question is answered below. The broader programme has no single completion criterion.

26.7. **Fyodorov–Hiary–Keating tightness**

Consider $[0,T]$ as a probability space, with normalised Lebesgue measure, and denote the corresponding variable by $\omega \in [0,T]$. Then, as $T \to \infty$, one has $\max_{h\in[0,1]} \log|\zeta(1/2 + ih + i\omega)| = \log\log T - \frac{3}{4}\log\log\log T + E$, where the error term E is bounded in probability as $T \to \infty$.

Source: Saksman and Seip, "Some open questions in analysis for Dirichlet series," Problem 6.1, pp. 9–10. The later Arguin–Bourgade–Radziwiłł result uses τ uniform on $[T, 2T]$ and $|h| \le 1$, so its interval and shift window do not literally match this statement.

Answer: Partial. **Evidence:** arXiv.

Arguin, Bourgade, and Radziwiłł, arXiv:2007.00988, Theorem 1, and arXiv:2307.00982, Theorem 1 and Corollary 1, state the Fyodorov–Hiary–Keating tightness claim. Their proof is written for τ uniform on $[T, 2T]$ and $|h| \le 1$, whereas the source formulation uses ω uniform on $[0,T]$ and $h \in [0,1]$; the literal one-unit lower-tail transfer is not verified here.

26.8. **Multiplicative chaos of the random prime field: supercritical maximum and freezing**

Let the field X be defined as in (8). Study the properties non-Gaussian chaos "$\exp(\beta X(x))$."

Source: Saksman and Seip, "Some open questions in analysis for Dirichlet series," Problem 6.2, p. 10. The wording above preserves the source's printed phrase, which omits "of the"; the random prime field is defined by equation (8).

Answer: Partial. **Evidence:** arXiv.

Saksman and Webb, arXiv:1604.08378, Theorems 4 and 5, construct subcritical and critical chaos for the direct random-zeta prime model. These results give a concrete partial analogue, while the source's broader programme about the random prime field remains open.

26.9. **Subcritical zeta-chaos convergence**

> "For $\beta \in (0,2)$ the random densities $(\log T)^{-\beta^2/4}|\zeta(1/2+ix+iT)|^\beta$, $x \in [0,1]$, converge in distribution to a constant multiple of the multiplicative chaos measure described in Theorem 1.8."

The source's printed formula has iT; the intended randomly shifted interpretation takes T to be ωT with ω uniform on $[0,1]$. Source: Saksman and Webb, "Multiplicative Chaos Measures for the Riemann Zeta Function," arXiv:1609.00027v3, Conjecture 1.10, p. 14.

Answer: Open. **Evidence:** Heuristic.

Gaussian multiplicative-chaos limits are established for the limiting field, but the actual randomly shifted zeta measure in this conjecture remains open.

26.10. **Critical zeta-chaos convergence**

The preceding conjecture concerns convergence of the positive measures built from the actual randomly shifted zeta function.

> "The previous conjecture holds for $\beta = \beta_c = 2$ as soon as one adds the normalizing factor $(\log\log T)^{1/2}$."

Source: Saksman and Webb, "Multiplicative Chaos Measures for the Riemann Zeta Function," arXiv:1609.00027v3, Conjecture 1.11, p. 14.

Answer: Expected yes. **Evidence:** Heuristic.

Harper announced a proof of the critical case, but no public proof manuscript was found; it remains an announcement rather than a checked resolution.

26.11. **Mesoscopic real zeta-chaos convergence**

> "There are mesoscopic analogues of the above conjectures and they can be formulated in a similar way as in Section 5."

The target is the pair of real positive-measure conjectures for the actual zeta function on the shrinking windows described in Section 5. Source: Saksman and Webb, "Multiplicative Chaos Measures for the Riemann Zeta Function," arXiv:1609.00027v3, Conjecture 1.12, p. 14.

Answer: Open. **Evidence:** Heuristic.

Mesoscopic convergence is proved for related Gaussian models, while the actual zeta-function analogues remain open.

26.12. **Helson better-than-square-root first moment**

> "For example, Helson [15] conjectured in the Steinhaus case one should have
>
> $$\mathbb{E}|\sum_{n\le x} f(n)| = o(\sqrt{x}),$$
>
> and observed that if this is true then a certain generalisation of Nehari's theorem from harmonic analysis is false."

Here f is a Steinhaus random multiplicative function; the source's reference [15] identifies Helson. Source: Harper, "Moments of Random Multiplicative Functions," arXiv:1703.06654v1, p. 2.

Answer: Answered: yes. **Evidence:** Published.

Harper, *Forum of Mathematics, Pi* 8 (2020), Theorem 1, proves that the Steinhaus first moment has order $\sqrt{x}/(\log\log x)^{1/4}$, which implies Helson's prediction.

26.13. **Square-root low-moment counter-conjecture**

> "They also conjectured, in opposition to Helson's conjecture, that $\mathbb{E}|\sum_{n\le x} f(n)|^{2q} \sim C(q)x^q$ as $x\to\infty$, for each fixed $0\le q\le 1$."

"They" refers to Harper, Nikeghbali, and Radziwiłł; the source discusses the Steinhaus model and then the analogous Rademacher model. Source: Harper, "Moments of Random Multiplicative Functions," arXiv:1703.06654v1, p. 3.

Answer: Answered: no. **Evidence:** Published.

Harper, *Forum of Mathematics, Pi* 8 (2020), Theorems 1 and 2, proves that for each fixed $0<q<1$ the Steinhaus and Rademacher $2q$ moments are $o(x^q)$ by a diverging log-log factor, refuting the positive-constant prediction.

26.14. **Non-Gaussianity of normalized Rademacher sums**

> "does one have a central limit theorem for $\sum_{n\le x} f(n)/\sqrt{\mathbb{E}|\sum_{n\le x} f(n)|^2}$?"

The source attributes the failure-of-CLT conjecture in the Rademacher case to Chatterjee. Source: Harper, "Moments of Random Multiplicative Functions," arXiv:1703.06654v1, p. 2, question (i), with the Rademacher attribution on p. 3.

Answer: Answered: no. **Evidence:** Published.

Harper, *Forum of Mathematics, Pi* 8 (2020), Corollary 1, proves that the mean-square-normalized sum tends to zero in probability in both the Steinhaus and Rademacher models, which precludes a nondegenerate Gaussian limit. The negative answer to the CLT question verifies Chatterjee's non-Gaussianity conjecture.

26.15. **A limiting law for long random multiplicative sums**

> "One now has an obvious and interesting follow-up question, namely what is the distribution of the properly renormalised sum $\sum_{n\le x} f(n)/(\sqrt{x}/(\log\log x)^{1/4})$?"

The source paragraph discusses both the Steinhaus and Rademacher models. Source: Harper, "Moments of Random Multiplicative Functions," arXiv:1703.06654v1, p. 10.

Answer: Partial. **Evidence:** Sketch.

Gorodetsky and Wong claim the normalized Steinhaus limit; the corresponding full Rademacher limit is not covered, so the two-model question is only partial.

26.16. **Sharp tail probabilities for random multiplicative sums**

> "It seems another interesting open question to determine the exact magnitude of the probabilities in Corollary 2."

Here the probabilities are tails of the random multiplicative sum in Corollary 2; Harper gives a fixed-λ prediction and a separate large-λ scale. Source: Harper, "Moments of Random Multiplicative Functions," arXiv:1703.06654v1, §1.2, p. 11.

Answer: Partial. **Evidence:** Sketch.

Gorodetsky and Wong, arXiv:2508.12956v3, Theorem 1.3, prove the limiting tail probability at each fixed threshold for the normalized Steinhaus sum. Harper, "High Moments of Random Multiplicative Functions," *Annales de l'Institut Henri Poincaré, Probabilités et Statistiques* 56 (2020), Corollary 1.3, gives upper estimates in a high-threshold range. Uniform sharp estimates across fixed and growing thresholds, and the corresponding Rademacher limit, remain open.

26.17. **Weighted circular GMC moment formula**

> "Conjecture 1. Let $\gamma\in(0,2)$. For all real p such that $p<4/\gamma^2$ we expect to have
>
> $$\mathbb{E}\left[\left(\frac{1}{2\pi}\int_0^{2\pi}|1-e^{i\theta}|^2 e^{\gamma X(e^{i\theta})/2}\,\mathrm{d}\theta\right)^p\right]=\frac{\Gamma(1-p\gamma^2/4)\Gamma(1+8/\gamma^2)\Gamma(1+4/\gamma^2-p)}{\Gamma(1-\gamma^2/4)^p\Gamma(1+4/\gamma^2)\Gamma(1+8/\gamma^2-p)}.$$
>
> and we can write again
>
> $$\frac{1}{2\pi}\int_0^{2\pi}|1-e^{i\theta}|^2 e^{\gamma X(e^{i\theta})/2}\,\mathrm{d}\theta \overset{\text{law}}{=} Y_\gamma X_2^{-1},$$

with Y_γ, X_2 independent and $X_2 \sim B(1+4/\gamma^2, 4/\gamma^2)$. ”

Here X is the source's circular log-correlated Gaussian field and $\mathrm{Beta}(a,b)$ is the standard beta law. Source: Guillaume Rémy, “The Fyodorov-Bouchaud Formula and Liouville Conformal Field Theory,” arXiv:1710.06897v3, Conjecture 1, equations (1.36)–(1.37), p. 10.

Answer: Answered: yes. **Evidence:** Published.

Rémy and Zhu, Communications in Mathematical Physics 395 (2022), Theorem 1.7 and Corollary 1.12, prove the inserted circular Gaussian multiplicative chaos moment formula and limiting law. Their normalization matches the source after the parameter change $\beta = -4/\gamma$.

26.18. **Divisor-restricted Gál form over square-free sets**

≪ Le résultat suivant, relatif aux éléments de $\mathcal{A}$, laisse supposer que la taille maximale de $Q(M)$ lorsque M décrit les sous-ensemble de $\mathbb{N}_1$ de taille N vaut $\Gamma^*(N)^{1/2+o(1)}$. Cette question demeure ouverte en l'état. Rappelons la définition des normes $Q(M)$ en $(1 \cdot 6)$. ≫

The text immediately preceding this passage defines the $Q(M)$ in §5 as the divisor-restricted quadratic form, and asks about all square-free sets of size N, not only $M \in \mathcal{A}$. We write $\mathcal{Q}(M)$ for that restricted form in the answer to distinguish it from the full Gál-form norm $Q(M)$ in the introduction. Here $\mathbb{N}_1$ denotes the square-free integers and $\Gamma^*(N)$ is their maximal normalized Gál sum. Source: Régis de la Bretèche and Gérald Tenenbaum, “Sommes de Gál et applications,” §5.1, before Lemma 5.2, version 8, p. 18 (published in *Proceedings of the London Mathematical Society* 119 (2019)).

Answer: Answered: yes. **Evidence:** Sketch.

Our proof establishes the conjectured square-root exponent for the divisor-restricted form on square-free sets, using the published Gál-sum asymptotics for matching bounds.
Answers, “Divisor-restricted Gál form over square-free sets”, p. 44.

26.19. **Erdős/Halász failure of a.s. square-root bound**

“On the other hand, Erdős [4] conjectured (in the Rademacher case) that almost surely $\sum_{n\le x} f(n) \neq O(\sqrt{x})$, whilst writing that ‘As far as I know nobody has a plausible guess for the true order of magnitude of $|\sum_{n\le x} f(n)|$’.”

The preceding paragraph states Halász's corresponding question for Steinhaus functions. Source: Harper, “Almost Sure Fluctuations of Random Multiplicative Functions,” arXiv: 2012.15809, p. 3.

Answer: Answered: yes. **Evidence:** Published.

Harper, *International Mathematics Research Notices* (2023), Theorem 1, proves that for each prescribed $V(x) \to \infty$ there are almost surely arbitrarily large x with $|\sum_{n\le x} f(n)| \ge \sqrt{x}(\log\log x)^{1/4}/V(x)$ in both models. In particular the proposed almost-sure $O(\sqrt{x})$ bound fails.

26.20. **Harper almost-sure logarithmic exponent**

“As we will explain, it now seems plausible that a bound of the form

$$\sqrt{x}(\log\log x)^{1/4+o(1)}$$

is sharp in this problem, and if so Theorem 1 is quite close to best possible.”

This predicts a matching upper exponent in the random-multiplicative-sum problem. Source: Harper, “Almost Sure Fluctuations of Random Multiplicative Functions,” arXiv:2012.15809, p. 3.

Answer: Answered: yes. **Evidence:** arXiv.

Harper, *International Mathematics Research Notices* (2023), Theorem 1, proves arbitrarily large lower fluctuations at the $1/4$ exponent for each prescribed diverging loss. Durkan and Pearce-Crump, arXiv:2607.29429v2, Theorem 1 and Corollary 1.1, prove the matching almost-sure upper exponent for both Steinhaus and Rademacher sums. Verreault, arXiv: 2608.21354v2, Theorem 1.1, proves another upper bound with a $\log_3 x$ factor. The exponent is determined, while the exact normalized limsup remains open.

26.21. **Smooth-number deterministic/random first-mean comparison for $x \le T^A$, $A \le 2$**

"For any fixed $A > 0$ and small $\epsilon > 0$, we have, very roughly speaking," followed by

$$\frac{1}{T}\int_{\substack{|t|\le T\\ t\not\approx 0}}\Bigg|\sum_{\substack{n\le x\\ P(n)\le x^{\epsilon}}} n^{-it}\Bigg|\,\mathrm{d}t \ll \mathbb{E}\Bigg|\sum_{\substack{n\le x\\ P(n)\le x^{\epsilon}}} f(n)\Bigg|, \qquad x \le T^A.$$

The source labels (1.2) a rough conjecture and does not quantify the exclusion $t \not\approx 0$. Our literal interpretation has no main-term subtraction and is false for every $A > 2$ and fixed $0 < \epsilon < 1$; that range is refuted by our written counterexample. Whether a precise version holds for $0 < A \le 2$ remains open. Source: Seth Hardy and Max Wenqiang Xu, "Helson's Conjecture for Smooth Numbers," arXiv:2511.03430v3, equation (1.2), p. 4.
Answer: Partial. **Evidence:** Sketch.

26.22. **Smooth numbers in square-root-length intervals**

"coming close to solving an old problem that, for any fixed $\epsilon > 0$, there exists an x^{ϵ}-smooth number in $[x, x + \sqrt{x}]$ when x is sufficiently large."

This is the older assertion mentioned by the authors after noting that Matomäki and Radziwiłł prove existence in intervals of length $h \gg_{\epsilon} \sqrt{x}$; the question asks for the exact coefficient one. Source: Seth Hardy and Max Wenqiang Xu, "Helson's Conjecture for Smooth Numbers," arXiv:2511.03430v3, Introduction, p. 3.
Answer: Partial. **Evidence:** arXiv.

Harman, "Numbers with a Large Prime Factor," *Acta Arithmetica* 91 (1999), Theorem 1, proves the exact interval assertion for $\epsilon > 1/(4\sqrt{e})$. Matomäki and Radziwiłł, *Annals of Mathematics* 183 (2016), Corollary 1, prove existence in $[x, x + C(\epsilon)\sqrt{x}]$ for every fixed $\epsilon > 0$. Neither result establishes existence in every exact interval $[x, x + \sqrt{x}]$ for all $\epsilon > 0$.

26.23. **Critical-chaos law of the iterated logarithm**

"Conjecture 1.4. Let f be a Steinhaus random multiplicative function, and let V_∞ be the associated random variable in [GW25, Theorem 1.3]. Almost surely,

$$\limsup_{x\to\infty} \frac{|M_f(x)|}{\sqrt{x}(\log_2 x)^{1/4}} = \sqrt{V_\infty}.$$

"

Here $M_f(x) = \sum_{n\le x} f(n)$ and $\log_2 x = \log\log x$. Source: William Verreault, "Almost Sure Upper Bound for Sums of Random Multiplicative Functions and Critical Chaos," arXiv: 2608.21354v2, Conjecture 1.4, §1.6, p. 5.
Answer: Partial. **Evidence:** arXiv.

Verreault's Theorem 1.1 proves an upper bound with an extra $(\log_3 x)^{1+\epsilon}$ factor; the exact limsup identity remains open.

26.24. **Keating–Snaith zeta moment leading constant**

"Then, using a random matrix analogy (now called the 'Keating-Snaith philosophy'), they make the following conjecture for the moments of the Riemann zeta function (see [16],[18]):

$$\lim_{T\to\infty} \frac{1}{T(\log T)^{\lambda^2}} \int_0^T \left|\zeta\left(\frac{1}{2} + it\right)\right|^{2\lambda} \mathrm{d}t = M(\lambda)A(\lambda),$$

where

$$M(\lambda) = \frac{G(1+\lambda)^2}{G(1+2\lambda)}, \qquad A(\lambda) = \prod_p \left(1 - \frac{1}{p}\right)^{\lambda^2} \sum_{m=0}^{\infty} \left(\frac{\Gamma(\lambda+m)}{m!\,\Gamma(\lambda)}\right)^2 p^{-m}.$$

"

The displayed conjecture is stated for real $\lambda \geq 0$; the paper attributes it to Keating and Snaith. Source: Ashkan Nikeghbali and Marc Yor, "The Barnes G-Function and Its Relations with Sums and Products of Generalized Gamma Convolution Variables," *Electronic Communications in Probability* 14 (2009), paragraph following Proposition 1.1, p. 397.
Answer: Partial. **Evidence:** Sketch.

The exact cases $\lambda = 0, 1, 2$ follow respectively from the normalization, the second moment, and Ingham's fourth-moment theorem. The conjectured leading constant for general real $\lambda \geq 0$ remains open.

26.25. **HLS almost-sure square-root logarithmic bound**

> "Can the above be sharpened to the statement that, almost surely, $\sum_{n=1}^{N} \chi(n) = O(\sqrt{N} \log N)$ as $N \to +\infty$?"

Here χ is a Haar-random multiplicative character; the preceding text gives the $O_\epsilon(N^{1/2+\epsilon})$ bound. Source: Håkan Hedenmalm, Peter Lindqvist, and Kristian Seip, "A Hilbert Space of Dirichlet Series and Systems of Dilated Functions in $L^2(0,1)$," §4.8, question after Corollary 4.8, p. 19; *Duke Mathematical Journal* 86 (1997).
Answer: Answered: yes. **Evidence:** Published.

Rachid Caich, "Almost Sure Upper Bound for Random Multiplicative Functions," *Acta Arithmetica* 224 (2026), 297–325, Theorem 1.1, proves almost surely

$$\sum_{n \leq x} f(n) \ll_\epsilon \sqrt{x} (\log_2 x)^{3/4+\epsilon}$$

for both models. At $\epsilon = 1/4$ this is $o(\sqrt{x} \log x)$, answering the cited question affirmatively for Steinhaus sums.

26.26. **Bounded partial sums for square-summable character coefficients**

> "The following conjecture seems reasonable: for almost all χ,
>
> $$\sum_{n=1}^{N} a_n \chi(n) = O(1) \qquad (N \to +\infty).$$

Here $\sum_{n=1}^{\infty} |a_n|^2 < \infty$ is fixed and almost all characters are with respect to Haar measure. Source: Håkan Hedenmalm, Peter Lindqvist, and Kristian Seip, "A Hilbert Space of Dirichlet Series and Systems of Dilated Functions in $L^2(0,1)$," Duke Mathematical Journal 86 (1997), Section 6, p. 30.
Answer: Answered: yes. **Evidence:** Published.

Hedenmalm and Saksman, "Carleson's Convergence Theorem for Dirichlet Series," Pacific Journal of Mathematics 208 (2003), Theorem 1.5, proves that $\sum_{n=1}^{\infty} a_n \chi(n)$ converges for almost every character whenever $(a_n) \in \ell^2$. Convergence implies bounded partial sums, so this proves the conjecture in its full stated scope.

26.27. **Montgomery's Dirichlet-polynomial moment conjecture**

> "One of Montgomery's conjectures states that for every $\varepsilon > 0$ and $p \in (2, 4)$ there exists a constant $C = C(\varepsilon)$ such that for all finite Dirichlet polynomials $f(s) = \sum_{n=1}^{N} a_m n^{-s}$ with $|a_n| \leq 1$ one has
>
> $$\int_0^T |f(it)|^p \, \mathrm{d}t \leq C N^{p/2+\varepsilon} (T + N^{p/2}) \quad \text{for } T > 1.$$
>
> "

The summand in this reported passage prints a_m where the coefficient is indexed by n; the formula uses the standard indexed expansion $f(s) = \sum_{n=1}^{N} a_n n^{-s}$. Søvik attributes the conjecture to Hugh L. Montgomery, "Ten Lectures on the Interface between Analytic Number Theory and Harmonic Analysis," CBMS Regional Conference Series in Mathematics 84, p. 129; the original book page was not independently checked. The quoted passage is

from Øistein Søvik, "Hankel Forms and Nehari's Theorem," section "The Hardy Space $\mathscr{H}^p$," discussion following the Dirichlet-series embedding question.
Answer: Partial. **Evidence:** arXiv.

Guth, arXiv:2503.07410, §3.4, Conjecture 3.4, states a related discrete bounded-coefficient ℓ^q estimate and says it is beyond current methods. Hough and Maynard, *Annals of Mathematics* 203 (2026), Theorem 1.1, improve large-value bounds in selected ranges, while their Conjecture 1.5 remains open; neither proves the continuous L^p bound above.

27. Composition operators

27.1. **Compactness of composition operators on $\mathscr{H}^2$**

Question. Saksman and Seip ask: "Characterize the compact composition operators on H^2." Problem 3.3, p. 5.
Answer: Partial. **Evidence:** Sketch.

The characteristic-zero case is known, and the general positive-characteristic case remains open. New sufficient classes include contact sets whose boundary mass has sufficiently high decay and certain mixed-cusp symbols; the known averaged counting necessity does not give an equivalence.

27.2. **Is the higher-dimensional target exponent optimal?**

Source passage. Bayart and Brevig write: "For $\dim(\varphi) \geq 3$, we conjecture that $(2^\alpha - 1)/d$ is not optimal." See the remark after Theorem 21. The inferred question concerns characteristic-zero Dirichlet polynomials of degree one or two with unrestricted range in dimension $d \geq 3$.
Answer: Partial. **Evidence:** Sketch.

The public sharpness note proves a strictly better target, so the stated exponent is not optimal. The exact best universal exponent is not determined by that improvement.
Answers, "Nonoptimality of a target exponent", p. 45.

Sharpness of composition-operator estimates

27.3. **Do Volterra boundedness classes depend on the exponent?**

Question. "Is $\mathbf{T}_g$ bounded on $\mathcal{H}^2$ if and only if it is bounded on $\mathcal{H}^p$ for every $p < \infty$?" Brevig–Perfekt–Seip, Question 1, p. 3.
Answer: Partial. **Evidence:** Sketch.

The exponent-independence question remains open for general symbols. The paper proves it for linear symbols; current work gives several additional sufficient classes but does not cover the logarithmic one-prime example or arbitrary symbols.

27.4. **Volterra symbols and the dual of $\mathscr{H}^1$**

Question. "Do we have $\mathcal{X}_2 \subset (\mathcal{H}^1)^*$?" Brevig–Perfekt–Seip, Question 2, p. 3.
Answer: Partial. **Evidence:** Sketch.

Brevig–Perfekt–Seip prove that boundedness on $\mathscr{H}^1$ implies membership in this dual. The converse reduction is now exact for finite-prime symbols, but it leaves a dimension-free dual estimate unresolved; the general inclusion remains open.

27.5. **Off-diagonal and endpoint rigidity on disc Hardy spaces**

Question. Laitila, Nieminen, Saksman, and Tylli ask: "Are there results corresponding to our main theorems for $C_\varphi \in \mathcal{L}(H^p, H^q)$ in the case $p \neq q$?" They also ask: "Is there an analogue of Theorem 1.5 for $p = 1$?" Problems 4.3(1)–(2), p. 17.
Answer: Answered: yes. **Evidence:** arXiv.

Shi–Li, Theorems A and C, state that they completely resolve both parts for $1 \leq p, q < \infty$. Their boundary-measure argument constructs fixed copies of the relevant function spaces when the contact set has positive measure.

27.6. **Strictly singular but noncompact composition operators**

Question. Laitila, Nieminen, Saksman, and Tylli ask in Problem 4.3(4): "Is there a Banach space E of scalar-valued analytic functions on $\mathbb{D}$ and an analytic map $\varphi : \mathbb{D} \to \mathbb{D}$, for which $C_\varphi \in \mathcal{S}(E) \setminus \mathcal{K}(E)$?"

Answer: Answered: yes. **Evidence:** Sketch.

An explicit weighted coefficient space and the symbol $\varphi(z) = z^2$ give such an example. The proof decomposes the coefficient space into dyadic exponent blocks and verifies strict singularity and noncompactness directly.

Answers, "A strictly singular noncompact composition operator", p. 46.

27.7. **A converse to the BMOA contact-set criterion**

Question. Laitila, Nieminen, Saksman, and Tylli ask: "Is the converse of Proposition 4.1(i) also true?" Problem 4.3(3), p. 17. The proposition concerns $E_\varphi = \{\zeta \in \mathbb{T} : |\varphi(\zeta)| = 1\}$.

Answer: Answered: no. **Evidence:** Sketch.

A Blaschke-product example has a null contact set but its composition operator is bounded below on a copy of ℓ^2 in BMOA. The construction uses thin zeros approaching the boundary, a feature not detected by contact-set measure.

Answers, "The BMOA contact-set converse", p. 46.

27.8. **Canonical symbols with large parameter**

Source passage. For $\varphi_\alpha(s) = 1/2 + \alpha(1 - 2^{-s})/(1 + 2^{-s})$, Brevig and Perfekt write: "We do not know whether the first two of these equalities hold also for $\alpha > \alpha_0$." Here α_0 solves $\alpha\zeta(1 + \alpha) = 2$; §4, p. 16.

Answer: Answered: no. **Evidence:** Published.

Brevig, Perfekt, and Pushnitski prove that the reproducing-kernel thesis holds exactly when $N(K_\alpha) = 0$; for the canonical symbol it fails when $\alpha \geq 2$. This disproves persistence of the first two equalities throughout the proposed range, although it does not locate every transition point. Sharpness of composition-operator estimates

27.9. **The reproducing-kernel thesis for affine symbols**

Question. Brevig and Perfekt ask: "If $\varphi(s) = c + r2^{-s}$ is in $\mathcal{G}$ and $r > 0$, does it hold that $\|C_\varphi\| = S_\varphi = S_{\varphi^*}$?" See §4, p. 16.

Answer: Answered: no. **Evidence:** Sketch.

An affine example in the public note separates the operator norm from the reproducing-kernel tests. The counterexample uses a compactness/eigenvector comparison and is recorded with a proof in the linked note. Affine composition-operator norms

27.10. **The exact norm near the boundary**

Source passage. Brevig and Perfekt write that it is "tempting to conjecture that $\|C_\varphi\| = S_\varphi = S_{\varphi^*}$ could hold at least for affine symbols $\varphi(s) = 1/2 + \xi(1 - 2^{-s})$ with sufficiently small $\xi > 0$." They add that this would imply $\|C_\varphi\| = \xi^{-1/2}$ for all sufficiently small $\xi > 0$. The inferred question is whether that consequence is true; §4, p. 16.

Answer: Answered: no. **Evidence:** Sketch.

The public affine-norm analysis gives a strict counterexample to the proposed equality along the small-parameter family. It also determines the relevant norm profile, so the asymptotic statement is not inferred from a leading-order estimate. Affine composition-operator norms

27.11. **Subordination between affine symbols**

Question. Brevig and Perfekt ask: "If φ_b and φ_c are two affine symbols with the same mapping properties, is it true that either C_{φ_b} is subordinate to C_{φ_c}, or C_{φ_c} is subordinate to C_{φ_b}?" See §7.2, p. 24.

Answer: Answered: no. **Evidence:** Sketch.

The public note gives an explicit pair of affine symbols for which neither operator is subordinate to the other. The paper's moment criterion for subordination makes the two failed inequalities transparent. Affine composition-operator norms

27.12. **Schatten-class criteria for composition operators**

Question. "Describe the symbols $\varphi \in \mathcal{G}_0$ such that $C_\varphi \in S_p$." Brevig and Perfekt, Problem 2, §8.2 of *A mean counting function for Dirichlet series and compact composition operators*, p. 35.

Answer: Partial. **Evidence:** Sketch.

Bayart–Kouroupis give an S_{2m} criterion for every symbol and a one-fold power criterion for bounded-imaginary-part symbols. For arbitrary symbols, our argument gives a scalar moment-series test for $0 < p < 2$; a local one-fold criterion and the unrestricted non-even range above 2 remain open.

27.13. **Compactness of characteristic-zero operators across p**

Source result. Brevig and Perfekt state: "$\mathscr{C}_\varphi : \mathscr{H}^2 \to \mathscr{H}^2$ is compact if and only if $\lim_{\operatorname{Re} w \to 1/2^+} \mathscr{M}_\varphi(w)/(\operatorname{Re} w - 1/2) = 0$." The inferred question is whether this criterion is independent of finite $p \geq 2$; the non-even sufficiency uses arXiv:2609.11560v2.

Answer: Answered: yes. **Evidence:** Sketch.

Conditional on: arXiv:2609.11560.

The compactness criterion is necessary for every finite $p \geq 2$, and sufficient at every exponent where the local embedding holds; in particular the converse is unconditional at even exponents. The non-even converse is conditional on the cited preprint.

Answers, "Compactness across Hardy exponents", p. 47.

27.14. **Operators bounded exactly at even exponents**

Question. Brevig, Ortega-Cerdà, and Seip ask in Problem 4.1: "Is there a linear operator, densely defined on $H^p(\mathbb{T}^\infty)$ for $1 \leq p \leq \infty$, which extends boundedly exactly when p is an even integer or $p = \infty$?"§4.1, p. 28.

Answer: Answered: yes. **Evidence:** arXiv.

Andersson's realization theorem constructs every set of exponents whose finite part is relatively closed in $[1, \infty)$. The even integers form such a set, so it gives the requested operator.

27.15. **Cyclicity of an identity plus a weighted shift**

Question. Fricain and Mau define T on $H^2 = \mathbb{C}e_0 \oplus H^2_0$ by the matrix with first diagonal entry 1 and forward-shift weights $\alpha_1, \alpha_2, \ldots$, where (α_n) is bounded, then ask: "Is T cyclic?" Question 5.9, p. 21 of their paper on weighted holomorphic Dirichlet series and composition operators with polynomial symbols.

Answer: Answered: cyclic exactly when all $\alpha_j \neq 0$ and $\sum_{j \geq 0} |\alpha_1 \cdots \alpha_j|^{-2} = \infty$. **Evidence:** Sketch.

The criterion agrees with the published nonzero-weight case and adds the necessary zero-weight exclusion. When it holds, $e_0 + e_1$ is cyclic; if a weight vanishes or the reciprocal-product series converges, the relevant adjoint eigenspace has dimension at least two.

27.16. **Positive-characteristic compactness and counting functions**

Question. Bayart asks: "On $\mathcal{H}^2$, can we get a necessary condition using some counting function without any extra assumption on $\varphi \in \mathcal{G}_{\geq 1}$? Or at least, if $\varphi \in \mathcal{G}_{\geq 1}$ is supported on a finite set of prime numbers, do we have C_φ is compact on $\mathcal{H}^2$ if and only if $\mathcal{N}_\varphi(w) = o(\operatorname{Re} w)$ as $\operatorname{Re} w \to 0$?" Question 3.6, p. 7.

Answer: Partial. **Evidence:** Sketch.

The unrestricted necessary condition is known in an averaged vertical-limit form, and the ordinary criterion is established for one-prime symbols. New work extends it to symbols

generated by a single integer frequency, including composite bases, and gives mixed-cusp compact examples; the general finite-prime converse remains open.

27.17. **Spectrum of a vertical translation on $\mathscr{H}^\infty_+$**

Question. "What is the spectrum of $\mathcal{C}_\varphi$ provided $\varphi(s) = s + i\tau$?" Bayart and Yao, Question 3.8, p. 15 of *Königs maps and commutants of composition operators on the Hardy–Hilbert space of Dirichlet series.*

Answer: Answered: $\{1\}$ if $\tau = 0$, and $\mathbb{T}$ otherwise. **Evidence:** Sketch.

The full argument identifies the dense point spectrum and separately proves that each remaining point of the unit circle is outside the range of $C_\varphi - \lambda I$. This range argument is needed because the spectrum of a continuous operator on a Fréchet space need not be closed.

Answers, "Spectrum of a vertical translation", p. 49.

Bayart–Yao questions and answers

27.18. **Cyclicity forces univalence on the half-plane**

Question. "The above proof shows that provided C_φ is cyclic, φ is univalent on $\mathbb{C}_{1/2}$. Is it also univalent on $\mathbb{C}_0$?" Bayart and Yao, Question 4.2, p. 18.

Answer: Answered: yes. **Evidence:** Sketch.

For every Gordon–Hedenmalm symbol of positive characteristic that is not univalent on $\mathbb{C}_0$, the range of C_φ has infinite codimension. The proof uses bounded evaluations of finite-prime truncations, extending the cyclicity implication beyond the region where point evaluations on $\mathscr{H}^2$ are bounded.

Answers, "Cyclicity and univalence", p. 49.

Bayart–Yao questions and answers

27.19. **Minimal commutants of diagonal operators**

Question. "Does there exist a diagonal operator associated to an injective sequence (λ_n) which fails to have a minimal commutant? If so, can we give a characterization of diagonal operators with a minimal commutant?" Bayart and Yao, Question 5.6, p. 22.

Answer: Answered: yes, with a criterion. **Evidence:** Sketch.

The criterion is that no nonzero $a \in \ell^1$ annihilates every moment $\sum_n a_n \lambda_n^k$, $k \geq 0$. A Vitali-disc packing gives an explicit annihilating measure and hence an injective diagonal example without a minimal commutant; the proof also recovers the classical thin-set cases.

Answers, "Minimal commutants of diagonal operators", p. 50.

Bayart–Yao questions and answers

27.20. **Removing a zero-free hypothesis in the disc Hardy space**

Question. Under Corollary 8.5's hypotheses, Bayart and Yao ask: "Can we dispense with the assumption that ψ does not vanish on $\mathbb{D}$ in Corollary 8.5?" Here $\varphi(z) = \lambda z^p \psi(z)$, $p \geq 2$, $\lambda \neq 0$, $\psi(0) = 1$, and $\|\varphi^{[n]}\|_\infty < 1$ for some $n \geq 1$; Question 8.7, p. 33.

Answer: Answered: yes. **Evidence:** Sketch.

A local univalent Böttcher coordinate, composed with a sufficiently deep iterate, gives a small global solution of $v \circ \varphi = v^p$. It produces a commuting composition operator outside the weak-operator closure of the polynomials in C_φ, proving failure of a minimal commutant without requiring ψ to be zero-free.

Answers, "Removing the zero-free hypothesis", p. 50.

Bayart–Yao questions and answers

27.21. **Lens maps and minimal commutants**

Question. For $\mathcal{C}(z) = (1+z)/(1-z)$ and $\varphi_\alpha(z) = (\mathcal{C}(z)^\alpha - 1)/(\mathcal{C}(z)^\alpha + 1)$, Bayart and Yao ask: "Does C_{φ_α} have a minimal commutant?" The source writes $0 < \alpha < \pi/2$; the self-map range is $0 < \alpha < 1$. See Question 8.11 and Remark 8.12, p. 36.

Answer: Answered: no. **Evidence:** Sketch.

For every $0 < \alpha < 1$, reflection $Rf(z) = f(-z)$ commutes with the lens composition operator but is not in the weak-operator closure of its polynomial algebra. A bounded functional built from a Cauchy transform separates reflection from that closure.
Answers, "Lens maps and minimal commutants", p. 50.

Bayart–Yao questions and answers

27.22. **Dense range for univalent Dirichlet-series symbols**

Question. Bayart and Yao ask: "Let $\varphi \in \mathcal{G}_{>0}$ be univalent. Under which conditions does C_φ have dense range?" Question 4.10, p. 20.
Answer: Partial. **Evidence:** Sketch.

Univalence on $\mathbb{C}_0$ is necessary, and a one-prime affine family has an exact criterion: density holds precisely when $|a| \log 2 \leq 1$ for $\varphi(s) = s + c + a2^{-s}$. A sufficient condition is also known from univalent extension and image containment; a general criterion for several prime frequencies remains open.

27.23. **Weight-independence on weighted polydisc Bergman spaces**

Question. "Let $d \geq 4$ and let $\varphi \in \mathcal{O}(\mathbb{D}^d, \mathbb{D}^d) \cap C^1(\overline{\mathbb{D}^d})$. Is it true that if C_φ is continuous on $A^2_\beta(\mathbb{D}^d)$ for some $\beta \geq d-3$, then it is continuous on $A^2_\beta(\mathbb{D}^d)$ for all $\beta \geq d-3$?" Bayart–Dorval, Question 4.2, p. 21.
Answer: Answered: yes. **Evidence:** arXiv.
Conditional on: arXiv:2606.29969v1.

Dorval's Theorem 1.1 (proved as Theorem 3.6) gives a necessary-and-sufficient boundary-jet criterion independent of β, for all $d \geq 2$. Its hypotheses include the stated C^1 regularity.

27.24. **Boundary jets and weighted Bergman boundedness**

Question. Bayart and Dorval ask in Question 3.2, p. 19: "For which $\beta \geq -1$ are there C^n self-maps φ, ψ with the same contact set and the same boundary jets through order n, for which C_φ is bounded and C_ψ is unbounded on $A^2_\beta(\mathbb{D}^d)$?" Here $d \geq 3$ and $n \in \mathbb{N}$.
Answer: Partial. **Evidence:** arXiv.
Conditional on: arXiv:2601.19308.

Their construction gives intervals of weights below $d-3$, while Dorval's weight-independent criterion rules out such pairs for $\beta \geq d-3$ and $n \geq 1$. Whether the construction intervals exhaust the remaining range is open, conditional on the cited preprint.

27.25. **Second-order norm asymptotics for a boundary symbol**

Source passage. For fixed $\sigma > 1/2$, Fang, Guo, Mishra, and Muthukumar state: "We also obtain a second-order expansion of the squared norm." See the abstract and Theorem 3.4 of *Critical Norm Profiles for Finite-Prime Composition Operators on the Hardy Space of Dirichlet Series* (arXiv:2608.26041). The inferred follow-up asks for the second-order limit along $\varphi_\xi(s) = 1/2 + \xi(1 - 2^{-s})$.
Answer: Answered: 0. **Evidence:** Sketch.

The written argument establishes the limit by a second-order expansion of the extremal Hankel norm. The cited theorem gives only the leading $1/\xi + O(1)$ term; this limit is an inferred follow-up, not a question stated by its authors.

27.26. **Numerical range of a weighted composition operator**

Question. For a non-constant c_0-symbol φ and $\psi \in \mathscr{H}^2$ with $\psi_1 = 0$ such that $C_{\psi,\varphi}$ is bounded on $\mathscr{H}^2$, Lahiri, Ojha, Halder, and Birbonshi ask: "Does 0 lie in the interior of the numerical range $W(C_{\psi,\varphi})$?" Question 2.13, p. 8; Question 3.9 asks a remaining affine-symbol case, p. 11.
Answer: Answered: no as stated; yes when $\psi \neq 0$. **Evidence:** Sketch.

The zero weight gives the counterexample $W(C_{0,\varphi}) = \{0\}$. For every nonzero such weight, a two-dimensional compression has an elliptical numerical range containing a nondegenerate disc about zero.

Answers, "Numerical range of a weighted composition operator", p. 51.

27.27. **Davis–Wielandt shells of affine composition operators**

Question. Lahiri, Ojha, Halder, and Birbonshi ask in Question 5.4: "Determine the Davis–Wielandt shell $\mathrm{DW}(C_\varphi)$." Here $\varphi(s) = cs + c_1$, c is a nonnegative integer, and boundedness requires $\operatorname{Re} c_1 \geq 0$ for $c \geq 1$ and $\operatorname{Re} c_1 > 1/2$ for $c = 0$; p. 17.

Answer: Answered: exact scalar support-function description. **Evidence:** Sketch.

For positive characteristic and positive real part, the shell is described by an explicit convergent scalar series and an arithmetic maximum over integers that are not c_0-th powers; the only unattained point of its closure is the origin. The remaining characteristic cases reduce to an ellipsoid or a convex hull, with the nonclosed boundary treated separately.

Answers, "Davis–Wielandt shells of affine composition operators", p. 51.

27.28. **Which boundedness sets can occur?**

Source passage. After Theorem 1.1 Andersson writes: "We are unaware of whether this completely characterizes all possible sets A, and find it an interesting question to determine precisely what sets A can occur." The inferred question asks for the exact boundedness sets among the common-domain operators in that theorem; §1 of *Exotic operators on the Hardy spaces of the infinite-dimensional torus*.

Answer: Answered: exactly those with $A \setminus \{\infty\}$ an F_σ set in $[1, \infty)$. **Evidence:** Sketch.

The proof shows the necessary and sufficient condition is F_σ, strengthening the relatively closed sufficient class in the source theorem. The argument uses the common dense-domain convention stated with the theorem.

Answers, "Which boundedness sets can occur?", p. 54.

27.29. **Are even exponents the only universal boundedness exponents?**

Source passage. Queffélec and Queffélec formulate the conjecture that even integers are the only p for which every characteristic-zero Gordon–Hedenmalm symbol gives a bounded operator on $\mathscr{H}^p$; see the Remark after Theorem 8.10.1, §8.10.2, pp. 273–274. The question considered here is whether that conjecture holds.

Answer: Answered: no, conditional for $p > 2$. **Evidence:** arXiv.

Conditional on: arXiv:2609.11560.

The cited local-embedding theorem, together with Bayart–Brevig's composition theorem, gives boundedness for every characteristic-zero symbol at every $p \geq 2$. For $p < 2$, local embedding fails and there are unbounded examples; thus the claimed even-integer threshold is false above 2.

27.30. **Approximation numbers of compact composition operators**

Source passage. Queffélec and Queffélec conclude: "The result of 1. is optimal, and that of 2. as well, up to the logarithmic factor." This follows Theorem 8.9.3, p. 272, in *Diophantine Approximation and Dirichlet Series*. Theorem 8.8.1, p. 268, gives the positive-characteristic lower bound, and Theorem 8.9.3 gives the critical finite-dimensional estimate. The inferred questions are whether the logarithmic loss in the first lower bound can be removed and what the exact approximation-number order is in the second case.

Answer: Answered: yes in (a), and $a_n \asymp n^{-(d-1)/2}$ in (b). **Evidence:** Sketch.

For (a), an antichain of input monomials gives the sharp polynomial lower bound without the logarithm. For (b), the logarithmic loss is removed by reducing the kernel to a half-plane Laplace embedding and summing operator-norm bounds over dyadic frequency blocks.

Answers, "Approximation numbers of composition operators", p. 54.

Sharpness of composition-operator estimates

27.31. **All-exponent BMOA necessity for Volterra operators**

Source passage. Brevig, Perfekt, and Seip write: "We will use this to prove a rather curious incomplete analogue to part (c) of Theorem 2.3." They explain that their argument yields

the BMOA conclusion only for rational p. The inferred question asks whether $T_g : H^p \to H^p$ bounded forces $g \in \mathrm{BMOA}(\mathbb{C}_{1/2})$ for every real $p > 0$; our answer uses the local-embedding claim in arXiv:2609.11560v2, Theorem 1.1.
Answer: Answered: yes. **Evidence:** Sketch.
Conditional on: arXiv:2609.11560.

The conditional argument lifts from p to $q = np > 2$ by the integer-power inclusions, then applies the local embedding and the BPS Carleson-measure argument. It gives the BMOA norm bound but does not prove $X_p = X_2$.

27.32. **Sharp gain for linear symbols**

Question inferred from Bayart–Brevig, §5, and the local embedding in arXiv:2609.11560v2. For a finite-prime linear symbol touching the boundary in effective dimension d, is $C_\varphi : \mathscr{H}^q \to \mathscr{H}^p$ bounded exactly when $p \le q(d+1)/2$ for real $q \ge 2$ and $p \ge 1$?
Answer: Answered: yes. **Evidence:** Sketch.
Conditional on: arXiv:2609.11560.

The contact-box mass has sharp exponent $(d+1)/2$, which yields the matching necessary and sufficient cross-exponent criterion. Failure above the threshold is unconditional; the full real-q sufficiency uses the unrefereed local-embedding theorem, while even q is unconditional.

28. Dirichlet series and frequencies

28.1. **Bohr's theorem and completeness**

Question. Defant and Schoolmann ask: "Is there any relation between the λ's satisfying Bohr's theorem and the λ's for which $\mathscr{D}_\infty(\lambda)$ is complete?" Problem 2.14, p. 10 of arXiv: 1902.02073v1.
Answer: Answered: yes. **Evidence:** Published.

Defant and Schoolmann later prove that, for every frequency, Bohr's theorem is equivalent to completeness of $\mathscr{D}_\infty(\lambda)$ (Theorem 5.1 of *Variants of a theorem of Helson on general Dirichlet series*). This gives the requested relation; completeness of the extended space is a separate issue.

28.2. **Hypercontractivity for general frequencies**

Question. Defant and Schoolmann ask: "Given $D = \sum a_n e^{\lambda_n s} \in H^p(\lambda)$, $1 \le p < \infty$, under which assumptions on λ is it true that $D_u = \sum a_n e^{\lambda_n u} e^{\lambda_n s} \in H^q(\lambda)$ for all $1 \le q < \infty$ and $u > 0$?" Problem 4.15, p. 21. For the positive translation $P_u D(s) = D(s+u) = \sum a_n e^{-\lambda_n u} e^{-\lambda_n s}$ used here, the displayed plus signs in the source question are misprints.
Answer: Answered: by an all-frequency operator criterion. **Evidence:** arXiv.

Theorem 1.1 of Carando, Defant, Marceca, Schoolmann, and Sevilla-Peris gives a necessary-and-sufficient criterion: it is enough to test every positive translation from $H^2(\lambda)$ to $H^{2k}(\lambda)$, for each integer $k \ge 1$. Their explicit additive-energy characterization requires an additional growth hypothesis; the later published survey poses a distinct, (LC)-restricted problem.

28.3. **Translated maximal inequalities beyond Bohr's condition**

Question. Defant and Schoolmann ask: "Does Theorem 4.8 hold for a wider class of frequencies than the class of λ's satisfying (BC)? Is it even true without any condition on λ?" Problem 4.19, p. 22.
Answer: Partial. **Evidence:** Published.

Bayart proves the strong translated estimate under (NC), a class strictly larger than (LC), and gives an all-frequency counterexample. Thus the unrestricted extension is false and a substantial wider class is covered; whether Bohr's condition alone suffices remains open.

28.4. **Maximal convergence abscissas for general frequencies**

Question. In arXiv:1902.02073v1, Defant and Schoolmann ask: "Given a frequency λ, give reasonable upper and lower estimates for $\sup_{D \in H_p(\lambda)} \sigma_c(D)$ and $\sup_{D \in H_p(\lambda)} \sigma_c(D)$, for example

in terms of $L(\lambda)$." Problem 4.17, p. 21; the printed second σ_c is corrected to σ_a in published Problem 4.21.

Answer: Answered: sharp bounds in terms of $L(\lambda)$ and p. **Evidence:** Sketch.

The optimal universal bounds are $L/2 \le C_p \le S_p \le L/p$ for $p \le 2$ and $L/p \le C_p \le S_p = L/2$ for $p \ge 2$. Block constructions attain the bounds, including joint profiles for the two maximal abscissas.

Answers, "Maximal convergence abscissas for general frequencies", p. 55.

28.5. **Unconditional bases in $\mathscr{H}^p$**

Question. Saksman and Seip ask: "Does H^p have an unconditional basis if $p \in (1,\infty)$ and $p \neq 2$?" Problem 2.5, §§2–3.

Answer: Open. **Evidence:** Heuristic.

The monomials are not an unconditional basis outside the Hilbert case, but that does not rule out a different basis. Recent results for canonical monomials and finite-dimensional subspaces do not settle the question for arbitrary bases.

28.6. **A Schauder basis for $\mathscr{H}^1$**

Question. Saksman and Seip ask in Problem 2.6: "Does H^1 have a basis?" Part (a), §§2–3.

Answer: Open. **Evidence:** Heuristic.

The monomials form a Markushevich basis, which is weaker than a Schauder basis. This does not decide whether some other Schauder basis exists.

28.7. **An unconditional basis for $\mathscr{H}^1$**

Question. Saksman and Seip ask in Problem 2.6: "Does H^1 have an unconditional basis?" Part (b), §§2–3.

Answer: Open. **Evidence:** Heuristic.

The question concerns arbitrary bases; finite-coordinate constructions and results for the canonical monomials do not provide an infinite-dimensional unconditional basis or an obstruction to one.

28.8. **Completeness of general Dirichlet-series spaces**

Question. Defant and Schoolmann ask: "Characterize completeness of $\mathscr{D}_\infty(\lambda)$ and $\mathscr{D}_\infty^{\mathrm{ext}}(\lambda)$." Problem 2.13, p. 10 of the published survey. Its Theorem 2.12 already reduces the unresolved part to whether completeness of the extended space implies completeness of the ordinary space.

Answer: Open. **Evidence:** Heuristic.

The published survey isolates the residual implication from completeness of $\mathscr{D}_\infty^{\mathrm{ext}}(\lambda)$ to completeness of $\mathscr{D}_\infty(\lambda)$. The cited results characterize the latter through Bohr's theorem but do not resolve the extended-space implication.

28.9. **Banach-valued maximal partial sums**

Question. Defant and Schoolmann ask: "Under which assumptions on the Banach space X and the frequency λ" Problem 5.7, p. 26. More precisely, do there exist such assumptions under which, for all $f \in H_p^\lambda(G,X)$ and $u > 0$,

$$\left\| \sup_N \left\| \sum_{n=1}^{N} \widehat{f}(h_{\lambda_n}) e^{-u\lambda_n} h_{\lambda_n} \right\|_X \right\|_{L^p(G)} \le C \|f\|_{H_p^\lambda(G,X)}?$$

They also ask which of λ, X, p, u determine C.

Answer: Partial. **Evidence:** Sketch.

The new estimate proves the source's suggested case under (NC), for every Banach space and $1 \le p \le \infty$, with a constant independent of X and p. A classification of all pairs (X,λ) remains open.

28.10. **Cesàro means in $\mathscr{H}^1$**

Question. Defant and Schoolmann ask: "Do the first-order Cesàro means of every $D \in \mathscr{H}^1$ converge to D in $\mathscr{H}^1$?" This is the final question in §8, p. 19; the authors answer it immediately afterward.

Answer: Answered: no. **Evidence:** arXiv.

The authors answer no immediately after the question: a positive answer would imply norm convergence for every analytic Fourier series in H^1, which is false.

28.11. **Dropping the bounded-argument hypothesis for an omitted curve**

Question. Dan and Guo ask: "In Theorem 7.1, can the assumption that γ has a bounded argument function be dropped?" Question 7.6, p. 42 of *The Periodic Dilation Completeness Problem.*

Answer: Answered: yes. **Evidence:** Sketch.

For a zero-free $F \in H^2(\mathbb{D}_2^\infty)$ whose range on $\mathbb{D}_2^\infty$ avoids the simple curve γ in Theorem 7.1, the answer is affirmative. The proof uses a conformal map onto the component of the complement containing the range and bounded approximate reciprocals.

Answers, "Cyclicity when the omitted curve has unbounded argument", p. 57.

28.12. **Kozlov's dilation-completeness conjecture**

Source passage. Dan and Guo write: "By Proposition 8.1, we conjecture that the dilation system $\mathcal{D}_\theta$ is complete only for $\theta \in \{1, 1/2, 2/3\}$." Remark 8.2, p. 45. The inferred remaining question is whether any irrational θ gives a complete system.

Answer: Partial. **Evidence:** arXiv.

Dan–Guo's later theorem classifies intervals with rational endpoints; for $(0, \theta)$ this gives exactly $\theta \in \{1, 1/2, 2/3\}$ among rational values. Irrational θ remains unresolved.

28.13. **Bohr's theorem and the Helson maximal estimate**

Question. Bayart asks: "It seems natural to ask whether (11) always follows from (10) or, equivalently, if any frequency λ satisfying Bohr's theorem also satisfies Helson's theorem." Question 4.8, §4.1.

Answer: Open. **Evidence:** Heuristic.

Bayart proves the estimate under his condition (NC), which covers a strict subfamily of the Bohr frequencies, and gives a counterexample without Bohr's theorem. Whether Bohr's theorem alone suffices remains open.

28.14. **Multiplier algebras of weighted Dirichlet-series spaces**

Question. Sahu asks: "For which weight sequences $\mathbf{w}$ is $M(H_{\mathbf{w}})$ isometrically isomorphic to $H^\infty(\operatorname{Re} s > \delta_{\mathbf{w}}/2) \cap \mathcal{D}$?" Question 2.2, p. 325.

Answer: Partial. **Evidence:** Published.

Sahu's Möbius-positivity condition is sufficient but not necessary as a full classification. Later published examples give equality and strict inclusion in additional cases; a general characterization is still open.

28.15. **Completeness for the square-root logarithm frequency**

Question. Defant and Schoolmann ask: "Is $\mathscr{D}_\infty((\sqrt{\log n}))$ complete?" Problem 2.15, p. 10 of arXiv:1902.02073v1.

Answer: Answered: yes. **Evidence:** Published.

For $\lambda_n = \sqrt{\log n}$, Landau's condition holds, hence Bohr's theorem does. Defant and Schoolmann's Theorem 5.1 equates Bohr's theorem with completeness of $\mathscr{D}_\infty(\lambda)$; their Corollary 5.5 gives this example explicitly.

28.16. **A condition weaker than Landau's condition**

Question. Defant and Schoolmann ask: "Find a reasonable condition on the frequency λ which is weaker than (LC), but is sufficient for the equality $\sigma_b^{\mathrm{ext}}(D) = \sigma_b(D)$ for all λ-Dirichlet series D." Problem 2.16, p. 11.

Answer: Answered: yes. **Evidence:** Published.

Bayart's condition (NC) is strictly weaker than (LC) and implies Bohr's theorem. Consequently $\sigma_b^{\mathrm{ext}}(D) = \sigma_b(D)$ for every λ-Dirichlet series D; the proof gives the stronger equality $\sigma_b^{\mathrm{ext}} = \sigma_b = \sigma_u$.

28.17. **Bass stable rank for multiplicative subsemigroups**

Question. Sasane conjectures: "Does $R(S) = \infty$ imply that the Bass stable rank of $\mathscr{H}_S^\infty$ is infinite?" Conjecture 6.1, §6, p. 11.

Answer: Answered: no. **Evidence:** Sketch.

The prime-power semigroup has infinite rank under the source's weak independence definition, but its algebra is isometric to $H^\infty(\mathbb{D})$, whose Bass stable rank is one. The stronger unique-generator hypothesis of the source's Theorem 6.2 is therefore essential to that theorem's conclusion.

Answers, "Bass stable rank of a Dirichlet-series algebra", p. 57.

28.18. **Rational dimension of logarithms in a short interval**

Source passage. Hedenmalm, Lindqvist, and Seip write: "The latter translates into the following basic number-theoretic problem: What is the multiplicatively rational dimension of the interval $N \le n < N + k$?" Here this dimension is the dimension over $\mathbb{Q}$ of the span of the logarithms; the inferred question asks for it as a function of N and k. §6, p. 30.

Answer: Partial. **Evidence:** Sketch.

For fixed length k, full rank is known for all $N \ge (100k)^{k+1}$, and the exceptional length-five starts are classified. A complete formula for the rank as both N and k vary is still open.

28.19. **An ordinary Dirichlet series with exactly one zero**

Question. Balazard asks: "Does there exist an ordinary Dirichlet series converging in some open half-plane, and having there one zero, and only one?" Problem-session item 8, p. 390 of the Oberwolfach Report 06/2014.

Answer: Open. **Evidence:** Heuristic.

A general Dirichlet series example does not answer the ordinary-series question. The purported ordinary-series construction of Hilberdink and Saias was withdrawn for an irreparable error; no unconditional ordinary example is established in the checked sources.

28.20. **Uniform translated partial sums in the Oberwolfach problem session**

Question. Bayart asks: "Is it the case that $\sigma_N \to 0$ as $N \to \infty$?" Oberwolfach Report 51/2017, problem 4, p. 33. Here $\sigma_N = \sup_{f \in H^\infty(\mathbb{C}_0)} \inf\{\sigma > 0 : \|S_N f\|_{H^\infty(\mathbb{C}_\sigma)} \le \|f\|_{H^\infty(\mathbb{C}_\sigma)}\}$.

Answer: Answered: no. **Evidence:** Sketch.

For every $N \ge 2$ there is a three-term polynomial for which the defining inequality fails for every $\sigma > 0$; the explicit construction is given in the answers entry.

Answers, "The translated partial-sum inequality", p. 58.

28.21. **A Dirichlet polynomial on the line** $\operatorname{Re} s = 1$

Question. Erdős and Ingham ask: "Is it true that $D(s) \neq 0$ for all $\operatorname{Re}(s) = 1$?" Here $D(s) = 1 + 2^{-s} + 3^{-s} + 5^{-s}$; Oberwolfach Report 51/2017, Problem 2, p. 31.

Answer: Partial. **Evidence:** Published.

The general assertion with an arbitrary summable set of terms is false, but the specific four-term polynomial above remains unresolved. A 2025 Oberwolfach report restates this finite case.

29. Vertical limits

29.1. An H^1 endpoint for vertical limits

Question. Andersson asks: "Can one take $E = \mathscr{C}_p(f)$ in Theorem 5.1 also in the case $p = 1$?" Question 1, §5, pp. 26–30.
Answer: Answered: yes. **Evidence:** Sketch.
Conditional on: The source's standing hypotheses, including the finite-mean assumption used for $p = 1$.

The endpoint identity holds on the full Carlson set by the linked proof. Its conditionality is inherited from the source's standing lemma; the result does not assert a new external theorem. The Hardy–Stein endpoint

29.2. One exceptional set for every target value

Question. Andersson asks: "Can one choose the set of full measure in Theorem 1.2 independently of $\zeta \in \mathbb{C} \setminus \{f(+\infty)\}$?" Question 2, pp. 26–30.
Answer: Open. **Evidence:** Heuristic.

The source proves the limit for each fixed target with a target-dependent full-measure set. Its ordered cutoff theorem and the related completed-class result take limits in a different order, so they do not imply the common-set boundary-first statement.

29.3. Counting functions in the completed Nevanlinna class

Question. Andersson asks: "Let $f \in \mathscr{N}$. Does there exist a set $E \subseteq \mathbb{T}^\infty$ of full measure such that if $\chi \in E$, then f_χ has an analytic continuation to $\mathbb{C}_0$ for which

$$\lim_{\sigma_0 \downarrow 0} \lim_{T \to \infty} \frac{\pi}{T} \sum_{\substack{s \in \mathscr{Z}(f_\chi - \zeta) \\ |\operatorname{Im} s| < T \\ \operatorname{Re} s > \sigma_0}} (\operatorname{Re} s - \sigma_0)$$

exists and is finite for all $\zeta \in \mathbb{C} \setminus \{f(+\infty)\}$?" Question 3, pp. 26–30; zeros are repeated according to multiplicity.
Answer: Answered: yes. **Evidence:** Sketch.
Conditional on: arXiv:2606.20293v2, Lemma 3.1.

The proof constructs one conull set controlling all targets through a countable dense set and analytic dependence, then passes to arbitrary targets by the completed-class counting estimate. It uses Lemma 3.1 of the source and preserves the order $T \to \infty$ followed by the boundary cutoff.
Answers, "Counting functions in the completed Nevanlinna class", p. 58.

29.4. The optimal H^p order-to-mean abscissa for $1 \le p \le 2$

Source passage. Brevig and Kouroupis state: "Fix $\beta > 0$. If an element f of $\mathscr{H}^2$ has an analytic continuation to $\mathbb{C}_0$ such that $\mu_f(\sigma) \le \beta$ for every $\sigma > 0$, then $\sigma_{\mathrm{m}}(f) \le \beta/(1 + 2\beta)$." Their Theorem 4 proves sharpness in $\mathscr{H}^2$. The inferred question is whether the same optimal bound holds in $\mathscr{H}^p$ for $1 \le p \le 2$.
Answer: Answered: $\beta/(1 + 2\beta)$, sharply. **Evidence:** Sketch.
Conditional on: arXiv:2609.35496.

The bound equals the H^2 value and is attained by the Brevig–Kouroupis extremal function. The proof uses Helson's inequality to transfer to H^2 and the source's sharp example; it relies on the source hypotheses and theorem.
Answers, "The optimal H^p order-to-mean abscissa for $1 \le p \le 2$", p. 59.

29.5. The order-to-mean abscissa for $p > 2$

Source passage. Brevig and Kouroupis state: "Fix $\beta > 0$. If an element f of $\mathscr{H}^2$ has an analytic continuation to $\mathbb{C}_0$ such that $\mu_f(\sigma) \le \beta$ for every $\sigma > 0$, then $\sigma_{\mathrm{m}}(f) \le \beta/(1 + 2\beta)$." The inferred question asks for the optimal analogue when $p > 2$.
Answer: Open. **Evidence:** Heuristic.

The source proves the sharp H^2 estimate $\beta/(1+2\beta)$. Power reduction yields bounds at even exponents, but does not identify the optimal general real-$p > 2$ value or its extremizers.

29.6. **The order-zero mean-value formula in H^p**

Source passage. Brevig and Kouroupis state: "If f in $\mathscr{H}^2$ has an analytic continuation to $\mathbb{C}_0$ of order 0, then

$$\|f\|_{\mathscr{H}^2}^2 = \lim_{\sigma\to 0^+} \lim_{T\to\infty} \frac{1}{2T} \int_{-T}^{T} |f(\sigma+it)|^2 \, \mathrm{d}t.$$

" Corollary 2, §2. The inferred question asks whether this ordered formula extends to every $1 \le p < \infty$.

Answer: Answered: yes. **Evidence:** Sketch.

Conditional on: arXiv:2609.35496 and arXiv:2401.09996.

For $p = 2k$, apply the H^2 formula to f^k. For general p, the proof combines hypercontractive shifts and the mean-value theorem for an even exponent above p, then lets the shift tend to zero. Both cited results are standing hypotheses.

Answers, "The order-zero mean-value formula in H^p", p. 60.

30. Optimal polynomial approximants and cyclicity

30.1. **A common zero-free compact set**

> Let $\{p_n^*\}$ be optimal approximants to $f \in D_\alpha \setminus \{0\}$. Is there a compact set $K \subset \mathbb{D}^2$ such that $p_n^*(z) \neq 0$ for $z \in K$ and all n? Similarly, if $\{p_n^*\}$ are optimal approximants to $f \in H_d^2$, is there a compact set $K \subset B_d$ such that $p_n^*(z) \neq 0$ for $z \in K$ and all n?

Sargent–Sola, *Optimal approximants and orthogonal polynomials in several variables*, arXiv: 2002.08790v2, Question 24, p. 21.

Answer: Answered: iff $f(0) \neq 0$ for nonempty K. **Evidence:** Sketch.

Our full written projection argument proves a common zero-free neighbourhood of 0 when $f(0) \neq 0$. If $f(0) = 0$, every approximant is identically zero, so no nonempty K exists. The source does not require K to be nonempty; allowing $K = \varnothing$ makes its literal question vacuous.

30.2. **When approximants are Taylor polynomials**

> Given $g \in H$ and a power series $\varphi(z) = \sum a_k z^k$, can we characterize $f \in H$ such that the nth optimal polynomial approximants to g/f are given by $T_n(\varphi)$ for all n greater than some $M > 0$?

Felder, *Optimal polynomial approximants and orthogonal polynomials*, arXiv:2003.10015v2, Question 1, p. 5.

Answer: Partial. **Evidence:** Sketch.

Felder's Theorem 3.8 classifies the normalized kernel-at-zero target. Our full written partial argument supplies Hardy-space and selected weighted polynomial or sparse-level examples. The classification for general g remains open; all-degree banded-recurrence rigidity theorems do not apply to sparse orthogonal levels.

30.3. **When the cyclic projection is a polynomial multiple**

> Given $g \in H$ and supposing p is a polynomial, can we characterize f such that $\Pi_{[f]}(g) = pf$?

Felder, *Optimal polynomial approximants and orthogonal polynomials*, arXiv:2003.10015v2, Question 2, p. 5.

Answer: Partial. **Evidence:** Sketch.

Felder's Theorem 3.8 classifies the normalized kernel-at-zero target. Our full written partial argument supplies Hardy-space and selected weighted polynomial or sparse-level examples. The classification for general g remains open; all-degree banded-recurrence rigidity theorems do not apply to sparse orthogonal levels.

30.4. **The extremal Jacobi norm**

What is M for the spaces D_α when $\alpha < 0$, $\alpha \neq -1$? What is the corresponding extremal function?

Bénéteau–Centner, *A survey of optimal polynomial approximants, applications to digital filter design, and related open problems*, arXiv:2102.01725, Open Question 1, p. 18.
Answer: Open. **Evidence:** Heuristic.

The source defines M as the infimum of $|z|$ over zeros of all optimal polynomial approximants in the space. For $\alpha < 0$, $M = 2/\|J\|$. Bénéteau–Khavinson–Liaw–Seco–Simanek, *Rev. Mat. Iberoam.* 35 (2019), Theorem 4.2 and Corollary 4.5, identify the extremal constant with the norm of a symmetric Jacobi matrix and prove existence of an extremizer. The case $\alpha = -1$ is explicit. These results do not evaluate the norm for the other negative parameters; no proposed value is supported here.

30.5. **Zero-free sectors at the circle**

Given a cyclic function $f \in H^2_\omega$, and given $\zeta \in \mathbb{T}$, is there a sector with vertex at ζ that is devoid of zeros of opas? More generally, are there regions or rays that the zeros avoid?

Bénéteau–Centner, *A survey of optimal polynomial approximants, applications to digital filter design, and related open problems*, arXiv:2102.01725, Open Question 2, p. 19.
Answer: Partial. **Evidence:** Sketch.

Our written argument gives an admissible block weight and a cyclic $f = (1-z)^2$ for which every sector at every $\zeta \neq 1$ contains zeros of approximants. The weight is not quasi-integral. Thus the quasi-integral setting motivating the source question, including D_α with $\alpha < 0$, remains open; the example also leaves $\zeta = 1$ untreated.

Full written argument.

30.6. **Rates for functions without analytic extension**

Given a function $f \in D_\alpha$ such that f does not have an analytic extension to the closed disk, find the rate of decay of $\|q_n f - 1\|_\alpha$.

Bénéteau–Centner, *A survey of optimal polynomial approximants, applications to digital filter design, and related open problems*, arXiv:2102.01725, Open Question 5, p. 22.
Answer: Partial. **Evidence:** Sketch.

Our full written partial argument constructs cyclic functions with arbitrarily slow decay, for every real α, and functions realizing prescribed subgeometric upper bounds. Consequently nonextension alone imposes no universal rate. A characterization of the rate for an individual f, as asked in the source, remains open.

30.7. **Boundary convergence for polynomial reciprocals**

Suppose f is a polynomial with zeros $Z(f)$ in the complement of $\mathbb{D}$ and let q_n be the n-th opa of $1/f$ in H^2_ω. For which weights $\{\omega_k\}_{k\in\mathbb{N}}$ is it true that $1 - q_n f$ converges to 0 uniformly on compact subsets of $\overline{\mathbb{D}} \setminus Z(f)$?

Bénéteau–Centner, *A survey of optimal polynomial approximants, applications to digital filter design, and related open problems*, arXiv:2102.01725, Open Question 6, p. 24.
Answer: Partial. **Evidence:** Sketch.

Bénéteau–Manolaki–Seco, *Constr. Approx.* 54 (2021), Theorems 1.8 and 6.1, treat simple zeros in Hardy and Bergman spaces and all-boundary simple zeros under monotone doubling weights. Our written argument gives the threshold $\beta < 1/m$ for $f = (1-z)^m$ and $\omega_k = e^{-ck^\beta}$, $c > 0$, $0 < \beta < 1$. Arbitrary admissible weights and mixed multiple zeros remain open.

30.8. **Boundary convergence beyond polynomials**

Does Theorem 5.5 hold for functions other than polynomials, and if so, for which ones?

Bénéteau–Centner, *A survey of optimal polynomial approximants, applications to digital filter design, and related open problems*, arXiv:2102.01725, Open Question 7, first question, p. 24.
Answer: Partial. **Evidence:** Published.

Bénéteau–Manolaki–Seco, *Constr. Approx.* 54 (2021), Theorem 1.8, proves the quoted convergence for polynomials with simple zeros. Bénéteau–Ivrii–Manolaki–Seco, *Simultaneous zero-free approximation and universal optimal polynomial approximants*, main theorem (arXiv: 1811.04308), published in *J. Approx. Theory* 256 (2020), 105389, gives a dense G_δ universality class. Baxter's theorem supplies nonpolynomial sufficient conditions in H^2, but these sufficient and generic results do not characterize either requested class.

30.9. **Characterizing approximant universality**

On the other hand, can we characterize the functions for which the universality behavior as in Theorem 5.4 occurs?

Bénéteau–Centner, *A survey of optimal polynomial approximants, applications to digital filter design, and related open problems*, arXiv:2102.01725, Open Question 7, second question, p. 24.
Answer: Partial. **Evidence:** Published.

Bénéteau–Manolaki–Seco, *Constr. Approx.* 54 (2021), Theorem 1.8, proves the quoted convergence for polynomials with simple zeros. Bénéteau–Ivrii–Manolaki–Seco, *Simultaneous zero-free approximation and universal optimal polynomial approximants*, main theorem (arXiv: 1811.04308), published in *J. Approx. Theory* 256 (2020), 105389, gives a dense G_δ universality class. Baxter's theorem supplies nonpolynomial sufficient conditions in H^2, but these sufficient and generic results do not characterize either requested class.

30.10. **Zero exclusion for Hardy approximants**

Let $1 < p < \infty$, $f \in H^p$, and $n \in \mathbb{N}$. If $f(0) \neq 0$, then $q_{n,p}[f, 1]$ is zero-free in $\overline{\mathbb{D}}$.

Centner, *Optimal polynomial approximants in L^p*, arXiv:2112.14002v3, Conjecture 7.1, p. 23.
Answer: Answered: no. **Evidence:** Sketch.

Our full written counterexample has a degree-one approximant with a zero inside $\mathbb{D}$, at an exponent $p \neq 2$. This refutes even open-disc zero exclusion. The classical Hilbert-space conclusion at $p = 2$ is unaffected.

Full written argument.

30.11. **Pairing spurious zeros with electrostatic partners**

Let $\{\xi_{n,k}\}$ be the set of spurious roots of φ_n and $\{s_{n,\ell}\}$ the set of the roots of the electrostatic partner S_n which are inside the disk and do not accumulate on $\mathbb{T}$. Then both sets has the same amount of elements and we can reorganize all of them in pairs $(\xi_{n,k}, s_{n,k})$ in such a way that

$$\xi_{n,k} = s_{n,k} - \frac{s_{n,k}}{n} + o(1/n),$$

for any of these pairs.

Orive–Sánchez-Lara–Seco, *Electrostatic interpretation of the zeros of optimal polynomial approximants*, arXiv:2507.15488, Conjecture 1, p. 20.
Answer: Answered: no. **Evidence:** Sketch.

Our full written counterexample uses a fixed cubic with three simple zeros on $\mathbb{T}$. Along a subsequence, two spurious zeros coalesce on the $1/n$ scale, whereas the two partner zeros have different limiting displacements. The example refutes the pairing clause; it does not refute the separate equal-cardinality assertion.

Answers, "Pairing spurious zeros with electrostatic partners", p. 60.

30.12. **Weak limits of approximant zeros**

It is conjectured that the same should hold for the OPA $\{p_n\}$ themselves.

Orive–Sánchez-Lara–Seco, *Electrostatic interpretation of the zeros of optimal polynomial approximants*, arXiv:2507.15488, unnumbered conjecture, p. 3.

Answer: Answered: Haar measure. **Evidence:** Sketch.

"The same" refers to convergence of normalized zero measures to uniform arc length on $\mathbb{T}$. Our full written deduction proves this for every critical polynomial and positive power of one, using the published asymptotics of Martínez-Finkelshtein–McLaughlin–Saff. The argument includes the bounded number of zeros that may escape after reversal.

Answers, "Weak limits of approximant zeros", p. 61.

30.13. **Removing an inner factor by a scalar**

Let $1 < p < \infty$ and $f \in H^p$ with $f(0) \neq 0$. If J is any non-constant inner function, then there exists a constant $c = c_f \in \mathbb{T}$ such that $\|1 - Jf\|_p > \|1 - cf\|_p$.

Bénéteau–Cheng–Felder–Khavinson–Manolaki–Maronikolakis, arXiv:2511.08000v2, Conjecture 4.3, p. 12.

Answer: Answered: no. **Evidence:** Sketch.

Our full written counterexample uses a scale parameter and defeats every unimodular scalar. It refutes the literal existential comparison, independently of whether the notation c_f is intended to prohibit dependence on J.

Full written argument.

30.14. **Bidisc approximant zeros near the origin**

It has been suggested by A. Dayan [7] that, without imposing symmetry, the zeros may approach the origin arbitrarily closely.

Navío–Seco, arXiv:2609.37180, Introduction after Theorem 1.1, p. 3.

Answer: Answered: yes. **Evidence:** Sketch.

The motivating passage concerns linear optimal approximants to reciprocals of stable bidisc polynomials. Our full written construction uses high powers of $(1 + z^2/2)(1 + w^2/2)(1 + zw/2)(1 + it(z - w))$. The normalized boundary mass concentrates at two symmetric torus maxima, and the resulting linear approximant has a zero tending to the origin as $t \downarrow 0$.

30.15. **Cyclic polynomials on the bidisc**

Characterize cyclic polynomials in the Dirichlet spaces D_α.

Dirichlet Series and Function Theory in Polydiscs, Oberwolfach Report 06/2014, problem session, Bénéteau, item 1, pp. 383–384.

Answer: Answered: classification. **Evidence:** Published.

Here $\|\sum a_{kl} z_1^k z_2^l\|_\alpha^2 = \sum (k+1)^\alpha (l+1)^\alpha |a_{kl}|^2$. Bénéteau–Knese–Kosiński–Liaw–Seco–Sola, *Trans. Amer. Math. Soc.* 368 (2016), 8737–8754, Main Theorem, gives the full classification. The thresholds depend on the irreducible factors and their boundary zero sets; in particular, $1 - z_1$ is cyclic exactly for $\alpha \leq 1$, whereas $1 - z_1 z_2$ is cyclic exactly for $\alpha \leq 1/2$.

30.16. **A real-variable proof of the Nyman equivalence**

Is there a real variable proof of the equivalence of (2) and (3)?

Dirichlet Series and Function Theory in Polydiscs, Oberwolfach Report 06/2014, Balazard's talk, Question 1, p. 350.

Answer: Open. **Evidence:** Heuristic.

In the source, (2) is $\pi(x) = \mathrm{li}(x) + O(x^{1/2+\varepsilon})$ for every $\varepsilon > 0$, and (3) says $\chi \in \overline{\mathrm{span}}\{\{t/\alpha\} : \alpha \geq 1\}$ in $L^2(0, \infty; t^{-2}\,\mathrm{d}t)$. Nyman's thesis (1950) proves the equivalence with the Riemann hypothesis, and Báez-Duarte (2003) restricts α to integers. These theorems do not provide the requested real-variable proof; the methodological restriction remains unspecified.

30.17. **Irrational local maxima of the autocorrelation**

Does there exist a positive irrational where A has a strict local maximum?

Dirichlet Series and Function Theory in Polydiscs, Oberwolfach Report 06/2014, Balazard's talk, Question 2, p. 350.

Answer: Answered: no. **Evidence:** Sketch.

Our full written proof gives a positive symmetric second difference at every positive irrational along a subsequence of continued-fraction scales. Thus no positive irrational is a local maximum, strict or otherwise. Báez-Duarte–Balazard–Landreau–Saias' published rational-maxima result then identifies all positive local maxima.

Answers, "Irrational local maxima of the autocorrelation", p. 61.

30.18. **The discrete Nyman invariant subspace**

Is it true that $V = \{D \in E^2,\ D(s)/\zeta(s) \text{ is holomorphic for } \Re s > 1/2\}$?

Dirichlet Series and Function Theory in Polydiscs, Oberwolfach Report 06/2014, Balazard's talk, Question 3, p. 352.

Answer: Open. **Evidence:** Heuristic.

Here E^2 consists of Dirichlet series with $D(s)/s \in H^2(\Re s > 1/2)$ and V is the closed span of $\zeta(s)\sum_{k\le n} c_k k^{-s}$ with $\sum c_k/k = 0$. Báez-Duarte's criterion proves the equality under the Riemann hypothesis. Bercovici–Foias and Balazard–Saias treat the continuous Nyman analogue; their invariant-subspace methods do not supply a theorem for the integer-dilation semigroup. The unconditional equality is open.

30.19. **Continuity of Nyman distances**

Is $D(\lambda)$ a continuous function of λ?

Dirichlet Series and Function Theory in Polydiscs, Oberwolfach Report 06/2014, problem session, Balazard, item 8, second question, p. 390.

Answer: Partial. **Evidence:** Sketch.

Here $D(\lambda) = \operatorname{dist}_L(\chi, \overline{\operatorname{span}}\{e_\alpha : 1 \le \alpha \le \lambda\})$, $L = L^2(0,\infty; t^{-2}\,\mathrm{d}t)$, $\chi(t) = [t \ge 1]$, and $e_\alpha(t) = \{t/\alpha\}$. Our full written partial argument proves left continuity, continuity at 1, and strict improvement over $D(1)$ for sufficiently large cutoffs. Right continuity at arbitrary $\lambda > 1$ and strict decrease on every interval remain open.

30.20. **Strict decrease of Nyman distances**

Is $D(\lambda)$ a strictly decreasing function of λ?

Dirichlet Series and Function Theory in Polydiscs, Oberwolfach Report 06/2014, problem session, Balazard, item 8, third question, p. 390.

Answer: Partial. **Evidence:** Sketch.

Here $D(\lambda) = \operatorname{dist}_L(\chi, \overline{\operatorname{span}}\{e_\alpha : 1 \le \alpha \le \lambda\})$, $L = L^2(0,\infty; t^{-2}\,\mathrm{d}t)$, $\chi(t) = [t \ge 1]$, and $e_\alpha(t) = \{t/\alpha\}$. Our full written partial argument proves left continuity, continuity at 1, and strict improvement over $D(1)$ for sufficiently large cutoffs. Right continuity at arbitrary $\lambda > 1$ and strict decrease on every interval remain open.